Imprecise beliefs, pure equilibria and city charges for competing delivery carriers

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Abstract

Competing parcel carriers share streets and customers while planning with travel-time data too thin to identify a law. We model them as players of a Cournot–congestion game: each carrier chooses a routing plan, interacts with rivals through zone aggregates of deliveries and vehicle-hours, and holds credal beliefs about traffic and rivals. Under exposure-based congestion pure imprecise equilibria exist at every belief width. Under vehicle-count congestion, deciding existence is strongly NP-complete below an explicit width; above a width that does not grow with the number of carriers, every mixed equilibrium rounds to an exact pure E-admissible equilibrium, and this bound is attained. We give an exact column-generation pricing algorithm for routes whose arc times are probability boxes with unknown dependence, embedded in branch-and-price, and a leader method, dual implementation pricing, that computes a city's zone charges by implementing a target social optimum with one best response per carrier per round instead of solving equilibria over a grid. On nine cities built from OpenStreetMap, gridded population and published parameters, dual implementation pricing made its target an exact equilibrium in every run and, where a full grid exists, reached 104–135% of the grid-optimal gain with at most 9.4% of its routing calls, mostly through zone-specific charges. The empirical price of anarchy is near one where carriers serve almost all demand and up to 1.94 where service is partial; among flat charges a per-delivery subsidy captures most of the gain, while a congestion charge alone captures little in all but one case.

Keywords: last-mile delivery, game theory, imprecise probabilities, vehicle routing, column generation, price of anarchy, smart cities

1. Introduction

Urban parcel delivery is carried out by several firms at once. Postal operators, integrators, retailer fleets and platforms send vans into the same neighbourhoods, sell to overlapping customers and slow one another down on the same streets. City authorities respond with instruments that act on all of them together: congestion and low-emission zones, kerbside and loading rules, and charges or subsidies per delivery [1, 2, 3, 4]. Designing such instruments requires a model of how competing carriers react to them, and the reaction is a routing decision taken under uncertainty about travel times and about what the rivals will do.

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Operations research has mature tools for each part of this problem in isolation. Stochastic, robust and distributionally robust vehicle routing treats one decision maker facing uncertain demand or travel times [5, 6, 7, 8]. Collaborative routing studies carriers who agree to exchange requests or share resources [9, 10, 11]. Platform economics studies competition through prices, wages and congestion, with routing reduced to queues [12, 13, 14]. Bilevel last-mile models put a platform or a public authority above carriers that do not interact with one another [15, 16]. The structured literature survey that preceded this work found four gaps where these streams fail to meet: no routing-level equilibrium model of competing parcel carriers under uncertainty (G1); no leader-follower model in which a public leader faces interacting carriers (G2); no data-driven or ambiguity-aware equilibrium computation for routing games (G5); and no public multi-carrier instances or protocol for reporting equilibrium quality (G6). We address parts of each, and we state below which parts.

Two features of the setting shape the model. First, a carrier rarely knows the probability law of travel times; it has a few weeks of observations and published congestion indices, which support bounds on distribution functions (probability boxes) and say nothing about the dependence between road segments. Second, a carrier's expectation about its rivals' activity in a zone is itself imprecise. Imprecise probability theory [17, 18] offers decision criteria for such credal beliefs, from the cautious Γ-maximin to the permissive E-admissibility. In a game these criteria lead to equilibrium notions whose existence and computation behave very differently, and that difference matters for a regulator who wants to predict the outcome of a charge.

Contributions. We claim the following, with the caveats on prior art given in Sections 2 and 4.

(i) *Model.* A Cournot–congestion game among carriers who choose routing plans from finite, nonconvex sets and interact through zone aggregates, with credal beliefs about traffic and rivals, four decision criteria, and two congestion regimes: exposure-based (D1) and vehicle-count (D2) (Section 3).

(ii) *Theory.* A credal absorption theorem: any mixed equilibrium can be rounded to a pure profile that is an exact E-admissible (hence maximal and interval-undominated) equilibrium once the belief width reaches an explicit radius whose bound does not grow with the number of carriers; the bound is attained. Below an explicit width, existence of pure imprecise equilibria is strongly NP-complete under D2 for all four criteria, while under D1 they exist at every width. The Γ-maximin equilibria form the equilibrium set of a precise game with a computable surcharge. These results, with tightness examples and price-of-anarchy transfers, are in Section 4; full proofs are in Appendix A. The discrete core statements, including the Shapley–Folkman lemma and the exact potential identity, are machine-checked in Lean 4 with no unproved steps; the full theorems, with their bounds and complexity claims, are not formalised (Section 4.8).

(iii) *Algorithms.* Budget-resource labelling (BRL), an exact pricing algorithm for routes whose arc times are given by probability boxes with unknown dependence: the worst-case expected convex piecewise-linear route loss becomes a shortest-path problem with one decision resource per kink, and labels are compared by a dominance rule restricted to a computable completion window; a branch-and-price method with this pricer makes

the carriers' integer routing step exact. Dual implementation pricing (DIP), a method for the city leader: it computes a target social optimum, starts from a closed-form tangent charge per zone derived from an exact identity between the carriers' potential and the social cost, and finds charges that make the target an exact equilibrium by cutting planes whose separation step is one best response per carrier at the target, so that no equilibrium is solved for the leader's decision. As a secondary method, zone-market clearing (ZMC) searches for good pure equilibria of the D1 game with one routing call per carrier per round, independently of the number of zones; on the city games it was not better than plain best response. These components are integrated into one equilibrium engine (Section 5).

(iv) *Evidence.* Nine city instances built from OpenStreetMap road networks, gridded population and web-sourced operating parameters, with every weak parameter flagged, and experiments on equilibria, city charges, empirical price of anarchy and stability, imprecision and algorithm performance over three random draws (seeds) of each city slice, including negative findings and the conclusions that change with the seed, together with tests of the theorems on the city equilibria and on vehicle-count variants of the games; the main game experiments use an exact integer routing step, and we quantify how the earlier inexact step changed the equilibria (Sections 6 and 7).

We do not claim a forecast for any city. The instances describe a representative slice of each city with proxy demand and several weak parameters, and the price-of-anarchy values are empirical, computed from the few equilibria the methods found.

2. Related work

Competing carriers and platforms. Figliozzi, Mahmassani and Jaillet studied carriers competing in sequential auctions for truckload requests, with routing-level opportunity costs and dynamic arrivals [19, 20]. Their setting has deterministic travel times and is evaluated by simulation, not equilibrium computation. Shared infrastructure among competing parcel firms has been analysed with stylised games: sharing of collection-delivery points between two express companies [21] and cooperative placement of parcel lockers [22]. Ride-hailing and on-demand platforms supply most of the equilibrium evidence on competition for workers and customers [12, 13, 14], but these models use queues rather than routes. Crowdsourced delivery has been reviewed and modelled in this journal [23, 24, 25, 26], mostly from the platform's side.

Leaders above carriers. Cerulli et al. [16] model a platform that assigns requests and sets compensation while each carrier solves its own profitable tour problem; the carriers do not interact, so the lower level is not a game. Catalano and Migliore [15] cast terminal design as a Stackelberg game with a public designer above a carrier equilibrium. Curb management supplies the most direct foundations for a public pricing leader: Xu and Sun [4] analyse on-street parking with delivery vehicles and find that calibrated parking prices reduce the need for dedicated delivery bays, and Liu et al. [27] optimise curbside prices for ride-hailing pick-ups. Policy-oriented work documents the restrictions cities use in practice [1, 2, 3]. None of these embeds interacting carriers with routing-level best responses.

Collaboration as a benchmark. Collaborative vehicle routing, with auctions, request exchange and cost allocation, is surveyed by Gansterer and Hartl [9], Guajardo and Rönnqvist [28] and Cleophas et al. [10]; auction-based collaboration was reviewed in this journal [29], and large-scale decentralised versions are studied by Los et al. [30, 11]. The savings reported there bound what cooperation could recover; they are not a price of anarchy, because the baseline carriers plan independently without competing for the same customers.

Uncertainty in routing. Robust and distributionally robust routing keeps the combinatorial structure of the deterministic problem when the ambiguity set has suitable structure [31, 6, 7, 32, 33]. Closest to our pricing algorithm, Ghosal et al. [8] show that a CVRP with ambiguous demand marginals and unknown dependence reduces, through a comonotone worst case, to a scenario CVRP solved by branch-and-cut. The comonotone worst case and the stop-loss decomposition of comonotone sums go back to Meilijson and Nádas [34] and are standard in actuarial science [35, 36]; worst-case marginals of a probability box under monotone losses are standard in probability-bound analysis [37]. Branch-price-and-cut with correlated travel times under a mean-covariance model [38] and path dominance by second-order stochastic dominance for independent arc times [39] address different dependence models. Dynamic and learning-based routing is reviewed by Soeffker et al. [40], Ulmer et al. [41] and Hildebrandt et al. [42]; data-driven distributionally robust optimisation is developed by Mohajerin Esfahani and Kuhn [43], with urban applications in this journal such as Wang et al. [44]. None of these papers adds competition among carriers.

Games, aggregation and ambiguity. Congestion and potential games guarantee pure equilibria in classes of finite games [45, 46], and selfish-routing theory bounds the inefficiency of nonatomic and atomic splittable flows and of competing congestible providers [47, 48, 49, 50, 51], and generalized Nash problems model shared capacities [52]. A carrier, however, chooses tours, which makes its strategy set finite and nonconvex. Liu et al. [53] prove, with a Shapley–Folkman argument in the tradition of Starr [54], that nonconvex aggregative games have approximate pure equilibria whose error does not grow with the number of players; the optimisation counterpart with Frank–Wolfe and rounding is in Aubin and Ekeland [55], Udell and Boyd [56] and Bonnans et al. [57]. Approximate pure equilibria in large games go back to Kalai [58] and Azrieli and Shmaya [59], and finding them is PPAD-complete in Lipschitz games [60]. Equilibrium under ambiguous beliefs has been studied as equilibrium in beliefs [61, 62], as pure equilibrium in beliefs with ambiguous payoffs [63], as Knightian robustness [64] and as robust-optimisation equilibrium [65, 66]; Crespi et al. [66] show that robust equilibria are approximate equilibria of the nominal game, which is the converse direction of our absorption theorem. Feik et al. [67] study convex aggregative games with an optimal-transport ambiguity ball around the aggregate and report the price of anarchy as a function of the radius; their strategy sets are convex and their criterion is worst-case. Christodoulou et al. [68] bound the price of anarchy of approximate equilibria as a function of the slack, and Bhaskar et al. [69] compute tolls that implement target equilibria using best-response queries. Deciding existence of pure equilibria is NP-complete in polymatrix and integer programming games [70, 71], and integer-programming methods select among pure equilibria [72].

Position of this paper. Relative to these streams, our model is an atomic, routing-level game with a public leader and credal beliefs. Its theory combines known ingredients (Shapley–

Folkman rounding, kernel reductions, potential games) into statements about imprecise equilibria that we did not find in the literature, and its algorithms adapt known machinery (labelling, column generation, price decomposition) to the structure of this game. Our searches were web and index searches with snippet-level reading for several neighbours; we list the closest neighbours of each result where it is stated.

3. Model

Throughout, $[n] = \{1, \dots, n\}$, $\Delta(S)$ is the probability simplex over a finite set S, and for $v \in \mathbb{R}^d$ and $w \in \mathbb{R}^d_{>0}$, $\|v\|_{w,\infty} = \max_k |v_k|/w_k$ and $\|v\|_M^2 = \langle v, Mv\rangle$.

3.1. Carriers, plans and coupling

Assumption A1 (Finite plans)**.** There are $N \geq 1$ carriers. Carrier i has a finite nonempty plan set $\mathcal{P}_i$. Each plan $p \in \mathcal{P}_i$ has a contribution $a_i(p) \in \mathbb{R}^d_{\geq 0}$ to the aggregate, an exposure $\varphi_i(p) \in \mathbb{R}^d_{\geq 0}$ to the aggregate, an exposure to traffic $t_i(p) \in \mathbb{R}^Z_{\geq 0}$, and a private profit $\pi_i(p) \in \mathbb{R}$. At a pure profile σ, $A(\sigma) = \sum_i a_i(\sigma_i)$ and $A_{-i}(\sigma) = A(\sigma) - a_i(\sigma_i)$.

Assumption A2 (Affine coupling)**.** There are a diagonal $M = \mathrm{diag}(M_1, \dots, M_d) \geq 0$ and $\gamma > 0$ such that, for a realised rival aggregate $m \in \mathbb{R}^d_{\geq 0}$ and traffic factor $\xi \in \mathbb{R}^Z_{\geq 0}$, carrier i's payoff from plan p is $u_i(p; m, \xi) = \pi_i(p) - \langle \varphi_i(p), Mm\rangle - \gamma\langle t_i(p), \xi\rangle$.

The application is the following instantiation. The city is divided into zones $z \in [Z]$, and $d = 2Z$ coordinates are indexed (z, Q) and (z, X). A routing plan of carrier i determines deliveries q_{iz}, vehicles n_{iz}, vehicle-hours τ_{iz} and a routing cost $C_i(p)$. Inverse demand in zone z is $\alpha_z - b_z Q_z$ with $b_z > 0$, and a vehicle-hour costs $\gamma(\xi_z + c_z X_z)$ with $c_z > 0$. The city charges s_z per delivery and θ_z per unit of the congestion variable. Carrier i's profit is

$$u_i = \sum_z \left[q_{iz}(\alpha_z - b_z Q_z) - \gamma\tau_{iz}(\xi_z + c_z X_z) - s_z q_{iz} - \theta_z x_{iz}\right] - C_i(p). \tag{1}$$

Two congestion regimes are distinguished:

- **D1 (exposure-based congestion):** $x_{iz} = \tau_{iz}$ and $X_z = \sum_i \tau_{iz}$;
- **D2 (vehicle-count congestion):** $x_{iz} = n_{iz}$ and $X_z = \sum_i n_{iz}$.

Equation (1) has the form of Assumption A2 with $a_{i,(z,Q)} = \varphi_{i,(z,Q)} = q_{iz}$, $M_{(z,Q)} = b_z$, $a_{i,(z,X)} = x_{iz}$, $\varphi_{i,(z,X)} = \tau_{iz}$, $M_{(z,X)} = \gamma c_z$, $t_{iz} = \tau_{iz}$ and $\pi_i(p) = \sum_z[\alpha_z q_{iz} - b_z q_{iz}^2 - \gamma c_z \tau_{iz} x_{iz} - s_z q_{iz} - \theta_z x_{iz}] - C_i(p)$. The game is *of type D1* if $\varphi_i = a_i$ for all carriers and plans, and *of type D2* otherwise. Welfare at the true mean traffic μ is

$$W(\sigma) = \sum_z \left[\alpha_z Q_z - \tfrac{b_z}{2} Q_z^2 - \gamma\mu_z T_z - \gamma c_z T_z X_z\right] - \sum_i C_i(\sigma_i), \qquad T_z = \textstyle\sum_i \tau_{iz}, \tag{2}$$

with charges treated as transfers. It counts consumer surplus, the congestion cost the carriers impose on each other and routing costs; Section 6 adds a proxy for the delay imposed on background traffic.

3.2. Imprecise beliefs

Assumption A3 (Traffic box)**.** $\Xi = \prod_z [\underline{\mu}_z, \overline{\mu}_z] \subset \mathbb{R}^Z_{\geq 0}$ with widths $\kappa_z = \overline{\mu}_z - \underline{\mu}_z$; a nominal mean $\hat{\mu} \in \Xi$ is fixed and the true mean μ lies in Ξ.

Assumption A4 (Credal sets, clipped and consistent)**.** Fix weights $w \in \mathbb{R}^d_{>0}$ and a width $\delta \geq 0$. For $c \in \mathbb{R}^d_{\geq 0}$ let $K^\delta(c) = \{m \in \mathbb{R}^d_{\geq 0} : |m_k - c_k| \leq \delta w_k \ \forall k\}$. At a pure profile σ, carrier i's credal set $\mathcal{M}_i^\delta(\sigma)$ is the set of Borel probability laws P on $\mathbb{R}^d_{\geq 0} \times \mathbb{R}^Z_{\geq 0}$ with finite first moments such that $\mathbb{E}_P[m] \in K^\delta(A_{-i}(\sigma))$ and $\mathbb{E}_P[\xi] \in \Xi$.

The box is centred at the realised rival aggregate, which is the consistency requirement of equilibrium in beliefs [61], and it is clipped at zero, so every belief is a feasible aggregate. The width δ measures how much a carrier is unsure about its rivals' activity, in units w_k.

Definition 1 (Decision criteria; 18)**.** For a finite option set $\mathcal{P}$ and a set $\mathcal{M}$ of laws, with $\underline{\mathbb{E}} = \inf_{P \in \mathcal{M}} \mathbb{E}_P$ and $\overline{\mathbb{E}} = \sup_{P \in \mathcal{M}} \mathbb{E}_P$, an option p is *E-admissible* if it maximises $\mathbb{E}_P[u(\cdot)]$ for some $P \in \mathcal{M}$; *maximal* if no p' has $\underline{\mathbb{E}}[u(p') - u(p)] > 0$; *interval-undominated* (ID) if $\overline{\mathbb{E}}[u(p)] \geq \max_{p'} \underline{\mathbb{E}}[u(p')]$; and Γ-*maximin* if $\underline{\mathbb{E}}[u(p)] = \max_{p'} \underline{\mathbb{E}}[u(p')]$.

Definition 2 (Imprecise credal Nash equilibrium)**.** For $C \in \{E, M, ID, \Gamma\}$ and $\delta \geq 0$, $\mathrm{ICNE}_C(\delta)$ is the set of pure profiles σ in which every σ_i is C-admissible for $(\mathcal{P}_i, \mathcal{M}_i^\delta(\sigma))$ with payoff u_i. The existence threshold is $\delta_0^C = \inf\{\delta \geq 0 : \mathrm{ICNE}_C(\delta) \neq \emptyset\}$.

3.3. The routing layer

A plan of carrier i is a set of depot-to-depot routes over its customer zones. Arc travel times are random. For every arc a the carrier knows only a probability box: CDFs $\underline{F}_a \leq \overline{F}_a$ on a bounded support $[0, B_a]$, for example a Dvoretzky–Kiefer–Wolfowitz (DKW) band around the empirical CDF of n observed days, $\underline{F}_a(x) = \max\{F_n(x) - \varepsilon, 0\}$ with $\varepsilon = \sqrt{\ln(2/\alpha)/(2n)}$ [73, 74]. The dependence between arcs is unknown. The joint credal set is the Fréchet class $\mathcal{M} = \{P : \text{the } a\text{-th marginal of } P \text{ lies between } \underline{F}_a \text{ and } \overline{F}_a \text{ for all } a\}$. A route of duration $T_r = \sum_{a \in r} X_a$ incurs the convex piecewise-linear loss

$$\phi(T) = c_0 + \gamma T + \sum_{j=1}^{H} \rho_j \, (T - \Theta_j)^+ , \qquad \gamma \geq 0, \ \rho_j > 0, \ 0 \leq \Theta_1 < \cdots < \Theta_H, \tag{3}$$

which in the experiments is $H = 1$: a fixed cost, an hourly cost and an overtime penalty after the shift length $\Theta_1 = T_{\max}$. The robust route cost is the upper prevision $\overline{\mathbb{E}}[\phi(T_r)] = \sup_{P \in \mathcal{M}} \mathbb{E}_P \phi(T_r)$, the Γ-minimax criterion for a loss, and it enters $C_i(p) = \sum_{r \in p} \overline{\mathbb{E}}[\phi(T_r)]$. The plan-level game of the previous subsections therefore contains a robust routing problem inside every best response, and the oracle of the next assumption is the interface between the two layers.

Assumption A5 (Oracles)**.** For each i there is an oracle that, for price vectors $\lambda^a, \lambda^\varphi \in \mathbb{R}^d$, $\nu \in \mathbb{R}^Z$ and a diagonal $Q \succeq 0$, returns a plan in $\arg\max_{p \in \mathcal{P}_i} \{\pi_i(p) + \frac{1}{2}\langle a_i(p), Q a_i(p)\rangle - \langle \lambda^a, a_i(p)\rangle - \langle \lambda^\varphi, \varphi_i(p)\rangle - \langle \nu, t_i(p)\rangle\}$.

A best response at belief (m, ξ) is the call $(0, Mm, \gamma\xi, 0)$. In the application the oracle is a prize-collecting robust routing problem with zone prices; Section 5 solves it by column generation with the BRL pricing algorithm.

Assumptions stated plainly. The coupling is affine in the aggregates (a BPR-type power congestion term would break the exact credal statements below; Remark 18). Charges are zone-specific and linear. Beliefs about rivals are boxes centred at the realised aggregate. The traffic factor ξ enters only through its mean, and in the experiments its uncertainty is carried by the probability boxes of the routing layer.

4. Theory

This section states the results and sketches the proofs; complete proofs are in Appendix A. Results labelled *classical* are known and cited.

4.1. Reduction to mean boxes and the criteria

Write $\bar{u}_i(p; m, \xi)$ for the payoff of Assumption A2 evaluated at the means (m, ξ) and $K_i^\delta(\sigma) = K^\delta(A_{-i}(\sigma)) \times \Xi$.

Lemma 3 (Reduction of the credal set to its mean box)**.** *Under Assumptions A1–A4: (a) for every $P \in \mathcal{M}_i^\delta(\sigma)$ and every p, $\mathbb{E}_P[u_i(p)] = \bar{u}_i(p; \mathbb{E}_P m, \mathbb{E}_P \xi)$; (b) $P \mapsto (\mathbb{E}_P m, \mathbb{E}_P \xi)$ maps $\mathcal{M}_i^\delta(\sigma)$ onto $K_i^\delta(\sigma)$; (c) hence each criterion of Definition 1 coincides with the same criterion evaluated over the compact box $K_i^\delta(\sigma)$, and infima and suprema are attained.*

Lemma 4 (Nesting and monotonicity)**.** *For every $\delta \geq 0$,*

$$\mathrm{ICNE}_E(\delta) \subseteq \mathrm{ICNE}_M(\delta) \subseteq \mathrm{ICNE}_{ID}(\delta) \quad \textit{and} \quad \mathrm{ICNE}_\Gamma(\delta) \subseteq \mathrm{ICNE}_M(\delta).$$

For $C \in \{E, M, ID\}$ and $\delta \leq \delta'$, $\mathrm{ICNE}_C(\delta) \subseteq \mathrm{ICNE}_C(\delta')$. For $C = \Gamma$ the monotonicity fails (Example 37 in Appendix A).

The inclusions between criteria are due to Troffaes [18]. Monotonicity in δ says that more imprecision can only enlarge the set of E-admissible, maximal or interval-undominated equilibria; Γ-maximin behaves differently because the worst case moves with δ.

Lemma 5 (Γ-maximin surcharge)**.** *If $\varphi_i(p), t_i(p) \geq 0$, the lower expectation over $K^\delta(c) \times \Xi$ is attained, for every plan simultaneously, at the upper corner $(c + \delta w, \overline{\mu})$ and is linear in mixtures. Consequently $\mathrm{ICNE}_\Gamma(\delta)$ is the set of pure Nash equilibria of the precise game G_δ^Γ with private profits*

$$\pi_i^\Gamma(p) = \pi_i(p) - \delta\langle \varphi_i(p), Mw \rangle - \gamma\langle t_i(p), \overline{\mu} \rangle,$$

the same coupling, and traffic set to zero.

The surcharge $\delta\langle \varphi_i(p), Mw \rangle$ is what a cautious carrier adds to the cost of activity in zone k because it fears up to δw_k more rival activity than it observes. Section 7 uses it to compute Γ-maximin equilibria of the city games.

4.2. Existence under exposure-based congestion

Proposition 6 (Existence under D1 at every width)**.** *Under D1, for every* $\hat{\xi} \in \Xi$*,*

$$\Psi_{\hat{\xi}}(\sigma) = \sum_i \left[\pi_i(\sigma_i) + \tfrac{1}{2}\|a_i(\sigma_i)\|_M^2 - \gamma\langle t_i(\sigma_i), \hat{\xi}\rangle\right] - \tfrac{1}{2}\|A(\sigma)\|_M^2$$

is an exact potential of the precise game at traffic $\hat{\xi}$*. Every maximiser lies in* $\mathrm{ICNE}_E(\delta)$ *for all* $\delta \geq 0$*, so* $\delta_0^E = \delta_0^M = \delta_0^{ID} = 0$*, and* $\mathrm{ICNE}_\Gamma(\delta) \neq \emptyset$ *for all* $\delta \geq 0$*.*

Sketch. For a unilateral change of carrier i from contribution a to a' with rival aggregate R, the quadratic terms give $\frac{1}{2}\|a'\|_M^2 - \frac{1}{2}\|a\|_M^2 - \frac{1}{2}\|R + a'\|_M^2 + \frac{1}{2}\|R + a\|_M^2 = -\langle a' - a, MR\rangle$, which equals the change of the coupling term when $\varphi_i = a_i$. A maximiser of an exact potential is a pure equilibrium [46], and the centre of every credal box is the realised aggregate. For Γ, G_δ^Γ is again of type D1.

4.3. Credal absorption

Let x^* be a mixed equilibrium of the precise game at traffic $\hat{\xi} \in \Xi$ (it exists by 75; under affine coupling, mixed equilibria are exactly the profiles in which every plan in the support is a best response to the mean rival aggregate, Proposition 38). Write B_j for carrier j's best-response set at x^*, $\bar{a}_j = \sum_p x_j^*(p) a_j(p)$, $\bar{A}_{-i}^* = \sum_{j \neq i} \bar{a}_j$, and define the spreads $s_{jk} = \max_{p,p' \in B_j} |a_{jk}(p) - a_{jk}(p')|$ and $r_{jk} = \max_{p \in B_j} |a_{jk}(p) - \bar{a}_{jk}|$, with $\bar{\sigma} = \max_{j,k} s_{jk}/w_k$, $\bar{\rho} = \max_{j,k} r_{jk}/w_k$, $d_L = \dim \operatorname{span} \bigcup_j (a_j(B_j) - a_j(B_j))$ and $k^\sharp = \min\{N, d_L\}$. Let $\sigma_{\max} = \max_{i,k} \operatorname{range}_{p \in \mathcal{P}_i} a_{ik}(p)/w_k$, so that $\bar{\rho} \leq \bar{\sigma} \leq \sigma_{\max}$. The *anchored radius* of a pure profile $p \in \prod_j B_j$ is $\hat{\delta}(p) = \max_i \|A_{-i}(p) - \bar{A}_{-i}^*\|_{w,\infty}$.

Theorem 7 (Credal absorption)**.** *Assume A1–A4 and let* x^* *be a mixed equilibrium at* $\hat{\xi} \in \Xi$*.*

(a) Absorption. *For every* $p \in \prod_j B_j$ *and every* $\delta \geq \hat{\delta}(p)$*,* $p \in \mathrm{ICNE}_E(\delta) \subseteq \mathrm{ICNE}_M(\delta) \subseteq \mathrm{ICNE}_{ID}(\delta)$*.*

(b) Radius. $\min_{p \in \prod_j B_j} \hat{\delta}(p) \leq \min\left\{ \max_i \max_k \sum_{j \neq i} r_{jk}/w_k,\ k^\sharp \bar{\sigma} + \bar{\rho},\ \bar{\sigma}\sqrt{k^\sharp \ln(2d)/2} + \bar{\rho} \right\}$*.*

(c) Uniform existence. $\delta_0^{ID} \leq \delta_0^M \leq \delta_0^E \leq \sigma_{\max} \min\{N - 1,\ d + 1,\ 1 + \sqrt{d \ln(2d)/2}\}$*.*

(d) Construction. *A profile achieving the second bound is obtained from* x^* *by one linear program (a vertex of the polytope of Shapley–Folkman decompositions) followed by an arbitrary choice for at most* $k^\sharp$ *carriers; one achieving the third bound by independent sampling of these carriers, with success probability at least* $1 - 2d\exp(-2t^2/(k^\sharp \bar{\sigma}^2))$ *per draw for target* $t + \bar{\rho}$*; one achieving the first bound is any* $p \in \prod_j B_j$*.*

Sketch. Every carrier in $p \in \prod_j B_j$ plays a best response to the *anchor* $\bar{A}_{-i}^*$, which is nonnegative. If $\delta \geq \hat{\delta}(p)$, the anchor lies in the clipped box around the realised aggregate $A_{-i}(p)$, so the Dirac law at the anchor belongs to the credal set and witnesses E-admissibility. The radius bounds follow from the decomposition $A_{-i}(p) - \bar{A}_{-i}^* = \sum_{j \in J}(a_j(p_j) - y_j) - (a_i(p_i) - \bar{a}_i)$, where J is the set of at most $k^\sharp$ carriers left mixed at a vertex of the Shapley–Folkman polytope (Theorem 36); the sum is bounded deterministically by $k^\sharp \bar{\sigma}$ or, after independent sampling, by Hoeffding's inequality [76] and a union bound. The own term $a_i(p_i) - \bar{a}_i$ cannot be removed (it is the "+1" in the bound of 53).

Corollary 8 (ε-Nash in the precise game; cf. 53)**.** *A profile p as in Theorem 7(a) is an ε-Nash equilibrium of the precise game at any $\mu \in \Xi$ with $\varepsilon_i \le \hat{\delta}(p) L_i(p) + \gamma \sum_z \kappa_z T_{iz}(p)$, where $L_i(p) = \max_{p'} \sum_k w_k M_k |\varphi_{ik}(p') - \varphi_{ik}(p_i)|$ and $T_{iz}(p) = \max_{p'} |t_{iz}(p') - t_{iz}(p_i)|$.*

Remark 9 (Novelty and regime)*.* Corollary 8 restates, in our notation, the N-independent approximate equilibrium of Liu et al. [53], who also use a Shapley–Folkman argument on the generators of a convexified equilibrium; we claim no novelty for it. What Theorem 7 adds is that under E-admissibility the rounding error is *absorbed* by the belief width: the rounded profile is an *exact* imprecise equilibrium, not an approximate one. The proof of part (a) is short; the content of the result lies in the bound, its attainment (Proposition 11), its failure for Γ-maximin (Remark 10), and the contrast with the complexity result below. In the city instances $N \le 6$ and $d = 2Z \ge 84$, so $k^\sharp = N$, the Shapley–Folkman step selects nothing, and the operative bound is the support-selection bound $(N-1)\bar{\rho}$. Shapley–Folkman matters only when the number of players exceeds the dimension, for example with per-vehicle or crowdsourced players.

Remark 10 (Γ-maximin)*.* By Lemma 5, $\mathrm{ICNE}_\Gamma(\delta)$ is the pure-equilibrium set of a *precise* game, so absorption does not apply to it, and under D2 that set can be empty (Proposition 11 at $\delta = 0$).

4.4. Tightness

Proposition 11 (The support-selection bound is attained)**.** *Let $N = 2$, one zone, D2, $M = \gamma c = 1$ and weight $w > 0$. Carrier 1 has plans $A_1 : (n, \tau, \pi) = (1, 4, 6)$ and $B_1 : (3, 1, 0)$; carrier 2 has $X : (1, 1, 0)$ and $Y : (3, 4, 6)$; payoffs are $\pi - \tau \cdot n_{\mathrm{rival}}$. Then the game has no pure Nash equilibrium, $\delta_0^E = \delta_0^M = 1/w$, and the bound $(N-1)\bar{\rho}$ of Theorem 7(b) equals $1/w$.*

In each of the four pure profiles exactly one carrier has a strict incentive to deviate at the realised count, and it becomes indifferent only at a belief that differs from the realised count by one vehicle.

Proposition 12 (Linear dependence on the number of zones is necessary)**.** *For every $Z \ge 1$ and $N \ge 2$ there is a D2 game with Z zones in which every pure profile has $\max_i \varepsilon_i \ge 3\lceil Z/2 \rceil$, while every profile of Theorem 7(a) satisfies $\hat{\delta}(p) L_i(p) \le 3Z$ ($w = 1$).*

4.5. Complexity under vehicle-count congestion

Theorem 13 (NP-completeness under D2)**.** *Consider the problem of deciding $\mathrm{ICNE}_C(\delta) \neq \emptyset$ for a game given by explicit plan lists with rational data, M, w, a degenerate traffic box and a rational $\delta \ge 0$. For each $C \in \{E, M, ID, \Gamma\}$ the problem is in NP. For every digraph D with maximum out-degree $\Delta^+ \ge 1$ one constructs in polynomial time a D2 instance with two plans per carrier and one zone per arc such that, for all $\delta \in [0, 1/(2\Delta^+ + 1))$, $\mathrm{ICNE}_E(\delta) = \mathrm{ICNE}_M(\delta) = \mathrm{ICNE}_{ID}(\delta) = \mathrm{ICNE}_\Gamma(\delta)$ is the set of profiles that play "in" exactly on a kernel of D. Hence the problem is strongly NP-complete for each criterion, already at $\delta = 0$, and for every fixed $\delta \in [0, 1/5)$.*

Sketch. Deciding whether a digraph has a kernel (an independent set that absorbs every other vertex) is NP-complete [77], also for planar digraphs with in- and out-degree at most two [78]. Each vertex becomes a carrier with an "out" plan of zero data and an "in" plan that

puts one vehicle on each out-arc zone and two vehicles on each in-arc zone, with exposure one on out-arcs and a small exposure on in-arcs. The gain of "in" is positive if no out-neighbour plays "in" and at most -1 otherwise, and the sign is constant on every credal box of width below $1/(2\Delta^+ + 1)$, so all four criteria select the realised strict best response.

Corollary 14 (Inapproximability of the existence threshold)**.** *Unless* P = NP*, no polynomial algorithm approximates δ_0^E of a D2 instance within any fixed factor, or within an additive error below* $1/5$*.*

Pure-equilibrium existence is known to be NP-complete in polymatrix games and integer programming games [70, 71]; Theorem 13 is a direct kernel reduction specialised to the rank-one D2 structure and extended to four credal criteria and a range of widths. Together with Proposition 6 it gives a complexity split between the two congestion regimes: charging or modelling congestion by exposure (vehicle-hours in a zone) yields a potential game with guaranteed pure imprecise equilibria, while counting vehicles does not. Under D2 the mixed equilibrium that Theorem 7 rounds is itself hard to compute: every bimatrix game embeds as a D2 instance with $N = 2$, so the step is PPAD-hard [79, 80] (Proposition 39).

4.6. Efficiency

Lemma 15 (Imprecise equilibria are approximate equilibria)**.** *If $\sigma \in \mathrm{ICNE}_C(\delta)$ and $\mu \in \Xi$, then σ is an ε-Nash equilibrium of the precise game with $\varepsilon_i \le \delta L_i(\sigma) + \gamma \sum_z \kappa_z T_{iz}(\sigma)$ for $C \in \{E, M\}$ and $\varepsilon_i \le 2\delta H_i + 2\gamma\Theta_i$ for $C = ID$, with $H_i = \max_p \sum_k w_k M_k \varphi_{ik}(p)$ and $\Theta_i = \max_p \sum_z \kappa_z t_{iz}(p)$.*

Proposition 16 (Monotone efficiency curves)**.** *For $C \in \{E, M, ID\}$ and $W > 0$, $\mathrm{PoA}_C(\delta) = W^* / \min_{\mathrm{ICNE}_C(\delta)} W$ is nondecreasing and $\mathrm{PoS}_C(\delta) = W^* / \max_{\mathrm{ICNE}_C(\delta)} W$ nonincreasing in δ, and both are step functions with at most $\prod_i |\mathcal{P}_i|$ steps. For $C = \Gamma$ neither monotonicity holds in general.*

Proposition 17 (Price of stability via the potential, D1)**.** *Under D1 with zero charges and $\mu = \hat{\mu}$, let σ^* maximise W. Every profile σ with $\Psi_{\hat{\mu}}(\sigma) \ge \max \Psi_{\hat{\mu}} - \eta$ satisfies $W(\sigma) \ge W^* - \sum_z \left[\frac{b_z}{2} \sum_i q_{iz}^{*2} + \frac{\gamma c_z}{2}(1 - \frac{1}{N}) T_z(\sigma)^2\right] - \eta$. In particular a potential maximiser, which lies in every $\mathrm{ICNE}_E(\delta)$, bounds $\mathrm{PoS}_C(\delta)$ for all δ and $C \in \{E, M, ID\}$.*

A smoothness-based price-of-anarchy bound also transfers to imprecise equilibria through Lemma 15 (Proposition 40), but it is conditional: we do not have smoothness constants for the Cournot–congestion game with arbitrary private routing costs. The price-of-anarchy values in Section 7 are therefore empirical.

4.7. Exact thresholds and the rounding algorithm

For any pure profile p, the exact E-threshold of carrier i,

$$A_i^E(p) = \min\{\delta : p_i \text{ is E-admissible at width } \delta\},$$

is the value of a linear program over the box, so $p \in \mathrm{ICNE}_E(\delta)$ if and only if $\delta \ge \max_i A_i^E(p)$ (Theorem 41). Theorem 7 and this threshold computation define an algorithm with four phases: relax (a mixed equilibrium by Frank–Wolfe on the concave relaxed potential under

D1, or by a double oracle under D2; 81), sparsify (a vertex of the Shapley–Folkman polytope), round, and compute the exact width. Under D2 its output satisfies the bounds of Theorem 7(b) (Theorem 42), but its relaxation phase is PPAD-hard in general. Under D1 the relaxation-and-rounding technique is classical [55, 56, 57]. The full rounding algorithm was not run on the city games, which use the D1 engine of Section 5; the exact thresholds of Theorem 41 were used in the tests of Section 7.7.

Remark 18 (Nonaffine congestion). With a power congestion term the expected payoff under mixed rival strategies is no longer a function of the mean aggregate; Lemma 3 and the exact credal statements fail, although relaxed equilibria still exist. We do not treat this case.

4.8. What is machine-checked

A Lean 4 file with Mathlib (`lean/CCU/LeanTargets.lean` in the repository) formalises the finite aggregative game with clipped credal boxes and E-admissibility. It compiles with no unproved steps (no `sorry`), and every theorem in it depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The following discrete core statements are machine-checked: E-admissibility implies maximality, and $\mathrm{ICNE}_E(\delta)$ is monotone in δ (Lemma 4, case E); the core of Theorem 7(a) (a profile whose plans are best responses to nonnegative anchors within the box is in $\mathrm{ICNE}_E(\delta)$); the Shapley–Folkman lemma for finite sets in a finite-dimensional real space, with the dimension of the space in place of d_L (Theorem 36; the formal proof lifts the sets with indicator coordinates and applies Carathéodory's theorem); the exact potential identity under D1 (Proposition 6); the combinatorial core of Theorem 13 (with integer-scaled gains, the equilibrium conditions hold exactly on kernels); the lower-bound core of Proposition 11; and the counting core of Proposition 12. Lean checks these core statements, not the full theorems: the radius bounds of Theorem 7(b)–(c), the Hoeffding step, the NP-completeness statement itself (Mathlib has no framework for polynomial-time reductions) and the algorithmic results are not formalised.

5. Algorithms

The equilibrium engine has two layers: an outer search over carriers' plans at common zone prices (ZMC) and, inside every oracle call, a robust routing problem solved by column generation with an exact pricing algorithm (BRL). Both are designed for the D1 instance of Section 3, which is the regime with guaranteed existence.

5.1. Budget-resource labelling (BRL)

Let $\overline{Q}_a(u) = \inf\{x : \underline{F}_a(x) \geq u\}$ be the upper quantile of arc a, $\bar{x}_a = \int_0^1 \overline{Q}_a(u)\,du$ and $\overline{\Pi}_a(d) = \int_0^1 \big(\overline{Q}_a(u) - d\big)^+ du$, the stop-loss transform of the lower CDF; $\overline{\Pi}_a$ is convex and nonincreasing with slopes in $[-1, 0]$.

Theorem 19 (Upper prevision and budget decomposition)**.** *For every route r,*

$$\overline{\mathbb{E}}[\phi(T_r)] = \int_0^1 \phi\Big(\sum_{a\in r} \overline{Q}_a(u)\Big)du = c_0 + \sum_{a\in r} \gamma \bar{x}_a + \sum_{j=1}^{H} \rho_j \min\Big\{\sum_{a\in r} \overline{\Pi}_a(d_a^j) :\ \sum_{a\in r} d_a^j \leq \Theta_j\Big\}.$$

The supremum is attained by the comonotone coupling of the lower CDFs; each minimum is attained, at an integer vector when quantiles and thresholds are integers.

The first equality and the stop-loss identity for comonotone sums with fixed marginals are due to Meilijson and Nádas [34] (see also 35, 36), and the choice of the lower CDF as worst marginal is standard [37]. We re-prove the statement in Appendix B because the algorithm relies on the inequality-budget form and on integrality. With a common quantile grid of L levels (weights ω_l), the route profile $S_{r,l} = \sum_{a \in r} q_{a,l}$ gives $\overline{\mathbb{E}}[\phi(T_r)] = c_0 + f_r(T_{\max})$ for $H = 1$, where $f_r(D) = \gamma \sum_l \omega_l S_{r,l} + \rho \sum_l \omega_l \, (S_{r,l} - D)^+$.

Theorem 20 (Budget-resource reformulation)**.** *Let y be dual prices and define the arc costs*

$$c_a(d^1, \ldots, d^H) = \gamma \bar{x}_a + \sum_j \rho_j \overline{\Pi}_a(d^j).$$

The dependence-robust pricing problem

$$\min_{r \in \mathcal{R}} \left\{ \overline{\mathbb{E}}[\phi(T_r)] - \sum_{i \in r} y_i \right\}$$

equals an elementary shortest path problem in which every arc chooses a consumption d_a^j of each of H budget resources, paid through the separable convex cost c_a and subject to the budget constraints $\sum_{a \in r} d_a^j \le \Theta_j$, $j \le H$. The number of resources is H; it does not depend on L, the sample size or the number of arcs.

A scenario view of the same problem, which is what the comonotone reduction of Ghosal et al. [8] gives, needs the L level sums $(S_{p,l})_l$ as resources and compares labels pointwise. BRL instead compares labels through their cost functions, but only where a completion can still reach. For a partial path (label) p ending at v with level profile S_p, collected duals $y(p)$, visited set $\mathcal{V}(p)$ and load $\ell(p)$, concatenation with a completion q is an inf-convolution, $f_{p \oplus q}(T_{\max}) = \min_D \{ f_p(D) + f_q(T_{\max} - D) \}$ (Lemma 43). Let $C_{\min}(v)$ be a lower bound on the duration of any completion from v and define the completion window $\mathcal{W}(p) = [\underline{D}(p), \overline{D}(p)]$ with $\overline{D}(p) = \min\{S_{p,L}, T_{\max} - C_{\min}(v)\}$ and $\underline{D}(p) = \min\{S_{p,1}, \overline{D}(p)\}$.

Definition 21 (Functional dominance)**.** Label p_1 functionally dominates p_2, both ending at v, if $\mathcal{V}(p_1) \subseteq \mathcal{V}(p_2)$, $\ell(p_1) \le \ell(p_2)$ and $(f_{p_1}(D) - y(p_1)) - (f_{p_2}(D) - y(p_2)) \le 0$ for all $D \in \mathcal{W}(p_2)$.

The difference is piecewise linear with breakpoints at the level sums, so the test needs at most $2L + 2$ evaluations.

Theorem 22 (Validity and exactness)**.** *Functional dominance implies that every completion of p_2 is a completion of p_1 with no larger reduced cost. The labelling algorithm that discards dominated labels (Algorithm 1) returns the minimum reduced cost exactly, with a route attaining it.*

The window is what makes the rule exact and strong at the same time. The slope of f_p lies in $[-\rho, 0]$ and that of $D \mapsto f_q(T_{\max} - D)$ in $[0, \rho]$, so an optimal split never lies above $\overline{D}(p)$, and below $S_{p,1}$ the difference of two label functions is nondecreasing (Lemma 44). Quantile-vector dominance and sector dominance are special cases of functional dominance, and neither converse holds (Proposition 45). Under independent arc times concatenation is a convolution of laws rather than an inf-convolution and the method does not apply. The

Algorithm 1 BRL pricing for the dependence-robust routing problem ($H = 1$)

Require: graph, demands, capacity; level weights ω; arc level profiles $q_{a,\cdot}$; $\gamma, \rho, T_{\max}, c_0$; duals y

$C_{\min}(v) \leftarrow$ shortest path from v to the depot with arc lengths $q_{a,1}$
buckets $\mathcal{B}_v \leftarrow \emptyset$; queue $\leftarrow$ {depot label}
while queue nonempty **do**
    pop p; record the closed route $p \oplus (v, 0)$ with reduced cost $c_0 + f_{p\oplus(v,0)}(T_{\max}) - y(p)$
    for each customer $j \notin \mathcal{V}(p)$ that fits the capacity **do**
        $p' \leftarrow$ extension of p to j ($S_{p'} = S_p + q_{(v,j),\cdot}$, $y(p') = y(p) + y_j$)
        if some $o \in \mathcal{B}_j$ functionally dominates p' **then** discard p'
        else delete the labels of $\mathcal{B}_j$ dominated by p'; insert and push p'
return recorded routes sorted by reduced cost

natural shortcut of fixing a single risk level for all routes gives a lower bound but is not exact (Proposition 46).

On the city instances the pricing runs over ng-routes [82]: the visited set is replaced by the ng-memory, and Theorem 22 holds verbatim, with the weaker ng-relaxation bound. Column generation [83, 84] with BRL pricing returns the exact LP bound of the dependence-robust routing problem when no negative reduced cost remains; an integer program over the generated columns (price-and-branch) gives a feasible plan but is not an exact integer method. Section 5.2 makes the integer step exact.

Prior art and the claim. The upper-prevision formula is Meilijson–Nádas; labelling with resources is standard; the comonotone scenario reduction for ambiguous marginals is in Ghosal et al. [8]. What we did not find in the literature is the per-arc budget used as a *decision resource* of the pricing problem (Theorem 20) and the completion-window functional dominance with its exactness proof (Theorem 22). A referee may reasonably describe BRL as Meilijson–Nádas plus resource-constrained labelling; our answer is that the speed of the method comes from the dominance rule (Section 7.6).

5.2. Branch-and-price with BRL pricing

Price-and-branch can miss the integer optimum because the columns generated at the root need not contain an optimal integer plan. We embed the BRL pricer in a branch-and-price method. The master is a set-partitioning model (every customer served once) for the routing benchmark, or a prize-collecting set-packing model for the carrier oracle of the game, with one column pool shared by all nodes; a column is usable at a node if all its arcs are allowed there. Each node solves its LP by column generation with the exact BRL pricer, restricted to the node's allowed arcs and with the completion bound $C_{\min}$ recomputed on them; functional dominance remains valid because the restrictions are local to the node. The pricer itself is unchanged.

- *Branching* is on the aggregated arc flow f_{ij}. The branch $f_{ij} = 0$ deletes the arc; $f_{ij} = 1$ deletes the other arcs leaving i and entering j. Integral arc flows imply an integral plan.

- *Fleet size.* When every customer must be served, the number of routes is at least the total demand divided by the capacity, rounded up. This row is added at the root, and a fractional number of routes is branched on before any arc; its dual enters every reduced cost as a constant.

- *ng-routes.* If a non-elementary column appears in a node solution, the ng-sets along its cycle are enlarged and the columns that became infeasible are removed. This only tightens the relaxation, so earlier bounds remain valid, and the reported plans are elementary.

- *Bounds.* The node LP value, and during column generation the Lagrangian bound $z + n_{\max} \min\{0, \min rc\}$, give a lower bound that is valid at every moment; a bound is marked invalid if a pricing call hits the label limit. Upper bounds come from integral nodes and from an integer program over the generated elementary columns. Nodes are selected by best bound, and the method stops when the relative gap is below 10^{-6}.

Because ng-routes relax elementary routes, the final bound is valid for the elementary problem and the incumbent is a feasible elementary plan, so a zero gap proves optimality. The scheme is standard branch-and-price machinery; its purpose here is to remove the inexact integer step, not to add a new branching rule.

5.3. Dual implementation pricing (DIP): the city's charges

The city leader chooses charges so that the carriers' equilibrium has low social cost. A direct approach searches a grid of charges and solves one equilibrium per grid point; with 63 points this costs about two thousand routing calls per city and seed (Section 7.2). DIP replaces the search by an implementation step: compute the profile the city wants, then compute charges under which that profile is an equilibrium, using the carriers' own routing oracle to separate candidate charges. No equilibrium is solved for the leader's decision.

Setting. We use the D1 instance of Section 3 in cost form. Carrier i serves a subset of its customer zones with all-or-nothing lots $d_{iz} > 0$, so $q_{iz} \in \{0, d_{iz}\}$ and $q_{iz}^2 = d_{iz} q_{iz}$; vehicle-hours are $\tau_{iz} = r\, q_{iz}$ for one ratio $r > 0$ (assumption (A3)); $R_i(p) \geq 0$ is the robust routing cost of plan p; $g_z = \gamma c_z$. The city posts s_z per delivery and t_z per vehicle-hour. Carrier i's cost and the social cost (charges are transfers) are

$$J_i(p; \sigma_{-i}) = R_i(p) + \sum_z \big[q_{iz}(-\alpha_z + b_z(q_{iz} + Q_{-i,z}) + s_z) + \tau_{iz}(g_z(\tau_{iz} + X_{-i,z}) + t_z) \big],$$

$$\mathrm{SC}(\sigma) = \sum_i R_i(\sigma_i) + \sum_z \big[-\alpha_z Q_z + \tfrac{b_z}{2} Q_z^2 + g_z X_z^2 \big].$$

Write $\zeta_z = g_z r^2$ (congestion slope in delivery units), $m_z = b_z + \zeta_z$ (the slope the carriers face) and $\bar{d}_z$ for the mean lot in zone z.

Proposition 23 (One instrument per zone)**.** *(a) For every plan, $\sum_z (s_z q_{iz} + t_z \tau_{iz}) = \langle u, q_i \rangle$ with $u_z = s_z + r t_z$, so the game depends on the charges only through $u \in \mathbb{R}^Z$. (b) Every u is realised by a uniform per-delivery charge $s = \min_z u_z$ over zones with lots and non-negative zone congestion charges $t_z = (u_z - s)/r$. (c) The grid of Section 7.2 is the two-parameter family $u = s\mathbf{1} + \vartheta\, r\, g \odot X^{\mathrm{NE}}$.*

Under u the game has the exact potential $\Phi_u(\sigma) = \sum_i R_i(\sigma_i) + \sum_z[-\alpha_z Q_z + \frac{m_z}{2}(Q_z^2 + \sum_i q_{iz}^2) + u_z Q_z]$ (Proposition 6).

Lemma 24 (Potential–welfare identity)**.** *Fix $Q^\circ \in \mathbb{R}^Z$, let $u_z^{\tan} = \zeta_z Q_z^\circ - \frac{m_z}{2}\bar{d}_z$ and $v = u - u^{\tan}$. For every pure profile σ,*

$$\Phi_u(\sigma) = \mathrm{SC}(\sigma) - \sum_z \tfrac{\zeta_z}{2}(Q_z - Q_z^\circ)^2 + L(\sigma) + \langle v, Q\rangle + \sum_z \tfrac{\zeta_z}{2}(Q_z^\circ)^2, \qquad L(\sigma) = \sum_{i,z} \tfrac{m_z}{2}(d_{iz} - \bar{d}_z) q_{iz}.$$

The *tangent charge* $u^{\tan}$ is the classical corrective tax of a Cournot market with an externality: the marginal external damage at the target aggregate minus a market-power subsidy [85, 86]. Two details are specific to all-or-nothing lots: the subsidy is half a lot, $\frac{m_z}{2}\bar{d}_z$ per delivery, and it includes the carriers' own congestion. With carrier-specific charges $U_{iz} = \zeta_z Q_{-i,z}^* - \frac{b_z}{2} d_{iz}$ every carrier's cost differences equal the social-cost differences, so every social optimum is an equilibrium; this is the classical marginal-contribution charge [87, 88] (Proposition 50 in Appendix C). A city cannot charge per carrier, and the next results measure what flat zone charges lose. In the convexified (mixed) game let x^* minimise the convexified social cost $\overline{\mathrm{SC}}$, with aggregate $\bar{Q}^*$.

Theorem 25 (Loss under asymmetric lots)**.** *Let $u = \zeta \odot \bar{Q}^* - \frac{m}{2} \odot \bar{d}$. Every convexified equilibrium x^u satisfies $0 \le \overline{\mathrm{SC}}(x^u) - \overline{\mathrm{SC}}(x^*) \le (1 + \varrho)E$, where $\varrho = \max_z \zeta_z/(b_z + \zeta_z) < 1$ and $E = \sum_{i,z} \frac{m_z}{2} |d_{iz} - \bar{d}_z|\, d_{iz}$ measures how unequal the lots in a zone are.*

Theorem 26 (Exact implementation with symmetric lots)**.** *If all lots in a zone are equal, the tangent charge makes x^* a convexified equilibrium, and every convexified equilibrium under it has aggregate $\bar{Q}^*$ and social cost $\overline{\mathrm{SC}}(x^*)$.*

Sketch. The identity holds for the convexified potential and social cost with the same algebra. Because $\overline{\mathrm{SC}}$ is linear plus a quadratic in $\bar{Q}$, its expansion around x^* is exact; comparing it with the identity at a minimiser of the potential bounds the aggregate deviation by the lot-asymmetry term L, and $|L| \le E$. With equal lots $L \equiv 0$.

The pure game. For a pure target σ^* and an admissible box $\mathcal{U}$ of charges let

$$P(\sigma^*) = \big\{u \in \mathcal{U} :\ \langle q_i(p) - q_i(\sigma_i^*), u\rangle \ge J_i^0(\sigma_i^*; \sigma_{-i}^*) - J_i^0(p; \sigma_{-i}^*)\ \ \forall i,\ \forall p \in \mathcal{P}_i\big\},$$

with J_i^0 the cost at zero charges. Since $J_i^u = J_i^0 + \langle u, q_i\rangle$, $P(\sigma^*)$ is exactly the set of charges under which σ^* is a pure Nash equilibrium: a polyhedron, possibly empty. Toll polyhedra of this kind are classical for nonatomic routing [89, 90] and inverse optimisation [91].

Theorem 27 (Separation and finite termination)**.** *(a) $u \in P(\sigma^*)$ if and only if no carrier has a strictly improving best response to σ_{-i}^* under u; an improving best response gives that carrier's most violated constraint. (b) With $\varepsilon = 0$, Algorithm 2 stops after at most $1 + \sum_i(|\mathcal{P}_i| - 1)$ rounds, either with $u \in P(\sigma^*)$ or with an infeasible LP, which happens only if $P(\sigma^*) = \emptyset$. Each round costs N oracle calls and no equilibrium solve.*

Proposition 28 (Single-level form)**.** *With optimistic selection, the leader's value is $V^\star = \min\{\mathrm{SC}(\sigma) : \sigma$ pure, $P(\sigma) \ne \emptyset\} \ge \min_\sigma \mathrm{SC}(\sigma)$. If Algorithm 2 ends in its main loop, the leader's loss is at most the suboptimality of the target, $\mathrm{SC}(\sigma^*) - \min_\sigma \mathrm{SC}(\sigma)$; in particular DIP is at least as good as any grid over any subfamily of charges when its target is a social optimum.*

Algorithm 2 DIP: dual implementation pricing

Target. Column generation on the convexified social cost (the carriers' oracle at social prices), rounding and coordinate polish $\Rightarrow$ pure profile σ^* with aggregate Q^*
Reference. $u^{\mathrm{ref}} \leftarrow \zeta \odot Q^* - \frac{m}{2} \odot \bar{d}$ (tangent charge); cuts $\mathcal{C} \leftarrow \emptyset$; $u \leftarrow u^{\mathrm{ref}}$
repeat
    for each i: $b_i \leftarrow$ best response to σ^*_{-i} under $u \triangleright N$ routing calls, no equilibrium solve
    for each i with $J_i^u(b_i) < J_i^u(\sigma_i^*)$: add $\langle q_i(b_i) - q_i(\sigma_i^*), u\rangle \geq J_i^0(\sigma_i^*) - J_i^0(b_i) + \varepsilon$ to $\mathcal{C}$
    if no cut was added: **return** u $\quad\triangleright$ σ^* is an exact pure equilibrium under u
    $u \leftarrow \arg\min\{\|u - u^{\mathrm{ref}}\|_1 : u \in \mathcal{U}$ satisfies $\mathcal{C}\}$ (LP); **if** infeasible: **break**
until
Fallback (only if $P(\sigma^*) = \emptyset$): best-response dynamics from σ^* under the least-violating charge and under u^{ref}; keep the better equilibrium
Instruments. $s = \min_z u_z$, $t_z = (u_z - s)/r$ (Proposition 23)

Theorem 29 (Loss of the fallback)**.** *For any charge u with $v = u - u^{\tan}$ and $Q^\circ = Q^*$, the equilibrium σ^u reached from σ^* by improving moves satisfies* $\mathrm{SC}(\sigma^u) - \mathrm{SC}(\sigma^*) \leq \sum_z \frac{\zeta_z}{2}(Q_z^u - Q_z^*)^2 + L(\sigma^*) - L(\sigma^u) + \langle v, Q^* - Q^u\rangle$.

Corollary 30 (Oracle calls)**.** *DIP uses the calls of the social target, N calls per cutting-plane round, and, only if $P(\sigma^*)$ is empty, the calls of the fallback dynamics. A grid of G points with an equilibrium solve of C_{eq} calls per point uses GC_{eq}.*

Theorem 31 (Imprecision shifts the charge)**.** *Under Γ-maximin credal boxes of width δ (Lemma 5), the charges that implement σ^* are $P(\sigma^*) - \hat{s}(\delta)$ with $\hat{s}_z(\delta) = \delta(b_z w_{(z,Q)} + r\, g_z w_{(z,X)})$. One run of DIP therefore gives $u(\delta) = u(0) - \hat{s}(\delta)$ for every width at no further oracle cost, and the leader attains the same social cost at every width as long as $u(\delta)$ is admissible. For E-admissibility, maximality and interval dominance, $u(0)$ works at every width.*

Proposition 32 (When one instrument suffices)**.** *A uniform per-delivery charge implements the convexified optimum under equal lots if the tangent charge is the same in every active zone. With the calibration of Section 6 ($\alpha_z = 1.5$ fee, $b_z = \alpha_z/(2Q_z^{\mathrm{full}})$) and n_z carriers with equal lots in zone z, the congestion-free tangent charge is $u_z = -0.375\,\mathrm{fee}/n_z$: a subsidy of 37.5% of the fee in single-carrier zones and 19% in two-carrier zones. The congestion charge contributes only spatial differentiation.*

Proofs are in Appendix C. Theorem 27(a) is what makes the method cheap: the separation oracle is one best response per carrier at the fixed target, not an equilibrium solve. Selection remains optimistic: the target is an exact equilibrium under the returned charges, but other equilibria may coexist.

Prior art and the claim. The tangent charge is the Pigou–Cournot corrective tax [85, 86]; carrier-specific charges are marginal-contribution utilities [87, 88]; toll polyhedra and implementing a target flow are classical [89, 90], including cost-balancing tolls for atomic games [92] and target-implementing tolls found by the ellipsoid method with an equilibrium oracle [69]; the loop is an inverse-optimisation cutting plane [91, 93, 94]; and best-response cutting

planes are used for pure equilibria of integer-programming games and related market problems [72, 95]. The closest leader-side method updates tolls by zeroth-order steps around an equilibrium solver in combinatorial congestion games, including freight carriers with hard tour problems [96]; like a grid, it needs an equilibrium solve per outer step. What remains new is modest: the exact potential–welfare identity of the lumpy Cournot–congestion game and the loss bounds it gives with the lot-asymmetry number E; the use of the carriers' own NP-hard best responses at the fixed target as the separation step, so that no equilibrium is solved; the translation of the implementing charges by the Γ-maximin surcharge; and the competitor-count rule.

5.4. Zone-market clearing (ZMC)

ZMC targets the D1 game in cost form. Let $c_i(p) = -\pi_i(p) + \gamma\langle t_i(p), \hat{\xi}\rangle$ (charges included), $\tilde{c}_i = c_i - \frac{1}{2}\|a_i\|_M^2$, and assume $M_k > 0$ on the active coordinates. Then $\Phi(\sigma) = \sum_i \tilde{c}_i(\sigma_i) + \frac{1}{2}\|A(\sigma)\|_M^2 = -\Psi_{\hat{\xi}}(\sigma)$ is an exact potential (Proposition 6). For a common price $\lambda \in \mathbb{R}^d$, the pricing oracle returns $O_i(\lambda) = \arg\min_p\{\tilde{c}_i(p) + \langle\lambda, a_i(p)\rangle\}$, which is the oracle of Assumption A5 with $Q = M$. Let $V_i(\lambda)$ be its value, $\mathcal{D}(\lambda) = \sum_i V_i(\lambda) - \frac{1}{2}\|\lambda\|_{M^{-1}}^2$ the price dual of the convexified potential, λ^* its unique maximiser, and $\bar{\Phi}^* = \max\mathcal{D} \le \min\Phi$ (Lemma 47).

Theorem 33 (Clearing identity)**.** *Let $\sigma_i \in O_i(\lambda)$ for all i and let $e = M^{-1}\lambda - A(\sigma)$ be the clearing error (supply minus demand at price λ). Then (a) $\Phi(\sigma) - \mathcal{D}(\lambda) = \frac{1}{2}\|e\|_M^2$, so $\Phi(\sigma) - \min\Phi \le \frac{1}{2}\|e\|_M^2$; (b) carrier i's Nash regret is at most $\frac{1}{2}\sum_{k\in\mathcal{Z}_i} M_k e_k^2$, where $\mathcal{Z}_i$ is the set of coordinates in which its plans differ; (c) σ_i is an exact best response to the belief $m_i = M^{-1}\lambda - a_i(\sigma_i) = A_{-i}(\sigma) + e$, hence E-admissible at every width $\delta \ge \max_{k\in\mathcal{Z}_i}|e_k|/w_k$ (if $m_{ik} \ge 0$ on $\mathcal{Z}_i$); (d) $e = 0$ implies that σ minimises Φ and is a pure Nash equilibrium; (e) conversely, exact clearing is attainable if and only if the convexification of the potential is tight at a pure profile.*

Part (a) is Fenchel–Young equality and a standard duality-gap fact. The game-specific content is in (b) and (c): the own-contribution term of the absorption bound cancels, because under D1 the own quadratic term turns a pricing-optimal plan into a best response to the anchor m_i. Finding a good pure equilibrium thus becomes a price search: find one λ at which the carriers' pricing answers nearly clear every zone. Because $\mathcal{D}$ is concave, the aggregate response is antitone in λ, and along each coordinate the sign of the residual $s_k = A_k(\lambda) - \lambda_k/M_k$ localises λ_k^* up to $M_k\chi_k$, where χ_k is the cross-zone variation: how far other coordinates' prices can move the response in coordinate k at a fixed own price.

Theorem 34 (Bracket invariant and dimension-free rate)**.** *Start from brackets $[0, M_k A_k^{\max}]$ and, after each round, move the lower (upper) end of bracket k to λ_k when $s_k > 0$ ($s_k < 0$). Then $\operatorname{dist}(\lambda_k^*, [l_k, u_k]) \le M_k\chi_k$ after every round. If the bracket midpoint is queried at least every other round, after T rounds (NT oracle calls, independent of d) every point of the bracket satisfies $|\lambda_k - \lambda_k^*| \le M_k A_k^{\max} 2^{-\lceil T/2\rceil} + M_k\chi_k$ for all k. If plan sets are separable across coordinates, $\chi = 0$.*

The floor $M_k\chi_k$ is genuine when zones are complementary (Example 48). Algorithm 3 alternates midpoint steps (for the rate) with secant steps on isotonic fits of each coordinate's response (for speed), then rounds over the collected plans and finishes with best-response polish.

Algorithm 3 ZMC: zone-market clearing for pure equilibria of the D1 game

Require: pricing and best-response oracles; rounds T

$l \leftarrow 0$, $u \leftarrow MA^{\max}$, $\lambda \leftarrow (l+u)/2$; plan pools $\mathcal{Q}_i \leftarrow \emptyset$

for $t = 0, \dots, T-1$ **do**

  for all i in parallel: $\sigma_i^t \leftarrow O_i(\lambda)$; $\mathcal{Q}_i \leftarrow \mathcal{Q}_i \cup \{\sigma_i^t\}$ ▷ N routing solves

  $s \leftarrow A(\sigma^t) - M^{-1}\lambda$; update every bracket by the sign of s_k (Theorem 34)

  if t odd: $\lambda \leftarrow (l+u)/2$; **else** for each k: secant step on an isotonic fit of (λ_k, A_k), clipped to $[l_k, u_k]$

$\sigma^{\text{price}} \leftarrow$ the pricing profile with the smallest $\frac{1}{2}\|e^t\|_M^2$

$\sigma^{\text{round}} \leftarrow$ restricted master over the pools and derandomised rounding (no oracle calls)

$\sigma \leftarrow$ the better of the two; sequential best responses until no carrier improves

return σ

Theorem 35 (Guarantee of ZMC)**.** *With exact oracles, ZMC uses NT pricing calls before the polish; the polish stops after finitely many best-response calls at a pure Nash equilibrium σ with $\Phi(\sigma) \le \bar{\Phi}^* + \frac{1}{2}\min_t \|e^t\|_M^2$; and $\sigma \in \mathrm{ICNE}_C(\delta)$ for all $\delta \ge 0$ and $C \in \{E, M, ID\}$. Run on the surcharged game of Lemma 5, it returns an exact Γ-maximin equilibrium at width δ.*

Charges translate the supply line: under charges t the equilibrium gross price is 1-Lipschitz in t and convexified aggregates respond monotonically (Proposition 49), so brackets and plan pools warm-start the next evaluation when a city leader searches over charges.

Prior art and limits. Price decomposition with a pricing oracle is classical (Dantzig–Wolfe, column generation and bundle methods; 97); learning separable piecewise-linear concave approximations from sampled slopes is SPAR [98]; Walrasian prices with indivisible goods are computed from demand oracles by Paes Leme and Wong [99]; and Duguet et al. [95] relate the regret of price-allocation pairs to duality and integrality gaps in combinatorial markets, which is the closest conceptual neighbour of Theorem 33(a)–(b). We claim novelty only for the own-term cancellation with credal anchors (Theorem 33(b),(c),(e)), the sign-bracket invariant with the cross-variation floor (Theorem 34) and the method they define. ZMC has no D2 version, does not bound the clearing error of pure answers a priori, and spends NT calls even on easy games.

5.5. Integration in the city experiments

In the experiments, carrier i's oracle at a price vector λ (zone prices on deliveries and vehicle-hours) chooses a set of node-disjoint routes from its depot over its customer zones. Each served zone contributes its whole lot (all-or-nothing) together with the prize, the own Cournot and congestion terms, the charges and λ as node costs; each route costs its BRL upper expectation under DKW probability boxes for nine sector multipliers. The oracle runs column generation with BRL pricing to LP convergence and then an integer packing over the generated columns. It returns the plan, its contribution $a_i(p) = (q_z, \tau_z)_z$ and the LP value as a lower bound on the oracle objective. Route costs do not depend on prices, so each carrier's column pool is reused across ZMC rounds and across the leader's charges. The approximations are: (A1) all-or-nothing lots; (A2) in the experiments of Sections 7.1–7.5 the integer step is branch-and-price (Section 5.2) with a time limit per call, and calls that

Table 1: City instances. Z: zones; population from GHS-POP inside the zones; major road and lane kilometres inside the zones; nodes of the major-road graph; N: number of carriers.

City	Z	pop. (M)	major road km	lane-km	graph nodes	N
New Delhi	120	25.0	5 556	8 590	138 542	5
Cairo	42	14.9	2 649	4 498	119 027	4
Moscow	160	12.6	4 475	12 325	146 234	5
Buenos Aires	208	13.3	5 764	12 032	130 526	4
New York	234	12.1	8 798	17 770	382 014	4
Mumbai	143	22.0	4 259	7 092	109 804	5
London	348	11.3	7 110	13 628	350 931	5
Paris	198	9.02	5 990	10 854	309 175	5
Berlin	593	4.65	5 490	11 147	192 577	6

are not proven optimal are counted and their remaining gap recorded; the sensitivity runs, the algorithm comparison, the ZMC checks, the theory tests and the tiny-game verification use the earlier price-and-branch step; (A3) zone exposure proportional to the lot, so that a congestion charge on hours and a delivery charge are collinear within a zone; (A4) linear (D1) congestion; (A5) oracle exactness at city scale is checked, not proved.

5.6. Optional quality statement

Two by-products are available at no extra oracle cost. Because BRL pricing is exact, the column-generation master value is a valid lower bound on the robust routing optimum, which gives every integer plan a valid optimality gap; branch-and-price closes that gap. In ZMC, the dual values $\mathcal{D}(\lambda^t)$ bracket the convexified optimum with the potential of the returned equilibrium, which gives an upper bound η_{rel} on the relative potential gap; with a heuristic routing oracle the LP bound of the oracle keeps the bracket valid. Neither is the contribution; Section 7 reports both.

6. Data and calibration

Road networks and zones. For each of nine cities (New Delhi, Cairo, Moscow, Buenos Aires, New York, Mumbai, London, Paris, Berlin) we use an OpenStreetMap extract [100] distributed by BBBike (replication timestamp 16 October 2026). The drivable major-road network (motorway to tertiary, with links) is kept as an undirected graph with class-based free-flow speeds and lane counts (OSM `lanes` tag where present, class defaults otherwise). Population comes from the GHS-POP R2023A grid, epoch 2025, at 30 arc-seconds [101]. Zones are blocks of 4×4 raster cells (about 3.7 km) inside the extract's bounding box with at least one inhabitant; zone-to-zone free-flow times are graph shortest paths between population-weighted centroids plus a fixed access time. Points of interest from the same extracts (warehouses, courier offices, post offices, wholesale) give a depot score per zone that is used to place carrier depots. Table 1 summarises the instances; the number of zones ranges from 42 to 593.

Operating parameters. Per-city parameters were collected from public web sources on 23 October 2026 and graded A (official or primary statistic), B (industry report or country-level proxy) or C (rough estimate or proxy from elsewhere). Table D.12 in Appendix D lists the

calibration used; the repository file `data/parameters/SOURCES.md` gives the source, URL, grade and note of each number. Congestion levels are those of the TomTom Traffic Index 2025 [102] and volume-delay curves use the BPR form with the conventional $\alpha = 0.15$, $\beta = 4$ [103].

Weak numbers. Several inputs are weak and are flagged as such. Parcels per capita are country-level figures for every city. Driver wages are minimum-wage floors or job-site averages, not courier payroll data. Drops per vehicle and van capacities outside the United States are proxies from India, London or a single company survey. Delivery fees in Cairo, Moscow and Buenos Aires are consumer or reference rates, not carrier revenue per parcel, and the European fees are regulator revenue per parcel. Moscow is absent from the 2025 TomTom index and uses a 2020 value. Market shares are available only for Paris and Berlin; elsewhere carriers have equal shares. The sensitivity of the equilibria to the five weakest parameters is reported in Table D.13.

Calibration rules. Each carrier serves 22 customer zones sampled around its depot with weight proportional to population times a gravity decay in travel time, so carriers overlap in a subset of zones; a lot is 10–50% of a vehicle's daily drops, proportional to the zone's parcel demand. Inverse demand has intercept $\alpha_z = 1.5\times$ fee and slope $b_z = \alpha_z/(2Q_z^{\text{full}})$, so that the price at full service is half the intercept. The congestion slope c_z is the derivative of the BPR curve at the TomTom congestion level, with capacity from lane-kilometres, and it is scaled by a representative-slice multiplicity m_z, the ratio of the city's real delivery vehicle-hours to the modelled ones, so that the modelled slice faces the congestion of the whole city. Travel-time multipliers of nine sectors follow a lognormal law matching the TomTom all-day and peak congestion, and each carrier observes 30 days, from which DKW probability boxes at confidence $1 - \alpha$ are built. The route loss (3) uses the vehicle-hour cost (wage plus fuel) per hour, twice that as fixed cost, three times that per overtime hour, and the shift length as deadline.

Proxies. The following are proxies and are labelled as such wherever they are used: the all-or-nothing lots, the sampled customer zones instead of whole-city demand, equal market shares in seven cities, the proportionality of vehicle-hours to deliveries within a zone, the slice multiplicity m_z, and a *broad* social cost that adds the delay imposed on background traffic (population times vehicle ownership times one hour per day, valued at the wage, times the added slowdown, divided by m_z). The broad objective is reported but not used for policy conclusions (Section 7.2).

7. Results

All numbers in this section are generated from the result files of the repository by `paper/make_tables.py`. The game experiments use three seeds (0, 1 and 2). A seed redraws the depots, the 22 customer zones of each carrier, the travel-time sectors and the 30 observed days, and therefore the probability boxes; the calibration inputs are the same in every seed. With at most three seeds we report the mean together with the minimum and maximum, or the values per seed, and read them as a rough indication of spread, not as a confidence interval. The seed variation does not cover the uncertainty in the calibration inputs, which is examined separately (Table D.13).

Oracle and coverage. The base equilibria, the leader's charges, the price of anarchy and the imprecision sweeps were computed with the exact branch-and-price integer step (Section 5.2) in the carrier oracle, with a limit of 20 seconds per call and a 120-second retry. Of 38,591 oracle calls, 38,408 (99.53%) were proven optimal; 183 remained unproven after the retry, with a remaining gap of at most 8.5 USD on plans worth hundreds to tens of thousands of USD. Coverage is partial, and Table D.11 in Appendix D lists it per city and seed:

- base equilibria: all 27 city-seeds;
- leader's charges: 24 city-seeds; the full 63-point grid only for New Delhi, Buenos Aires, London. For Cairo, Moscow, New York, Mumbai, Paris, Berlin only 7 to 17 grid points per seed were solved exactly, chosen by screening on the earlier price-and-branch grid, so the leader's gains in these cities are lower bounds on the gains over the full grid (marked † in Table 3);
- price of anarchy: 24 city-seeds (missing: Moscow seed 2, New York seed 2, Berlin seed 2); in New York, Moscow and Berlin the social-optimum benchmark of the broad objective was still computed with price-and-branch;
- imprecision: 22 city-seeds; all 12 widths for New Delhi, Buenos Aires, Mumbai, London, and 6 widths (zero, base and widest of each sweep) for Cairo, Moscow, New York, Paris, Berlin (missing city-seeds: Cairo seed 2, Moscow seed 2, New York seed 2, Paris seed 2, Berlin seed 2).

The sensitivity runs (Table D.13), the algorithm comparison (Table 9), the ZMC checks (Table D.17), the theory tests of Section 7.7 and the tiny-game verification were not re-run and use the earlier price-and-branch oracle. Every equilibrium reported below is a pure Nash equilibrium of the game with the exact oracle up to unproven calls; Section 7.8 reports the independent checks, and Section 7.6 the effect of the earlier inexact integer step.

7.1. Unpriced equilibria

Table 2 reports the unpriced equilibria computed by ZMC with eight rounds and best-response polish. The cities fall into two groups. In Cairo, Moscow, New York, Paris, Berlin the carriers serve at least 85% of customer demand. In New Delhi, Buenos Aires, Mumbai, London they serve between 18% and 45%, because the delivery fee is low relative to the vehicle-hour cost or the shift is short. The added travel-time slowdown caused by delivery vehicles ranges from 0.77% to 17.6% (Moscow). Figure 1 shows the equilibrium zone prices of seed 0 on the city grids. The bound η_{rel} on the relative potential gap is below 1% in 3 cities and reaches 44% in the low-service cities, where the potential is close to zero and the dual bound is loose; a small proven gap therefore holds only where service is high.

7.2. The city leader's charges

A city leader chooses a congestion charge per vehicle-hour, $t_z = \vartheta \gamma c_z X_z^{\mathrm{NE}}$ with multiplier $\vartheta \in \{0, 0.5, 1, 2, 4, 8, 16\}$, and a per-delivery charge s expressed as a fraction of the fee in $\{-0.6, \ldots, 0.6\}$; negative values are subsidies. Each grid point requires one warm-started equilibrium. The grid serves as the baseline for the leader's charges, which are computed without a grid in Section 7.3, and for the comparison of one and two instruments. Table 3

Table 2: Unpriced equilibria with the exact oracle (mean over three seeds; seed standard deviation in parentheses). Served: share of customer demand delivered; slowdown: added mean travel-time slowdown from delivery vehicles; price: mean zone price (USD per parcel); veh.-h: vehicle-hours per day; welfare W in USD per day of the modelled slice; calls: oracle calls; regret^{LP}: guaranteed upper bound on any carrier's gain from deviating (USD), nonzero only where an oracle call was not proven optimal; η_{rel}: upper bound on the relative potential gap.

City	N	Z	served	slowdown %	price	veh.-h	welfare	calls	regret^{LP}	η_{rel}
New Delhi	5	120	0.26 (0.01)	4.2	0.803	49.1	82.7 (7.0)	48	$7.1 \cdot 10^{-15}$	0.12
Cairo	4	42.0	1.0 (0)	3.6	0.885	125	1 066 (20)	39	0.0020	$-2.4 \cdot 10^{-4}$
Moscow	5	160	0.99 (0.004)	18	2.76	170	6 756 (154)	47	1.2	0.0036
Buenos Aires	4	208	0.36 (0.03)	0.77	0.941	43.6	272 (19)	39	$2.1 \cdot 10^{-14}$	0.24
New York	4	234	0.99 (0.002)	7.8	6.82	112	20 003 (1 441)	39	4.8	0.0050
Mumbai	5	143	0.45 (0.02)	4.6	0.751	83.1	153 (10)	47	$1.8 \cdot 10^{-14}$	0.046
London	5	348	0.18 (0.03)	6.7	4.67	24.8	570 (101)	48	$1.1 \cdot 10^{-13}$	0.44
Paris	5	198	0.85 (0.04)	5.0	4.66	105	5 332 (437)	50	0.76	0.039
Berlin	6	593	0.89 (0.01)	3.7	3.36	127	8 619 (256)	58	0.90	0.029

Table 3: City leader, main objective, exact oracle (3 seeds). Gain of the best charge over no charge (% of |social cost|), mean [min, max] over seeds, with the congestion charge only, the delivery charge only, or both. Delivery share: gain with the delivery charge alone divided by the gain with both, per seed, shown where the combined gain is at least 0.5%. Subsidy: seeds in which the best point with both instruments has $s < 0$. Served share and city revenue (USD/day; negative is a net subsidy) at that point. †: only a subset of the grid solved exactly, so gains are lower bounds and shares are ratios of lower bounds (a share above one is an artefact of the reduced grid).

City	cong. only	deliv. only	both	deliv. share by seed	subsidy	served	revenue
New Delhi	0	10.3 [5.0, 17]	14.5 [12, 17]	0.41 / 0.97 / 0.65	3/3	0.39 [0.36, 0.42]	-54.7 [-64, -48]
Cairo†	0	0	0	–	0/3	1.0	0
Moscow†	0	0.128 [0.13, 0.13]	0.128 [0.13, 0.13]	–	2/2	1.0	-3 243 [-3 375, -3 112]
Buenos Aires	2.49 [0.43, 6.5]	35.4 [27, 40]	35.4 [27, 40]	1.0 / 1.0 / 1.0	3/3	0.74 [0.65, 0.79]	-413 [-528, -226]
New York†	0	0.393 [0.33, 0.46]	0.393 [0.33, 0.46]	–	2/2	1.0	-1 994 [-2 097, -1 891]
Mumbai†	0	2.09 [0.37, 3.3]	2.16 [0.81, 3.3]	0.46 / 1.4 / 0.79	3/3	0.55 [0.49, 0.58]	-64.4 [-91, -20]
London	0.446 [0, 1.3]	49.3 [31, 67]	59.7 [34, 90]	0.92 / 0.89 / 0.74	3/3	0.49 [0.39, 0.54]	-1 359 [-1 803, -808]
Paris†	0.0412 [0, 0.12]	3.47 [2.8, 4.7]	3.57 [2.8, 4.7]	1.0 / 0.91 / 1.0	3/3	0.95 [0.89, 0.99]	-2 712 [-4 160, -849]
Berlin†	0	1.70 [1.6, 1.8]	1.77 [1.6, 1.9]	0.93 / 1.0	2/2	0.93 [0.93, 0.93]	-2 372 [-2 822, -1 921]

reports the reduction of social cost relative to no charge, as a percentage of its absolute value, for the best exact grid point using only the congestion charge, only the delivery charge, or both.

Congestion charge alone. The congestion charge alone gains little in all but one of the 24 city-seed cases: in 23 of them it gains less than 1.7% of social cost, the largest being 1.34% (London, seed 0). The exception is Buenos Aires, seed 0, with 6.51%. An earlier price-and-branch run, which gave 1.67% there, had suggested that this gain stays below about 1.7% everywhere; that reading does not survive. The difference comes from the baseline, not from the charge: with the exact oracle the unpriced equilibrium selected in Buenos Aires seed 0 is a worse one (welfare 252.2 against 266.7 USD per day), while the best congestion-charge equilibrium is about as good as before. In Buenos Aires the three seeds give 6.5 / 0.43 / 0.53%. Whether a congestion charge alone is worth anything can therefore depend on which unpriced equilibrium the market is in.

Per-delivery instrument. In 17 city-seed cases the combined gain is at least 0.5% of social

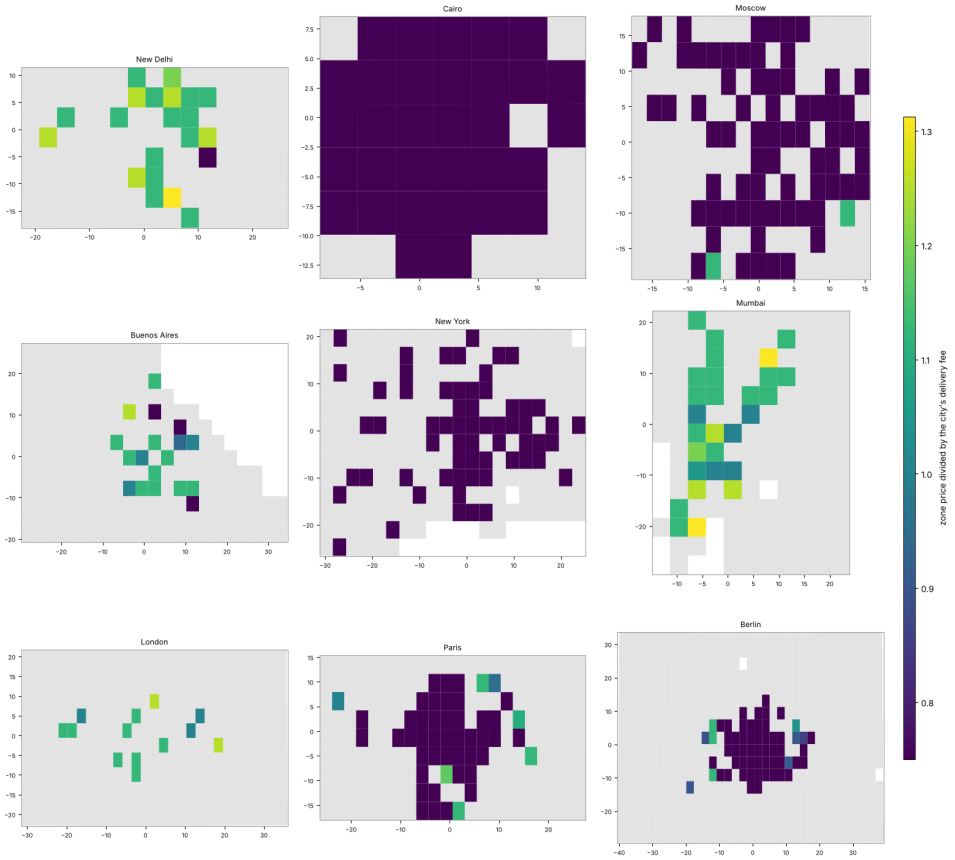


Figure 1: Unpriced equilibrium zone prices divided by the city's delivery fee, exact oracle, seed 0 (light grey: zones without deliveries; only carriers' customer zones are active).

cost. Qualitatively the delivery charge dominates: it achieves at least half of the combined gain in 15 of these cases, and the congestion charge alone is negligible in all but the Buenos Aires case above. The exceptions are Mumbai seed 0 (0.46), New Delhi seed 0 (0.41). In the cities with the full exact grid the shares are New Delhi 0.41–0.97, Buenos Aires 1.0–1.0, London 0.74–0.92. In the cities with a reduced grid the shares are ratios of lower bounds and are not reliable to two digits; in Mumbai seed 1 the share exceeds one, which is an artefact of the reduced grid. Mumbai's seed-0 share was 0.04 with price-and-branch and is 0.46 with the exact oracle, and New Delhi's seed-0 share fell from 0.57 to 0.41. In 17 of 17 material cases the best point with both instruments has a negative delivery charge, a subsidy.

Where the gain is large. The leader's gain is material only where the unpriced equilibrium under-delivers. In Cairo, Moscow, New York, where service is nearly complete, the gain is at most 0.46% in every seed; in Cairo none of the exact grid points improves on no charge, so the small gains found earlier with price-and-branch were artefacts of the inexact integer step. In the low-service cities the size of the gain varies widely between seeds: in London the combined gain ranges from 34.5% to 90.1% and the delivery-only gain from 30.8% to 67.1%; in New Delhi from 12.2% to 17.4% and from 5.03% to 16.9%. The three cities with gains above 10% (London, Buenos Aires and New Delhi) are the same in every seed, but the

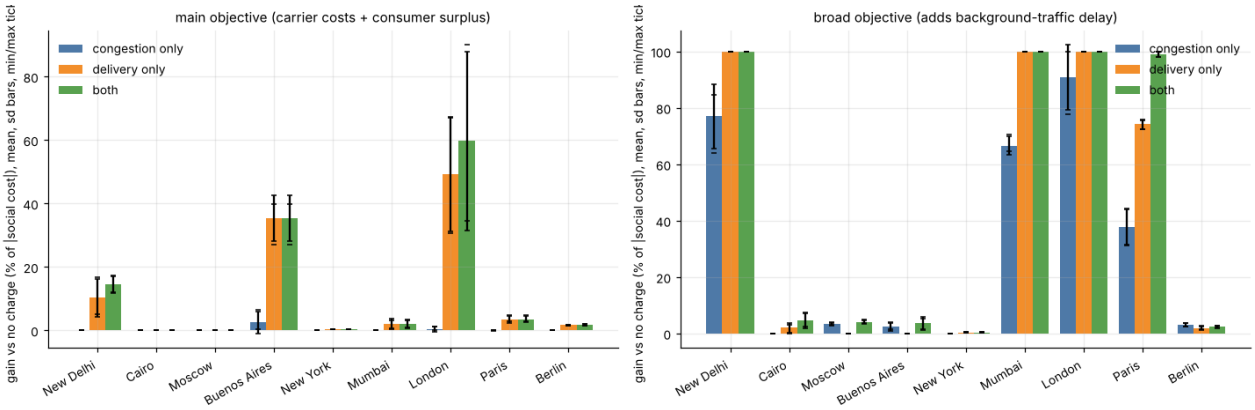


Figure 2: Gain of the city leader's best exact charge over no charge, by instrument: mean over the seeds computed, with standard-deviation bars and minimum/maximum ticks. Left: main objective. Right: broad objective with the background-delay proxy, whose shut-down optima are an artifact of the proxy. Cairo, Moscow, New York, Mumbai, Paris, Berlin use a reduced exact grid (lower bounds).

magnitudes are not pinned down.

The mechanism is visible in the model: the main inefficiency is under-provision caused by fixed route costs and market power, which a subsidy corrects, while the congestion externality among carriers is small. Within a zone the two instruments are collinear (A3), so the congestion charge can add only spatial differentiation. Subsidies cost the city money (negative revenue in the table); in the welfare definition charges are transfers, so this fiscal cost is not part of the social cost. Figure 2 shows the gains for both objectives.

With the broad objective, which adds the proxy for delay imposed on background traffic, the leader's best response in New Delhi, Mumbai, London is to stop deliveries altogether in every seed, and in Paris to cut service to at most 8.5% of demand (Table D.14). This is an artifact of the proxy: when the slice is scaled by m_z, the imputed delay cost per delivery vehicle-hour exceeds the carriers' surplus. The broad-objective results are also not robust to the seed: in Buenos Aires the combined gain is 4.6 / 5.4 / 1.1% in seeds 0, 1 and 2. We report these results and draw no policy conclusion from them.

7.3. Charges by dual implementation pricing

DIP (Section 5.3) computes the leader's zone charges without a grid. It was run with the exact oracle on all nine cities and three seeds (27 runs) and compared with the grid of Section 7.2 for the same city and seed (Table 4, Figure 3). The gain share is DIP's social-cost gain over the unpriced equilibrium divided by the gain of the best grid point; calls count every routing-oracle query of each method, with the same counter.

Gain and cost. In the three cities with a full exact grid (New Delhi, Buenos Aires, London, 9 runs), DIP reaches 104% to 135% of the grid-optimal gain (mean 115%) with 4.9% to 9.4% of the grid's oracle calls (mean 6.6%). Over all runs with a positive gain DIP reaches at least 0.999 of the best evaluated grid point, and it uses at most 9.4% of the calls of a 63-point grid. In Cairo no charge improves on the unpriced equilibrium, and DIP returns charges that leave it in place. The priced equilibrium attains the social cost of the target: its empirical price of anarchy is between 0.995 and 1.000 (values below one occur where DIP's multi-start target is

Table 4: DIP against the leader's grid, per city and seed (0 / 1 / 2). Grid pts: exact grid points available (pb: the 63-point price-and-branch grid, used where no exact grid exists for that seed). Gain share: DIP gain divided by the best grid gain (n/a: no charge improves on the unpriced equilibrium). Calls: DIP's oracle calls as a percentage of the calls of the grid run, and of a 63-point grid at the same calls per point. Rounds: cutting-plane rounds (1: the tangent charge implemented the target without any cut). Exact: runs in which the target is an exact pure equilibrium under the returned charges. PoA of the grid's best point and of DIP's priced equilibrium.

City	grid pts	gain share	calls % (grid run)	calls % (63 pts)	rounds	exact	PoA grid	PoA DIP
New Delhi	63 / 63 / 63	1.23 / 1.35 / 1.14	5.6 / 4.9 / 5.5	5.6 / 4.9 / 5.5	2 / 1 / 2	3/3	1.02 / 1.05 / 1.02	1.00 / 1.00 / 1.00
Cairo	13 / 17 / 13	n/a / n/a / n/a	11 / 8.5 / 12	2.3 / 2.3 / 2.4	1 / 1 / 1	3/3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00
Moscow	12 / 10 / 63^{pb}	1.10 / 1.00 / 1.00	13 / 13 / 2.9	2.6 / 2.1 / 2.9	1 / 1 / 1	3/3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00
Buenos Aires	63 / 63 / 63	1.09 / 1.05 / 1.08	6.3 / 6.3 / 5.4	6.3 / 6.3 / 5.4	1 / 1 / 1	3/3	1.03 / 1.01 / 1.02	1.00 / 1.00 / 1.00
New York	7 / 7 / 63^{pb}	0.999 / 0.999 / 1.04	24 / 23 / 2.9	2.6 / 2.6 / 2.9	1 / 1 / 1	3/3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.000
Mumbai	15 / 14 / 14	3.25 / 1.20 / 1.28	24 / 31 / 26	5.6 / 6.9 / 5.8	1 / 2 / 2	3/3	1.02 / 1.00 / 1.00	0.997 / 0.997 / 0.995
London	63 / 63 / 63	1.04 / 1.31 / 1.06	7.2 / 9.4 / 8.7	7.2 / 9.4 / 8.7	3 / 3 / 3	3/3	1.01 / 1.08 / 1.02	1.00 / 1.00 / 0.995
Paris	14 / 13 / 13	1.02 / 1.06 / 1.05	22 / 23 / 21	4.9 / 4.8 / 4.4	3 / 1 / 2	3/3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00
Berlin	13 / 13 / 63^{pb}	1.03 / 1.03 / 1.13	17 / 15 / 4.1	3.5 / 3.2 / 4.1	1 / 1 / 1	3/3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00

slightly better than the social-optimum benchmark of Section 7.4), against 1.01 to 1.08 for the grid's best point in the full-grid cities.

Exactness. In all 27 runs the cutting-plane loop ended with charges under which the target is an exact pure equilibrium; the regret against a fresh exact oracle is at most $9.1 \cdot 10^{-13}$ USD, and the fallback was never needed. The tangent charge alone implemented the target in 18 runs; the others needed at most 3 rounds and 4 cuts. The uniform part of the returned charges is a subsidy of 0.37 to 0.50 of the fee, and the zone charges follow the competitor-count rule of Proposition 32 (correlation 0.98 to 1.00 with $-0.375\,\text{fee}/n_z$ in the seed-0 runs of the full-grid cities, as reported in `theory/new_algorithm_C/results.md`). Figure 4 shows the charges for London.

Qualifiers. Several limits apply.

- Gains above 100% of the grid come mainly from zone-specific charges. Restricted to the grid's two-parameter family, DIP reaches on average 1.0 of the grid gain, with the worst case 0.50 (Mumbai, seed 0).
- Against the reduced exact grids of six cities, which have only 7–17 screened points, DIP uses more than 10% of their calls in most runs (Cairo 8.5–12%, Moscow 13%, New York 23%, Mumbai 24–31%, Paris 21–23%, Berlin 15–17%), although at most 9.4% of a 63-point grid.
- Wall-clock time is not always lower: the median ratio to the grid's time is 0.22, but in London seed 0 DIP took 247 s against 157 s, because the social-pricing calls of column generation are harder branch-and-price instances than warm-started equilibrium calls.
- Selection is optimistic. The target is an exact equilibrium under the returned charges but need not be the only one. The equilibrium that the grid's own protocol reaches from the unpriced prices under DIP's charges is the target in 21 of 27 runs, and in all runs its gain is at least 0.999 of the grid's. In 210 enumerated tiny games the equilibrium under DIP's charges was unique in only 35% of the games.

Table 5: Empirical price of anarchy and stability, main objective, exact oracle (3 seeds; missing: Moscow seed 2, New York seed 2, Berlin seed 2). W_{SO}: rounded social optimum (USD/day), mean [min, max]; #NE: distinct equilibria found; PoA of the ZMC equilibrium, of the worst equilibrium found, and PoS (best equilibrium found), per seed; priced gap: shortfall of the equilibrium under the leader's best charges from W_{SO} (%), per seed.

City	W_{SO}	#NE	PoA (ZMC eq.)	PoA (worst NE)	PoS	priced gap %
New Delhi	97.7 [86, 106]	4–5	1.15 / 1.23 / 1.16	1.15 / 1.24 / 1.20	1.12 / 1.23 / 1.16	2.4 / 4.9 / 1.6
Cairo	1 066 [1 046, 1 086]	2–3	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	0 / 0 / 0
Moscow	6 837 [6 744, 6 929]	2	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00	0.012 / 0
Buenos Aires	375 [361, 390]	3–6	1.43 / 1.42 / 1.29	1.43 / 1.54 / 1.31	1.32 / 1.41 / 1.27	2.5 / 1.3 / 1.7
New York	20 035 [18 604, 21 466]	1–2	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00	0 / 0
Mumbai	158 [148, 170]	4–6	1.02 / 1.04 / 1.04	1.11 / 1.08 / 1.04	1.02 / 1.04 / 1.04	1.5 / 1.2 / 0.39
London	927 [882, 991]	5–6	1.57 / 1.45 / 1.94	1.61 / 1.67 / 1.98	1.52 / 1.42 / 1.94	1.4 / 7.4 / 2.2
Paris	5 530 [5 023, 5 879]	3–5	1.03 / 1.03 / 1.05	1.03 / 1.03 / 1.05	1.03 / 1.02 / 1.05	0.047 / 0.19 / 0.23
Berlin	8 632 [8 570, 8 694]	2–4	1.02 / 1.02	1.02 / 1.02	1.02 / 1.02	0.054 / 0.054

- The result can be no better than the social target, whose quality is limited by the rounding gap of the social solver. In the tiny games DIP reached the exact leader optimum over all zone charges in 93% of the games, and in 100% when given the exact social optimum as target, so every miss there was a miss of the social solver, not of the implementation step. DIP used 23 to 64 calls per tiny game against 1 323 to 1 794 for the grid.

- The closed forms need D1 and one hours-per-delivery ratio; without the ratio the per-zone instrument is two-dimensional, although the polyhedron and the cutting plane still apply.

Imprecision. By Theorem 31 the charges for Γ-maximin carriers are the same charges shifted by the surcharge. At $\delta = 0.2$ and $\delta = 0.8$ the shifted charges made the target an exact equilibrium in every run (largest regret $9.1 \cdot 10^{-13}$ USD) without further oracle calls, so the welfare loss that imprecision causes without charges (Section 7.5) is removable by charges in the model. At $\delta = 0.8$, however, the required uniform subsidy is 0.98 to 3.6 times the fee, which is not a realistic policy.

7.4. Empirical price of anarchy and stability

The social optimum is approximated by column generation on the convexified welfare problem followed by derandomised rounding and coordinate polish; this gives a feasible pure profile W_{SO} and, from the relaxation, an upper bound. The set of equilibria consists of those found by ZMC, by best response from the empty and the full-service profile, and from three random price vectors, which yields between 1 and 6 distinct equilibria per city and seed. The price of anarchy and stability in Table 5 are therefore *empirical*: ratios of W_{SO} to the equilibria found, not bounds over all equilibria.

In Cairo, Moscow, New York, Mumbai, Paris, Berlin the empirical price of anarchy of the ZMC equilibrium is at most 1.05 in every seed computed. It is larger in New Delhi, Buenos Aires, London in every seed, and the ordering London > Buenos Aires > New Delhi holds in each seed; across seeds it ranges from 1.45 to 1.94 in London, 1.29 to 1.43 in Buenos Aires and 1.15 to 1.23 in New Delhi. The size of London's price of anarchy is therefore not pinned down. Compared with the earlier price-and-branch run, city means move by at most 0.022 and the ordering of the nine cities by price of anarchy, price of stability and worst equilibrium is

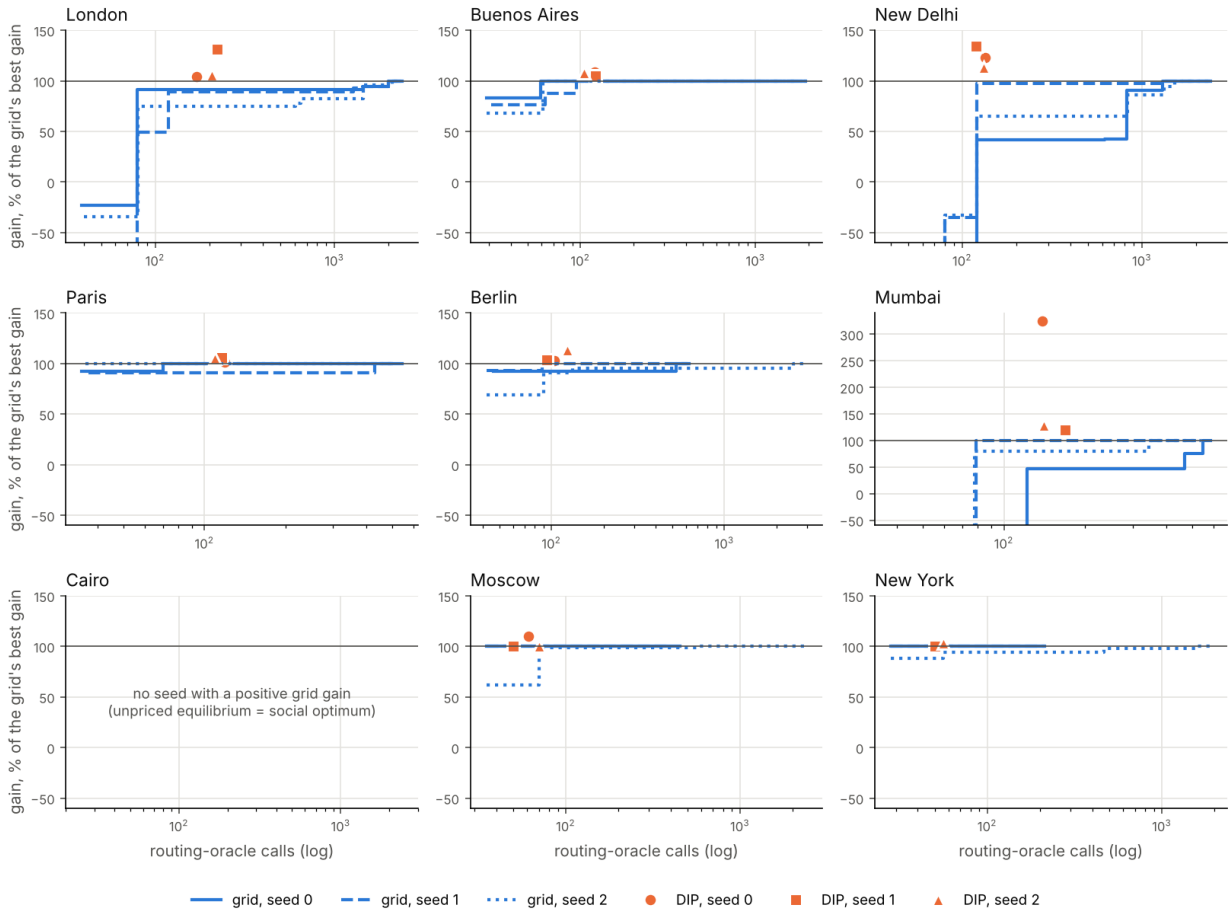


Figure 3: Leader's social-cost gain against routing-oracle calls: best grid point found so far (lines, exact grid rows in solve order) and DIP (one marker per run); 100% is the best grid point. Six cities have reduced exact grids, and seed 2 of Berlin, Moscow and New York uses the price-and-branch grid.

unchanged; the largest single change is Buenos Aires seed 0 (1.349 to 1.429), where a different unpriced equilibrium is selected, and Berlin seed 0 moves from 1.014 to 1.020. For the worst equilibrium found, Mumbai is borderline (up to 1.11 in seed 0).

Equilibrium selection matters in some games. In 5 of the 24 city-seed games the price of anarchy of the worst equilibrium found exceeds the price of stability by more than 0.05: Mumbai seed 0 (PoS 1.02, worst 1.11), Buenos Aires seed 0 (PoS 1.32, worst 1.43), Buenos Aires seed 1 (PoS 1.41, worst 1.54), London seed 0 (PoS 1.52, worst 1.61), London seed 1 (PoS 1.42, worst 1.67). In the others the equilibria found differ little in welfare.

With the leader's best grid charges the equilibrium comes much closer to the rounded optimum, but not uniformly: the shortfall is at most 0.4% in every seed in Cairo, Moscow, New York, Paris, Berlin, and reaches 7.4% in London, seed 1 (priced PoA up to 1.08) and 4.9% in New Delhi, seed 1. Two further caveats bound these statements: the relaxation bound

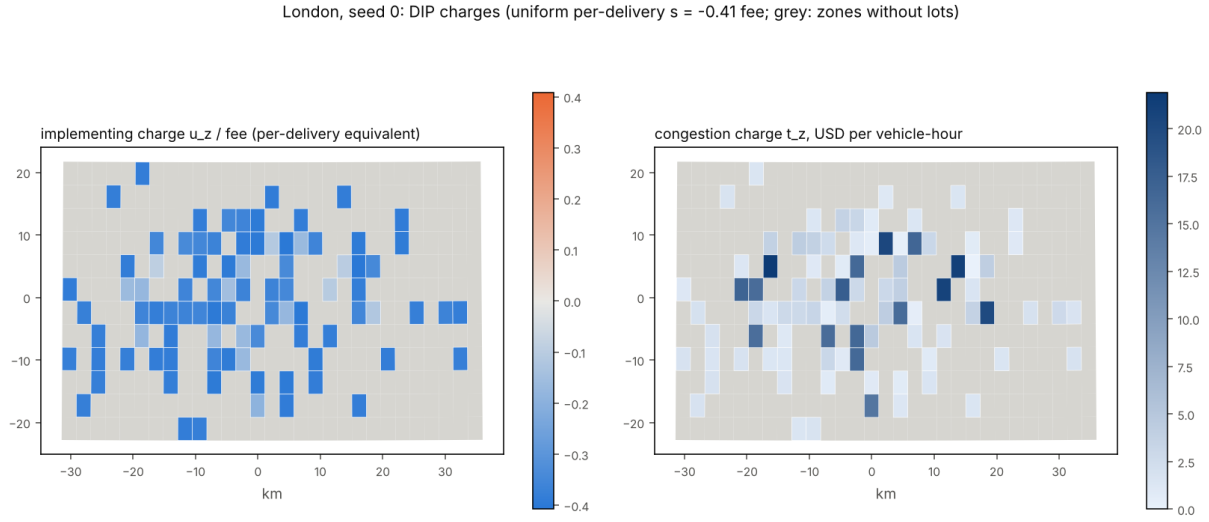


Figure 4: DIP charges for London, seed 0. Left: implementing charge u_z per delivery, in units of the fee (negative: subsidy). Right: the zone congestion charge per vehicle-hour that, with the uniform per-delivery subsidy, realises u. Grey: zones without lots.

exceeds the rounded optimum by up to 25% (London), so the true price of anarchy could be larger by that margin, and all values refer to the lumpy all-or-nothing lots of (A1). Figure 5 shows the same quantities.

7.5. Imprecision

Two kinds of imprecision are varied (Table 6, Figure 6). The first is the width of the travel-time probability boxes, through the DKW confidence level, from $\varepsilon = 0$ to $\varepsilon = 0.41$; it acts through the BRL route costs. Wider boxes raise route losses and cut service. At the widest box, over all cities and seeds computed, deliveries fall by between 0% and 78% and welfare by between 2.7% and 82%. The grouping of cities is the same in every seed: Cairo loses no deliveries, Moscow and New York lose a few per cent, Paris and Berlin somewhat more, Mumbai more than these, and New Delhi, Buenos Aires and London lose about half or more; the magnitudes in the low-service cities vary by seed (London: -60.0 / -56.1 / -78.2% in seeds 0, 1 and 2). The second is the credal width about rivals' aggregates under Γ-maximin, through the surcharge of Lemma 5 with w equal to the unpriced equilibrium aggregate, from $\delta = 0$ to $\delta = 0.80$. At the widest setting deliveries fall by between 31% and 100% and welfare by between 22% and 100%. Relative to the earlier price-and-branch run, the city means of these widest-width effects change by at most 1.8 percentage points; single seeds change by up to 6.0 points.

Deliveries are not always monotone in the width. Of the 44 (city, seed, sweep) series computed, 3 contain at least one rise, by up to 50 deliveries, all in Buenos Aires; all series are monotone in 8 cities. In Cairo, Moscow, New York, Paris, Berlin only three widths per sweep were solved exactly, which leaves little room for a rise to show. Proposition 16 does not apply here: for Γ-maximin no monotonicity holds in general, and for the probability-box sweep the reported equilibrium is a single one reached by ZMC, which can switch between equilibria as

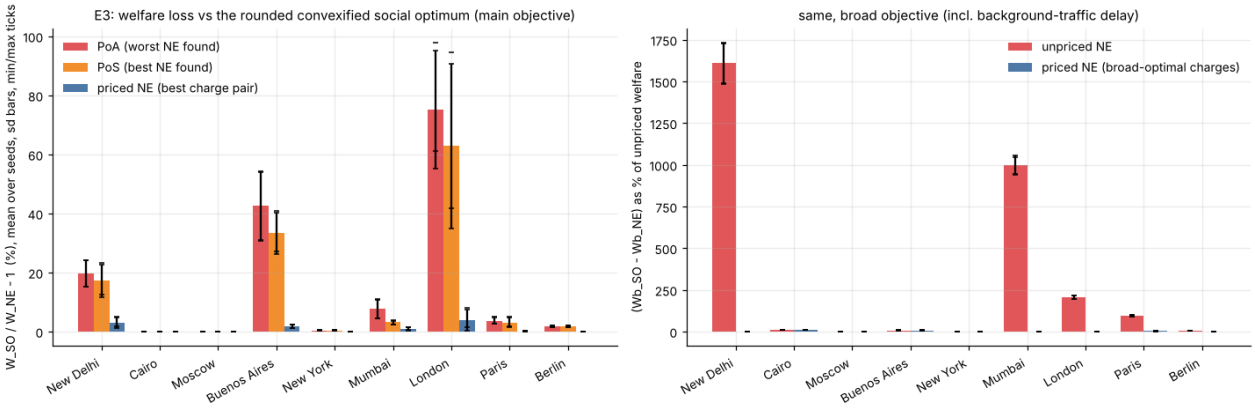


Figure 5: Welfare loss relative to the rounded social optimum, exact oracle, mean over the seeds computed with standard-deviation bars and minimum/maximum ticks. Left: main objective, worst and best equilibria found and the priced equilibrium. Right: broad objective (proxy).

Table 6: Imprecision, exact oracle (3 seeds; mean [min, max]). Left block: widest probability box ($\varepsilon = 0.41$) against $\varepsilon = 0$, change in deliveries and welfare (%). Right block: widest Γ-maximin width ($\delta = 0.80$) against $\delta = 0$. Monotone series: (seed, sweep) series in which deliveries never rise with the width. †: three exact widths per sweep.

City	Δdeliv. %	ΔW %	Δdeliv. %	ΔW %	monotone series
New Delhi	-53.5 [-55, -51]	-60.4 [-61, -59]	-62.2 [-75, -49]	-51.2 [-65, -39]	6/6
Cairo†	0	-2.81 [-3.0, -2.7]	-30.9 [-31, -31]	-22.3 [-23, -22]	4/4
Moscow†	-1.51 [-2.1, -0.95]	-10.5 [-11, -9.9]	-64.8 [-68, -62]	-56.6 [-61, -53]	4/4
Buenos Aires	-52.3 [-57, -49]	-52.9 [-57, -47]	-64.9 [-65, -64]	-54.6 [-56, -52]	3/6
New York†	-2.53 [-3.0, -2.1]	-10.9 [-11, -11]	-66.2 [-69, -63]	-57.9 [-61, -55]	4/4
Mumbai	-29.4 [-34, -23]	-30.3 [-32, -27]	-59.0 [-62, -56]	-44.7 [-48, -41]	6/6
London	-64.8 [-78, -56]	-74.8 [-82, -70]	-89.5 [-100, -81]	-86.5 [-100, -76]	6/6
Paris†	-9.08 [-11, -6.9]	-17.0 [-18, -16]	-73.5 [-74, -73]	-63.5 [-64, -63]	4/4
Berlin†	-8.64 [-9.2, -8.1]	-17.6 [-18, -17]	-71.3 [-72, -70]	-59.8 [-61, -58]	4/4

the width changes. Where monotonicity holds, it is a property of the equilibrium reached, not a guarantee for the game. In every case a pure equilibrium was found (Γ-maximin equilibria exist under D1 by Proposition 6), every run converged, and the largest regret against the pooled exact oracle was $9.1 \cdot 10^{-13}$.

Imprecision acts like a tax on activity: it reduces deliveries, vehicle-hours and congestion together, at the cost of welfare. Because vehicle-hours are proportional to deliveries in a zone (A3), the model cannot separate a congestion effect from a delivery effect here.

7.6. Algorithm performance

BRL pricing. The upper-prevision formula agrees with a linear program over all joint laws on 40 random instances to a relative error of $4.5 \cdot 10^{-14}$; for scale, the comonotone worst case exceeds the value under independent worst marginals by 4.5% on average and up to 14.1%. The pricing equals complete enumeration on every test instance, including 18 of 18 sub-instances built from the nine cities. Table 7 compares labelling rules on random instances with $L = 33$ levels and nine sectors. BRL takes between 0.004 and 0.065 seconds, while

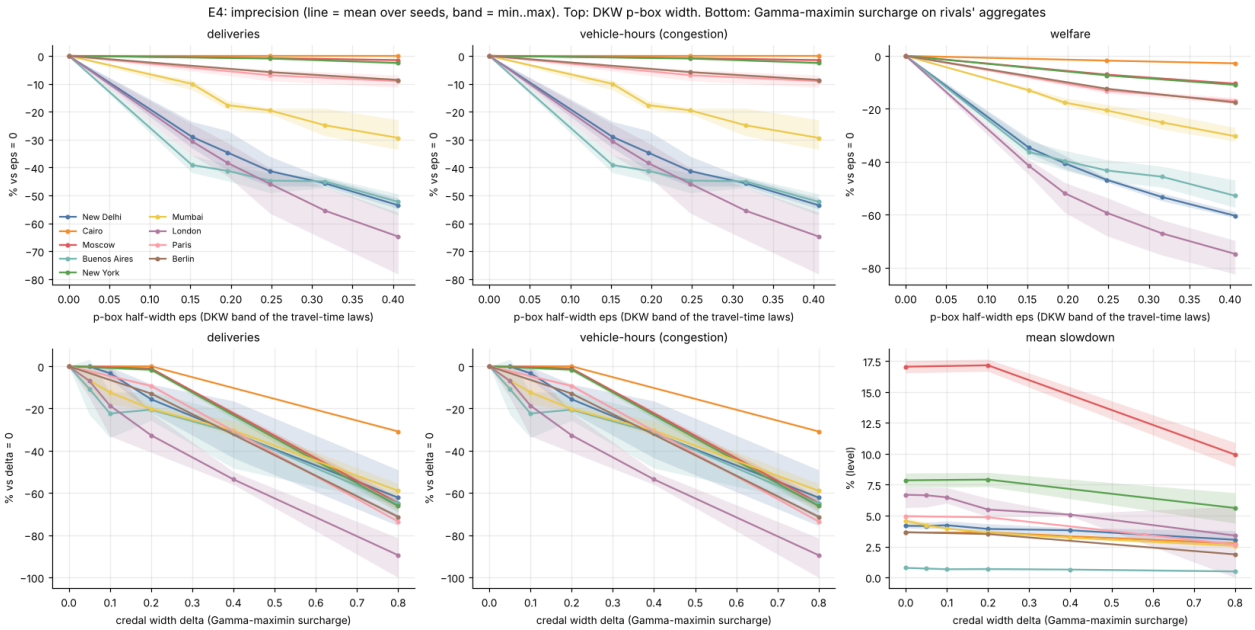


Figure 6: Imprecision, exact oracle, mean over the seeds computed (line) with the minimum–maximum band. Top: probability-box half-width (route costs through BRL). Bottom: Γ-maximin credal width (surcharge on rivals' aggregates).

sector dominance times out from the smallest size and quantile-vector and scenario labelling time out from $n = 15$. With dual prices scaled by 2.5, BRL prices $n = 25$ in 1.8 s while the risk-neutral labelling timed out; this reflects deadline pressure in those instances, where windows collapse to points, not a general complexity property. Fixing one risk level for all routes is suboptimal in 9 of 20 small exact comparisons, by up to 6.42%.

Branch-and-price on the metro instances. On the nine metro routing instances (22 customer zones, $L = 51$, ng-routes with eight neighbours, seed 0) branch-and-price with BRL pricing proved optimality (gap 0) in 9 of nine cities (Table 8), in at most 18 minutes on one core (Moscow, with 7487 nodes); pricing takes about 64% of the total time and the rest is spent in master LPs and bookkeeping. Against complete enumeration on 800 random formulations (partitioning and packing, up to ten customers), branch-and-price returned the optimum in 800 of 800, while price-and-branch over the root columns was strictly suboptimal in 39 of them, by up to 7.3%. An earlier version of this benchmark, which used price-and-branch, found the BRL plan worse than a heuristic optimised on the sixty observed days (SAA) in London, Paris and Berlin. That deficit came from the inexact integer step. With branch-and-price the London optimum has the same robust value as the SAA plan, which was therefore already optimal for the robust objective; the optimum is strictly better than the best of three heuristics under the robust objective in Moscow (-1.1%), Buenos Aires (-2.4%), New York (-10%), Mumbai (-0.32%), Paris (-0.10%), Berlin (-0.92%); and in New Delhi, Cairo, London the best heuristic was already optimal. The remaining limits are that the instances are small (capacity allows three to six customers per route), Moscow's congestion is imputed, and larger instances would likely need cutting planes.

The robust optimum is not the best plan under the calibrated law that generated the data.

Table 7: BRL pricing on random instances (nine sectors, $L = 33$ levels, capacity 6, two instances per size). Time in seconds (labels kept); TO: 60 s limit. The risk-neutral labelling solves a different (expected-value) problem and is shown for reference; SAA scenarios optimise the sample-average objective.

n	risk-neutral	BRL	sector (K)	quantile (L)	SAA scenarios
10	0.0050 (19)	**0.0040 (16)**	TO	2.2 (4946)	5.4 (7904)
10	0.017 (36)	**0.0080 (16)**	TO	2.1 (4391)	4.7 (7715)
15	0.17 (264)	**0.019 (48)**	TO	TO	TO
15	0.029 (69)	**0.010 (24)**	TO	TO	TO
20	0.39 (714)	**0.049 (97)**	TO	TO	TO
20	1.4 (1730)	**0.047 (89)**	TO	TO	TO
25	0.46 (650)	**0.059 (94)**	TO	TO	TO
25	0.49 (693)	**0.065 (109)**	TO	TO	TO

Table 8: Branch-and-price with BRL pricing on the nine metro routing instances (seed 0, 22 customer zones, robust objective in driver-hours). Root LP: column-generation bound; P&B plan: price-and-branch plan of the earlier pipeline; B&P optimum and remaining gap; nodes and time on one core; difference of the optimum from the best of three heuristics under the robust objective, and from the SAA heuristic plan under the calibrated law. *Moscow congestion imputed.

City	root LP	P&B plan	B&P optimum	gap %	nodes	time s	vs best heur. %	vs SAA, true law %
New Delhi	63.03	63.18	63.18	0	65	1.63	0	0
Cairo	40.39	40.42	40.42	0	215	5.30	0	0
Moscow*	42.01	43.65	43.17	0	7487	1 063	-1.1	-0.57
Buenos Aires	47.29	48.37	48.13	0	1659	166	-2.4	7.0
New York	46.10	46.10	46.10	0	1	8.44	-10	4.8
Mumbai	53.82	54.08	53.91	0	177	3.59	-0.32	-0.31
London	51.04	52.03	51.38	0	59	21.2	0	0
Paris	37.73	38.32	38.25	0	4305	715	-0.10	-0.026
Berlin	51.85	52.67	52.06	0	3	5.63	-0.92	6.8

Evaluated under that law (20,000 draws), the branch-and-price plan is worse than the SAA plan in Buenos Aires (+7.0%), New York (+4.8%), Berlin (+6.8%), and equal or slightly better elsewhere. The p-box worst-case cost of the robust optimum lies between 10% and 45% above its expected cost under the calibrated law. This conservatism is a property of the criterion (sixty observed days, a support cap, unknown dependence), not of the algorithm: a user who trusts an estimated law should not adopt the dependence-robust criterion.

Effect of the inexact integer step on the equilibria. The game experiments were first computed with price-and-branch in the carrier oracle. Tested against a fresh exact oracle, those earlier base equilibria were exact equilibria in New Delhi, Buenos Aires, Mumbai, London in every seed, but not in 10 of the 27 city-seeds: Cairo (3 of 3 seeds, up to 0.82 USD); Moscow (2 of 3 seeds, up to 0.83 USD); New York (1 of 3 seeds, up to 4.0 USD); Paris (2 of 3 seeds, up to 2.0 USD); Berlin (2 of 3 seeds, up to 3.5 USD). In those cases a carrier could gain a few USD per day with a plan the pooled integer program had not generated. The effect on the reported quantities is small except where a different equilibrium is selected. City-mean welfare of the base equilibria changes by at most 0.14% outside Buenos Aires, and by -1.4% in Buenos Aires, where seed 0 moves to a different equilibrium (-5.4%); single seeds elsewhere change by up

Table 9: Methods on the nine city games (seed 0, earlier price-and-branch oracle). d_{eff}: active coordinates. Oracle calls (left block) and relative potential gap to the convexified optimum, in % (right block), for best response (BR), ZMC with eight rounds, Frank–Wolfe and column generation with eight rounds (FW, CG8) and column generation to convergence (CG); all methods end with the same best-response polish.

City	N	Z	d_{eff}	BR	ZMC	FW	CG8	CG	BR	ZMC	FW	CG8	CG
New Delhi	5	120	64	14	45	45	50	65	13.8	10.2	10.2	10.2	10.2
Cairo	4	42	32	8	36	24	16	16	$1.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-5}$	$1.0 \cdot 10^{-5}$
Moscow	5	160	71	12	45	45	45	60	0.220	0.220	0.220	0.220	0.220
Buenos Aires	4	208	62	10	39	37	40	91	17.9	16.3	16.6	16.0	16.0
New York	4	234	64	8	36	36	36	76	0.451	0.451	0.437	0.437	0.437
Mumbai	5	143	55	14	48	48	48	60	4.60	1.98	1.98	1.98	1.98
London	5	348	89	15	49	49	50	70	20.0	18.0	18.0	18.0	18.0
Paris	5	198	68	13	50	50	50	108	2.74	2.78	2.74	2.74	2.74
Berlin	6	593	82	17	65	59	59	143	2.32	2.43	2.32	2.46	2.46

to 0.53%, with both signs, so the exact oracle changes which equilibrium is reached rather than improving welfare systematically. The changes in the leader's gains, the price of anarchy and the imprecision effects are reported in Sections 7.2–7.5. The exact oracle takes three to twelve times longer per equilibrium than price-and-branch.

ZMC. On synthetic D1 games with non-separable plan sets (Table D.15), ZMC's call count is flat in Z and, at an equal budget of oracle calls, its pure equilibria have the smallest potential gap in 17 of 20 instances; the advantage over Frank–Wolfe and column generation with eight rounds grows with Z, while full column generation reaches somewhat better equilibria with several times more calls.

On the city games the picture is different (Table 9). ZMC uses between 36 and 65 calls, about ten per carrier, and its potential gap coincides with those of Frank–Wolfe and column generation with eight rounds to three digits in most cities; the number of cities in which it is better than all of them by more than 10^{-3} is 0. Plain best response needs only 8 to 17 calls and is as good as ZMC (within 10^{-3} relative gap) in Cairo, Moscow, New York, Paris, Berlin; it is worse in New Delhi, Buenos Aires, Mumbai, London. ZMC therefore does not beat best response on these games. We see three reasons: only 32 to 89 coordinates are active because each carrier serves 22 zones, so the dimension that favours ZMC in the synthetic instances is absent; the calibrated games are mildly nonconvex; and carriers share few zones. The dimension-independence of ZMC is therefore not tested at city scale by these instances.

7.7. Theory meets data

The city games are of type D1, so the profile returned by ZMC with best-response polish is an exact pure equilibrium; its own E-threshold is zero and Theorem 7 holds trivially for it. To test the theorems in a non-trivial way we computed the exact thresholds of Theorem 41 (E-thresholds by Kelley cuts on the linear program, maximal and interval-dominance thresholds by bisection), with belief weights w equal to the realised aggregate. These tests were run with the earlier price-and-branch oracle; because that oracle is exact only up to a small gap, each threshold is relaxed by the largest improvement observed at a verified equilibrium. Table 10 and Figure 7 summarise the results.

Table 10: Theorems on the city games (three seeds). Mixed: seeds with a non-trivial mixed (relaxed) equilibrium. For those seeds (means): $\bar{\rho}$; the anchored three-term bound of Theorem 7(b); the N-uniform bound of Theorem 7(c); the largest empirical E-threshold over the tested profiles; and the ratio of the latter to the anchored bound. Γ-maximin (Lemma 5): drop in the share of demand served from the precise to the surcharge equilibrium at $\delta = 0.1$ and $\delta = 1$, and the maximality threshold of the surcharge equilibrium divided by δ at $\delta = 1$.

City	mixed	$\bar{\rho}$	anchored bd.	N-unif. bd.	max δ_E	ratio	drop δ=0.1	drop δ=1	M-thr./δ
New Delhi	3/3	1.0	1.61	10.7	0.64	0.41	0.0075	0.18	0.89
Cairo	0/3	–	–	–	–	–	0	0.47	0.99
Moscow	0/3	–	–	–	–	–	0.0021	0.73	0.97
Buenos Aires	1/3	0.96	1.76	8.80	0.83	0.47	0.082	0.31	0.93
New York	1/3	0.82	1.58	3.68	0.82	0.52	0.0090	0.71	0.98
Mumbai	2/3	0.74	1.23	10.6	0.53	0.43	0.049	0.34	0.92
London	3/3	0.85	1.56	13.6	0.82	0.53	0.033	0.17	0.76
Paris	3/3	0.88	1.39	8.43	0.69	0.49	0.036	0.74	0.96
Berlin	1/3	0.94	0.937	7.97	0.41	0.44	0.050	0.77	0.97

Credal absorption. Non-trivial mixed equilibria were constructed from pairs of distinct pure equilibria (found by best-response dynamics from perturbed starts) by solving the two-plan indifference conditions and checking with the oracle that every support plan is a best response over all plans. This succeeded in 14 of the 27 city-seed games; Cairo, Moscow have none, because almost all demand is served and carriers do not conflict. On all 274 tested profiles from the products of the supports (159 with the realised-aggregate weights and the rest with a second weighting), no statement of Theorem 7 was violated: the empirical E-threshold never exceeded the anchored radius, the minimum anchored radius never exceeded the three-term bound, the existence threshold never exceeded the N-uniform bound, and the order of the interval-dominance, maximality and E thresholds held (0 violations). The bound is loose by about a factor of two on the worst profile: the largest empirical threshold is 0.28 to 0.60 times the anchored bound (median 0.45), and 0.032 to 0.22 times the N-uniform bound. As Remark 9 predicts for $N \leq d$, the support-selection term $(N-1)\bar{\rho}$ is the smallest of the three terms in every case. In this small sample the ratio shows no dependence on N and a mild increase with d; with 24 anchors this is exploratory. Without the best-response polish, the rounded ZMC profile is E-admissible at a finite width in only 10 of 27 games, because some carrier plays a strictly dominated plan; the polish is what makes absorption apply to the returned profiles. Two limits apply: mixed equilibria were found only with two-plan supports, so not all relaxed equilibria were tested, and thresholds are exact only up to the oracle gap.

Γ-*maximin.* In all 108 (city, seed, δ) cases the equilibrium of the surcharge game of Lemma 5 differs from the precise equilibrium, and in 106 of them the precise equilibrium is not Γ-admissible. The surcharge equilibrium is maximal at width δ, with a maximality threshold below δ in every case with a finite threshold (at $\delta = 1$ between 0.76 and 0.99 of δ in the city means; 2 Cairo cases are infinite because of oracle slack). Γ-maximin is costly in service: averaged over seeds, the share of demand served drops by 0 to 0.082 at $\delta = 0.1$ and by 0.17 to 0.77 at $\delta = 1$.

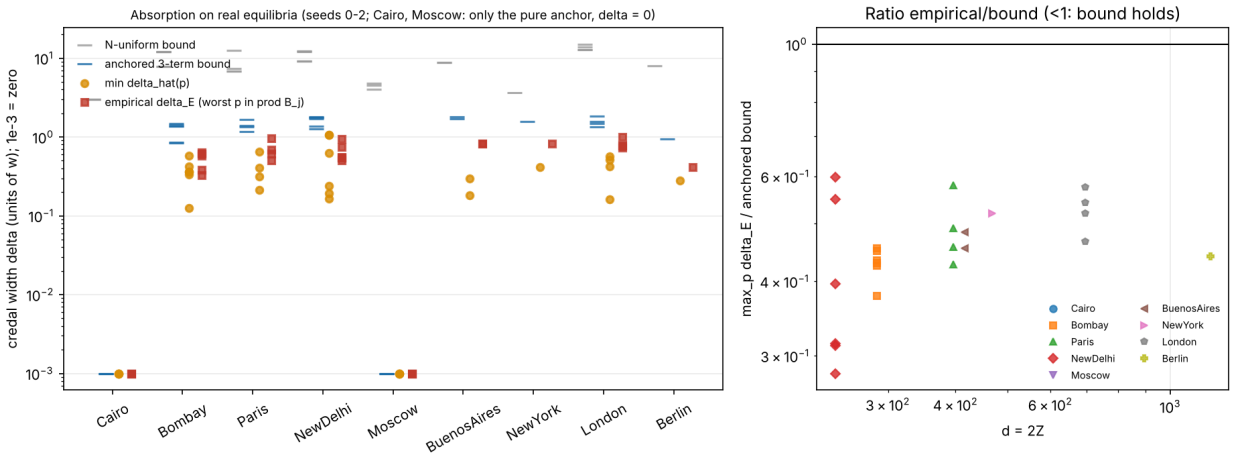


Figure 7: Credal absorption on the city equilibria (three seeds). Left: N-uniform bound, anchored bound, smallest anchored radius and largest empirical E-threshold per anchor (Cairo and Moscow: pure anchors only, plotted at 10^{-3} for zero). Right: ratio of the largest empirical E-threshold to the anchored bound against $d = 2Z$; values below one mean the bound holds.

Vehicle-count congestion. The gadgets behave as proved: the example of Proposition 11 has no pure equilibrium and existence threshold exactly $1/w$ (3 of 3 weights), and the kernel construction of Theorem 13 reproduces the kernel sets for all four criteria on 223 of 223 digraphs. Among random bilinear D2 games with three players, 6 of 400 have no pure equilibrium, against 0 of 150 D1 controls. Small routing-integrated D2 games, by contrast, always had a pure equilibrium (no pure equilibrium in 0 of 540 game-scale cases): with exposure proportional to service and plans that add over zones, routing plans did not produce the pattern of Proposition 11 in these instances. On the city games of Cairo, Mumbai and Paris (three seeds each) with a vehicle-count externality scaled by factors 1, 10, 100, 1 000, 10 000, all 180 best-response runs converged to a pure equilibrium (0 cycles; Figure D.8). Vehicle-count congestion did not prevent pure equilibria on the city instances; non-existence required the adversarial plan menus of the gadgets. The D2 calibration of the city games is an assumption introduced for this test.

7.8. Verification

On 40 tiny games (two or three carriers, three or four customer zones) solved by exhaustive enumeration, the integrated ZMC–BRL profile is a pure equilibrium of the enumerated game in every case and reaches the global potential minimum in 40 of 40; best response reaches it in 35, Frank–Wolfe with rounding in 38 and column generation in 39 (Table D.16). These instances support correctness more than they separate the methods.

At city scale, every base equilibrium, every verified leader best point (24 city-seeds) and every equilibrium of the price-of-anarchy experiment has regret at most $9.1 \cdot 10^{-13}$ USD against an independent fresh exact oracle (elementary routes, empty column pool); all grid points of the leader and imprecision experiments have zero regret against the pooled exact oracle up to rounding (largest $9.1 \cdot 10^{-13}$). Because 183 oracle calls were not proven optimal, the guaranteed regret bound (incumbent of the verifying call against its proven lower bound) is nonzero for 9 profiles, at most 4.84 USD (a New York base case with welfare of about 20,000

USD per day). These checks rely on the same branch-and-price method as the oracle; the pricer itself was checked against complete enumeration (Section 7.6). The clearing identity of Theorem 33(a) held to $3.4 \cdot 10^{-12}$ in every round of every city in the earlier price-and-branch run (Table D.17); this check was not repeated with the exact oracle.

8. Managerial and policy implications

The implications below are conditional on the model and on the proxies of Section 6; they indicate directions and magnitudes to be tested with city data, not recommendations for any particular city.

For city authorities. First, the efficiency loss from competition among carriers is concentrated where service is incomplete. Where carriers already serve almost all demand, the empirical price of anarchy is close to one and charges change little; a city in that situation gains more from managing kerb space and background traffic than from pricing deliveries. Second, where service is partial, the lever with the largest effect in our instances is a per-delivery instrument, and in every city-seed case with a material gain its sign is that of a subsidy. How large the gain is, however, depends strongly on the sampled slice of the city (in London between 34.5% and 90.1% of social cost across three seeds), so the direction of the recommendation is better supported than its size. A congestion charge on delivery vehicle-hours alone recovers little in all but one city-seed case, because in these instances the congestion that carriers impose on each other is small relative to the under-provision caused by fixed route costs and market power; the exception (Buenos Aires, seed 0) arose because the unpriced market sat in a worse equilibrium, which shows that the value of a charge can depend on which equilibrium the market is in. Third, adding the congestion charge to the subsidy raised the gain noticeably only in some low-service cases, and which cases these are changed with the seed and with the integer method; a city should not count on a second instrument without local evidence. Fourth, even the best grid charges did not bring every equilibrium close to the social optimum (a remaining shortfall of up to 7.4% in one London case), and any subsidy has a fiscal cost that a welfare criterion with transfers does not count, and its distributional effect across zones is uneven: access remains unequal in the low-service cities even at the best charges. Fifth, an objective that counts delay to other road users can reverse the recommendation; in our instances it did so only because of a crude proxy, which is the strongest argument for measuring that delay rather than imputing it.

Computing charges. The city does not need to search over charges. With DIP it can compute zone charges that make a chosen target an equilibrium of the carriers' game at a small fraction of the routing calls of a grid, and the charges have a readable structure: a per-delivery subsidy of roughly 37.5% of the fee divided by the number of carriers present in the zone, corrected for congestion. A zone-differentiated scheme of this kind did better than the best flat two-parameter scheme in the full-grid cities. It is also less transparent and harder to administer than a flat subsidy, and the target equilibrium need not be the only equilibrium under the charges; a city should weigh these points before adopting zone-specific charges.

For carriers. Imprecision is costly. Wider probability boxes on travel times and wider credal sets about rivals both reduce the volume a carrier finds worth serving. Under the dependence-robust criterion the planned cost is well above the expected cost under the simulated law, so

a carrier with enough data to estimate a law, or with knowledge of the dependence between road segments, should use it. Conversely, a carrier that plans robustly behaves like one facing a tax, which lowers congestion as a side effect.

For modelling. The theory gives a practical rule for choosing a congestion measure. Charging or modelling congestion through exposure (vehicle-hours in a zone) yields a potential game in which pure equilibria exist for every degree of imprecision and can be computed with a routing oracle. Counting vehicles instead makes the question whether any pure imprecise equilibrium exists NP-complete in general. On our city instances this worst case did not materialise: with a vehicle-count externality, best-response dynamics always reached a pure equilibrium. The hardness result is therefore a warning about what can happen with adversarial plan menus, not a prediction that vehicle-count rules make city outcomes unpredictable. The absorption theorem quantifies how much belief imprecision is enough to make a predicted pure outcome internally consistent.

Human-centred smart-city management science. The call for this special issue asks how data-driven optimisation can serve the people who live in and run cities. Three features of this work bear on that question. The model places the decision-relevant uncertainty where it is experienced: carriers' dispatchers plan with short travel-time records and with guesses about competitors, and the credal formulation represents what they know rather than imposing a law they do not have. The welfare accounting separates consumer surplus, carriers' costs, the delay carriers impose on each other, the city's fiscal position and the spatial distribution of access, so that a planner can see who gains and who pays under each instrument. And the evidence is produced from open data, with every weak parameter flagged and the code public, so that a city can replace our proxies with its own counts and rerun the analysis. The limits of that evidence are set out in the next section.

9. Limitations and future work

Lumpy lots. Carriers serve a customer zone entirely or not at all. The resulting Cournot game is lumpy, and part of the measured inefficiency may be an artifact of indivisibility; divisible lots or several lot sizes per zone would test this.

Exactness of the routing oracle and coverage. The game experiments use branch-and-price with a time limit per call; 183 of 38,591 calls were not proven optimal, so for 9 profiles the guaranteed regret bound is positive (at most 4.84 USD). The exact re-run is incomplete: the leader's grid is exact in full only for New Delhi, Buenos Aires, London, so the leader's gains in the other six cities are lower bounds and their delivery-only shares are not reliable to two digits; the price-of-anarchy experiment lacks Moscow seed 2, New York seed 2, Berlin seed 2; the imprecision sweeps use three widths and two seeds in Cairo, Moscow, New York, Paris, Berlin. The sensitivity runs, the algorithm comparison, the ZMC checks, the theory tests and the tiny-game verification still use price-and-branch, whose equilibria were not exact in 10 of 27 city-seeds. Branch-and-price itself was tested only on instances with 22 customer zones.

Proxies and weak parameters. Demand is sampled, market shares are equal in seven cities, vehicle-hours are proportional to deliveries within a zone, and the congestion coupling is scaled by a slice multiplicity. Most operating parameters are country-level or rough estimates; over their plausible ranges, delivery fee, drops per vehicle and the number of carriers move welfare by tens of percent (Table D.13). The sign of the best per-delivery instrument depends on the fee-to-cost ratio and could change with better data.

Vehicle-count congestion and the theory tests. The main city experiments are in the D1 regime. The D2 city runs of Section 7.7 use a calibration introduced for that test and found no game without a pure equilibrium, so they do not test Theorem 7 under D2; the full rounding algorithm of Section 4.7 was not run on real data. The absorption tests rely on mixed equilibria with two-plan supports and on thresholds that are exact only up to the oracle gap. The Lean formalisation covers the discrete core statements listed in Section 4.8, not the full theorems with their bounds and complexity claims.

Dual implementation pricing. DIP's result is no better than its social target, which comes from column generation with rounding; its selection is optimistic, since other equilibria may coexist under the returned charges; its closed forms need D1 and one hours-per-delivery ratio; and its advantage in oracle calls is smaller against the reduced exact grids, while wall-clock time was not lower in every run. The charges needed under strong ambiguity aversion are unrealistically large subsidies.

Broad social cost. The background-delay proxy drives degenerate shut-down optima and is not a basis for conclusions. A measured value of the delay caused by delivery vehicles is needed.

Seeds and scope of the evidence. Three seeds per city give a rough indication of spread, not a confidence interval, and they vary only the sampled slice (depots, customer zones, sectors, observed days), not the calibration inputs, so they understate total uncertainty. Several readings of an earlier single-seed, price-and-branch analysis did not survive the second and third seeds or the exact oracle: the role of the congestion charge in Mumbai, a priced equilibrium within 2.4% of the optimum everywhere, a price of stability always within 0.05 of the price of anarchy, and a congestion-only gain below about 1.7% in every case. Some effects hinge on which equilibrium is selected, as the Buenos Aires seed-0 case shows. Leader optima are grid optima on 63 points. Price-of-anarchy values are empirical, from at most six equilibria per city and seed, and the social optimum is a rounded convexified solution with a relaxation bound up to 25% higher. ZMC's dimension-independence was not stressed at city scale. Novelty searches were web and index searches, partly at the level of abstracts; the closest neighbours (53, 63, 67 for the theory; 8 for BRL) should be compared in full text.

Future work. Open theoretical questions include the complexity of finding a pure E-admissible equilibrium above the guaranteed width (we conjecture PPAD-hardness), smoothness constants for the Cournot–congestion game, and an absorption result for nonaffine congestion. On the empirical side, carrier-level data on parcel volumes, stops and travel times, and measured kerbside delay, would turn the instances from a model of a slice into a calibrated planning tool.

10. Conclusion

We modelled competing delivery carriers as players of a Cournot–congestion game with routing-level strategies and credal beliefs. Under exposure-based congestion the game has a potential and pure imprecise equilibria at every width; under vehicle-count congestion their existence is strongly NP-complete below an explicit width, while above a width that does not grow with the number of carriers any mixed equilibrium rounds to an exact E-admissible equilibrium. An exact pricing algorithm for routes under probability boxes with unknown dependence, embedded in branch-and-price, makes the D1 game computable with an exact routing oracle, and dual implementation pricing computes the city's zone charges by implementing a target social optimum with one best response per carrier per round; on the city games it made its target an exact equilibrium in every run and reached at least 0.999 of the best grid point's gain at a small fraction of the grid's routing calls. On nine city instances with three random draws each, the empirical price of anarchy is close to one where service is complete and larger, with a wide spread across draws, where it is partial; a per-delivery subsidy captures most of what a city leader can gain in most city-seed cases with a material gain; a congestion charge alone captures little in all but one case, the exception arising from the selection of a worse unpriced equilibrium; and imprecision acts like a tax, although the response of the equilibrium reached is not always monotone in the width. These results were computed with an exact integer routing step; the earlier price-and-branch step had left some equilibria inexact by a few USD per day. The zone-market search did not outperform best response on these instances. The exact pricing algorithm outperformed scenario and quantile labelling on random pricing instances, and with branch-and-price it gives proven optimal robust routing plans on all nine metro instances; under the calibrated law those plans are not always better than a sample-average plan, which reflects the conservatism of the robust criterion. On the city equilibria the absorption bounds held in every test, with a slack of about a factor of two, and vehicle-count congestion did not prevent pure equilibria.

Data and code availability

Code, processed data, parameter sources, result files, figures and the scripts that generate the tables of this paper are available at `https://github.com/vinhqdang/Competing-Carriers-under-Uncertainty`. Raw OpenStreetMap extracts are distributed under the Open Database License; GHS-POP is distributed by the European Commission Joint Research Centre.

Declaration of competing interest

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Appendix A. Proofs of the results of Section 4

The proofs in this appendix are taken from the theory notes in the repository (file `theory/main.tex`), with the notation of Section 3.

Appendix A.1. Classical tools

Theorem 36 (Shapley–Folkman, constructive vertex form; 54)**.** *Let $S_1, \dots, S_N \subset \mathbb{R}^d$ be finite and nonempty, $L = \operatorname{span} \bigcup_i (S_i - S_i)$ and $d_L = \dim L$. Index the elements of S_i by $\mathcal{P}_i$ through an onto map a_i. For $\bar{x} \in \sum_i \operatorname{conv} S_i$, the polytope $\Lambda(\bar{x}) = \{\lambda \in \prod_i \Delta(\mathcal{P}_i) : \sum_i \sum_p \lambda_i(p) a_i(p) = \bar{x}\}$ is nonempty, and at every vertex λ the set $J(\lambda) = \{i : |\operatorname{supp} \lambda_i| \geq 2\}$ has $|J(\lambda)| \leq \min\{N, d_L\}$.*

Proof. Nonemptiness: $\bar{x} = \sum_i y_i$ with $y_i \in \operatorname{conv} S_i$, and each y_i is a convex combination of points $a_i(p)$. $\Lambda(\bar{x})$ is a bounded polyhedron in standard form $\{\lambda \geq 0 : H\lambda = h\}$ with columns $c_{i,p} = (a_i(p), e_i) \in \mathbb{R}^d \times \mathbb{R}^N$ and $h = (\bar{x}, \mathbf{1})$. A feasible λ is a vertex if and only if the columns $\{c_{i,p} : \lambda_i(p) > 0\}$ are linearly independent. Fix a vertex and pick $p_i^0 \in \operatorname{supp} \lambda_i$ for each i. The family $\{c_{i,p_i^0}\}_i \cup \{c_{i,p} - c_{i,p_i^0} : p \in \operatorname{supp} \lambda_i \setminus \{p_i^0\}\}$ is obtained from the positive columns by an invertible (unitriangular) transformation, so it is linearly independent. Its second part consists of the vectors $(a_i(p) - a_i(p_i^0), 0) \in L \times \{0\}$, which number $\sum_i (|\operatorname{supp} \lambda_i| - 1) \geq |J(\lambda)|$. Linearly independent vectors in $L \times \{0\}$ number at most d_L. Hence $|J| \leq d_L$, and trivially $|J| \leq N$. □

Appendix A.2. Section 4.1

Proof of Lemma 3. (a) $u_i(p; m, \xi)$ is affine in (m, ξ) with fixed coefficients, and P has finite first moments; linearity of the integral gives the identity. (b) Into: by definition. Onto: for $k = (m, \xi) \in K_i^\delta(\sigma)$, the Dirac law at (m, ξ) is supported in $\mathbb{R}^d_{\geq 0} \times \mathbb{R}^Z_{\geq 0}$ because $m \geq 0$ and $\xi \in \Xi \subset \mathbb{R}^Z_{\geq 0}$; its means are (m, ξ), so it belongs to $\mathcal{M}_i^\delta(\sigma)$. (c) Each criterion depends on P only through the numbers $\mathbb{E}_P[u(p)]$ and $\mathbb{E}_P[u(p') - u(p)]$, which are affine functions of the mean by (a). By (b), ranging over P is the same as ranging over $k \in K_i^\delta(\sigma)$, a product of compact intervals; continuous functions attain their extrema on it. □

Proof of Lemma 4. For a finite $\mathcal{P}$ and the compact convex box K with continuous affine payoffs: $E \subseteq M$. If p is optimal at k_0, then for every p', $\bar{u}(p'; k_0) - \bar{u}(p; k_0) \leq 0$, so $\min_k[\bar{u}(p') - \bar{u}(p)] \leq 0$. $M \subseteq ID$. If p is maximal, then for each p' there is k with $\bar{u}(p; k) \geq \bar{u}(p'; k)$; hence $\max \bar{u}(p; \cdot) \geq \bar{u}(p; k) \geq \bar{u}(p'; k) \geq \min \bar{u}(p'; \cdot)$. $\Gamma \subseteq M$. If $\min_k[\bar{u}(p'; k) - \bar{u}(p; k)] > 0$, then $\min \bar{u}(p') > \min \bar{u}(p)$, so p is not Γ-maximin. These inclusions hold carrier by carrier and give the inclusions between equilibrium sets. *Monotonicity.* $K^\delta(c) \subseteq K^{\delta'}(c)$ for every c, and the realised centre $A_{-i}(\sigma)$ does not depend on δ. E: a witness belief stays admissible. M: a dominance strict on the larger set is strict on the smaller one, so non-domination on the smaller set implies it on the larger one. ID: $\overline{\mathbb{E}}$ increases and every $\underline{\mathbb{E}}$ decreases as the set grows. □

Example 37 (Γ-equilibria are not nested)**.** Take one carrier with a fixed external aggregate $c = (1, 2)$, $d = 2$, $M = I$, $w = (2, 1)$, no traffic term, and two plans with $\pi = 0$, $\varphi(A) = (1, 0)$, $\varphi(B) = (0, 1)$. The lower values are $-(1 + 2\delta)$ for A and $-(2 + \delta)$ for B, so $\mathrm{ICNE}_\Gamma(0) = \{A\}$ and $\mathrm{ICNE}_\Gamma(2) = \{B\}$.

Proof of Lemma 5. A minimum of affine functions of the mixture x is concave, and

$$\min_k \textstyle\sum_p x(p)\bar{u}_i(p; k) \geq \sum_p x(p) \min_k \bar{u}_i(p; k)$$

because the left side picks one k for all p. With $\varphi, t \geq 0$ and $M \geq 0$, $\bar{u}_i(p; m, \xi)$ is coordinatewise nonincreasing in (m, ξ). The box $K^\delta(c)$ has the coordinatewise maximum $c + \delta w$, which lies in it (clipping only removes points below 0), and Ξ has the maximum $\overline{\mu}$. So $k^+ = (c + \delta w, \overline{\mu})$ minimises every $\bar{u}_i(p; \cdot)$ and the lower expectation is linear in the mixture. Substituting k^+ with $c = A_{-i}(\sigma)$ gives

$$\bar{u}_i(p; k^+) = \pi_i^\Gamma(p) - \langle \varphi_i(p), M A_{-i}(\sigma) \rangle,$$

the payoff of G_δ^Γ; Γ-maximin is then best response in G_δ^Γ. □

Appendix A.3. Sections 4.2 and 4.3

Proposition 38 (Relaxed equilibria are mixed equilibria)**.** *Under Assumption A2, x is a mixed Nash equilibrium of the finite game with payoffs $\bar{u}_i(\sigma_i; A_{-i}(\sigma), \hat{\xi})$ if and only if $\operatorname{supp} x_i \subseteq \arg\max_p \bar{u}_i(p; \bar{A}_{-i}(x), \hat{\xi})$ for all i. Such x exist.*

Proof. Under the product law x_{-i}, the expected payoff of a pure plan p is $\mathbb{E}[\bar{u}_i(p; A_{-i}, \hat{\xi})] = \bar{u}_i(p; \mathbb{E}A_{-i}, \hat{\xi})$ because $\bar{u}_i$ is affine in A_{-i}, and $\mathbb{E}A_{-i} = \bar{A}_{-i}(x)$ by linearity. A mixed profile is a Nash equilibrium if and only if each x_i is supported on pure best responses to x_{-i}. Existence follows from Nash [75]. □

Proof of Proposition 6. For a unilateral change $p \to p'$ of carrier i, put $a = a_i(p)$, $a' = a_i(p')$ and $R = A_{-i}$. Then $\Psi(p')-\Psi(p) = \pi_i(p')-\pi_i(p)+\frac{1}{2}\|a'\|_M^2-\frac{1}{2}\|a\|_M^2-\frac{1}{2}\|R+a'\|_M^2+\frac{1}{2}\|R+a\|_M^2-\gamma\langle t_i(p')-t_i(p),\hat{\xi}\rangle$, which equals $\pi_i(p')-\pi_i(p)-\langle a'-a, MR\rangle-\gamma\langle t_i(p')-t_i(p),\hat{\xi}\rangle = u_i(p')-u_i(p)$, using $\varphi_i = a_i$. A maximiser σ is a pure equilibrium [46]. The point $(A_{-i}(\sigma),\hat{\xi})$ lies in $K_i^\delta(\sigma)$ for every δ, so $\sigma \in \mathrm{ICNE}_E(\delta)$. For Γ, Lemma 5 shows that $\mathrm{ICNE}_\Gamma(\delta)$ is the equilibrium set of G_δ^Γ, which is again D1 with modified private profits and therefore has a potential and a pure equilibrium. □

Proof of Theorem 7. (a) Fix i and $\delta \geq \hat{\delta}(p)$. The anchor $\bar{A}^*_{-i}$ is a sum of convex combinations of nonnegative vectors, hence nonnegative. By definition of $\hat{\delta}(p)$, $|\bar{A}^*_{-i,k}-A_{-i,k}(p)| \leq \hat{\delta}(p)w_k \leq \delta w_k$ for all k, so $(\bar{A}^*_{-i},\hat{\xi}) \in K_i^\delta(p)$. By Lemma 3(b) the Dirac law at this point belongs to $\mathcal{M}_i^\delta(p)$; under it the expected payoff is $\bar{u}_i(\cdot;\bar{A}^*_{-i},\hat{\xi})$, which $p_i \in B_i$ maximises. Hence p_i is E-admissible for every i, and the inclusions follow from Lemma 4.

(b), first term. For any $p \in \prod_j B_j$, $A_{-i}(p)-\bar{A}^*_{-i} = \sum_{j\neq i}(a_j(p_j)-\bar{a}_j)$ and $|a_{jk}(p_j)-\bar{a}_{jk}| \leq r_{jk}$.

(b), second term. $\bar{A}^* \in \sum_j \operatorname{conv} a_j(B_j)$ because $\operatorname{supp} x_j^* \subseteq B_j$. Apply Theorem 36 with $\mathcal{P}_j$ replaced by B_j: a vertex λ of $\Lambda(\bar{A}^*)$ exists and can be found by linear programming. Let $J = J(\lambda)$, $|J| \leq k^\sharp$, and $y_j = \sum_p \lambda_j(p)a_j(p)$. For $j \notin J$, λ_j is a point mass at some $p_j \in B_j$ and $a_j(p_j) = y_j$; for $j \in J$ choose any $p_j \in \operatorname{supp}\lambda_j \subseteq B_j$. Since $\sum_j y_j = \bar{A}^* = \sum_j \bar{a}_j$,

$$A_{-i}(p)-\bar{A}^*_{-i} = \textstyle\sum_{j\in J}(a_j(p_j)-y_j)-(a_i(p_i)-\bar{a}_i). \tag{A.1}$$

For $j \in J$ both $a_{jk}(p_j)$ and y_{jk} lie in $[\min_{B_j} a_{jk}, \max_{B_j} a_{jk}]$, so the sum is at most $k^\sharp\bar{\sigma}w_k$ in coordinate k, and the last term is at most $\bar{\rho}w_k$.

(b), third term. If $J = \emptyset$ or $\bar{\sigma} = 0$, the sum vanishes. Otherwise draw $p_j \sim \lambda_j$ independently for $j \in J$. For each k the variables $(a_{jk}(p_j)-y_{jk})/w_k$, $j \in J$, are independent, have mean zero and range in an interval of length at most $\bar{\sigma}$. By Hoeffding's inequality and a union bound over k, $\Pr(\|E\|_{w,\infty} \geq t) \leq 2d\exp(-2t^2/(k^\sharp\bar{\sigma}^2))$, where E is the sum in (A.1). This is below one for $t > t_0 = \bar{\sigma}\sqrt{k^\sharp \ln(2d)/2}$; outcomes are finitely many, so some outcome has $\|E\|_{w,\infty} \leq t_0$, and $\hat{\delta}(p) \leq t_0 + \bar{\rho}$ for it.

(c) Relaxed equilibria exist (Proposition 38). Use $r_{jk} \leq s_{jk}$, $\bar{\rho} \leq \bar{\sigma} \leq \sigma_{\max}$, $k^\sharp \leq d$ and $\sum_{j\neq i} r_{jk}/w_k \leq (N-1)\sigma_{\max}$; the order of thresholds follows from Lemma 4. *(d)* This was shown in (b); linear programming finds a vertex in time polynomial in $\sum_j |B_j|$ and d. □

Proof of Corollary 8. Let $m^* = \bar{A}^*_{-i}$ and $m = A_{-i}(p)$. For any p', $\bar{u}_i(p';m,\mu)-\bar{u}_i(p_i;m,\mu) = [\bar{u}_i(p';m^*,\hat{\xi})-\bar{u}_i(p_i;m^*,\hat{\xi})]-\langle\varphi_i(p')-\varphi_i(p_i), M(m-m^*)\rangle-\gamma\langle t_i(p')-t_i(p_i),\mu-\hat{\xi}\rangle$. The bracket is nonpositive because $p_i \in B_i$; each term $|M_k(\varphi_{ik}(p')-\varphi_{ik}(p_i))(m_k-m_k^*)|$ is at most $w_kM_k|\Delta\varphi_{ik}|\hat{\delta}(p)$, and $|\mu_z-\hat{\xi}_z| \leq \kappa_z$. □

Appendix A.4. Sections 4.4 and 4.5

Proof of Proposition 11. For carrier 1, $\bar{u}(A_1;n)-\bar{u}(B_1;n) = 6-3n$, so A_1 is optimal iff $n \leq 2$; for carrier 2, $\bar{u}(Y;n)-\bar{u}(X;n) = 6-3n$, so Y is optimal iff $n \leq 2$. With two plans and a convex belief set, E-admissibility and maximality coincide: some n in the box makes the plan weakly better. In (A_1,X) carrier 2 sees $n = 1$ and needs a belief ≥ 2; in (A_1,Y) carrier 1 sees $n = 3$ and needs a belief ≤ 2; in (B_1,Y) carrier 2 sees $n = 3$ and needs ≤ 2; in (B_1,X)

carrier 1 sees $n = 1$ and needs ≥ 2. Each requirement holds iff $\delta w \geq 1$, and the other carrier best-responds at the realised count. Hence $\mathrm{ICNE}_E(\delta) = \mathrm{ICNE}_M(\delta) = \emptyset$ for $\delta < 1/w$ and both equal the full profile set at $\delta = 1/w$; there is no pure Nash equilibrium. The unique mixed equilibrium makes each carrier indifferent, so the rival's mean count is 2, both mix $\frac{1}{2}$–$\frac{1}{2}$, $r_j = 1$, $\bar{\rho} = 1/w$ and $(N-1)\bar{\rho} = 1/w$. □

Proof of Proposition 12. Take two carriers with plan sets $\{A_1, B_1\}^Z$ and $\{X, Y\}^Z$, the one-zone data of Proposition 11 in each zone and no cross-zone interaction, and $N - 2$ carriers with a single null plan. Payoffs are sums over zones and plan sets are products, so a carrier's regret is the sum of its zone regrets. In each zone and each of the four zone profiles, exactly one of the two carriers has regret 3 and the other 0 (e.g. in (A_1, X) carrier 2 gains $2-(-1) = 3$ by switching to Y). So $\varepsilon_1 + \varepsilon_2 = 3Z$ with both multiples of 3, and $\max \geq 3\lceil Z/2 \rceil$. For the upper bound, the zone-wise mixed equilibrium has $\bar{\rho} = 1$, every $p \in \prod B_j$ has $\hat{\delta}(p) \leq 1$, and $L_i(p) \leq \sum_z 1 \cdot 1 \cdot |\Delta\tau_z| = 3Z$. □

Proof of Theorem 13. Membership. The witness is a profile σ. With K the clipped box and Ξ a point: E-admissibility of σ_i is feasibility of the linear system $\{m \in K : \bar{u}_i(\sigma_i; m) \geq \bar{u}_i(p'; m)\ \forall p'\}$; maximality requires $\max_{m \in K}[\bar{u}_i(\sigma_i; m) - \bar{u}_i(p'; m)] \geq 0$ for each p', a linear function maximised over a box coordinatewise; ID and Γ require maxima and minima of $\bar{u}_i(p; \cdot)$ over the box, also coordinatewise.

Construction. Let $V(D) = [n]$ and $\Delta^- = \max(1, \max \text{ in-degree})$. Create one zone per arc $e = (u, v)$, using only its vehicle coordinate, with $M_e = \gamma c_e = 1$, $w_e = 1$ and no deliveries. Set $\eta = 1/(2\Delta^- + 2)$. Carrier i has the plan *out* with all data 0, and the plan *in* with $\pi_i = 1$ and, for each out-arc $e = (i, j)$, vehicles $n_{ie} = 1$ and exposure $\tau_{ie} = 1$; for each in-arc $e = (h, i)$, vehicles $n_{ie} = 2$ and exposure $\tau_{ie} = \eta$; zero elsewhere. This is D2, since $\varphi = \tau \neq a = n$.

Payoff gap. The gain of *in* at belief m is $D_i(m) = 1 - \sum_{e=(i,j)} m_e - \eta \sum_{e=(h,i)} m_e$. At the realised aggregate of σ^S, $m_e = 2 \cdot \mathbf{1}[j \in S]$ for $e = (i, j)$ and $m_e = \mathbf{1}[h \in S]$ for $e = (h, i)$, so $D_i^{\mathrm{real}} = 1 - 2o_i - \eta h_i$ with $o_i = |N^+(i) \cap S|$ and $h_i = |N^-(i) \cap S| \leq \Delta^-$. If $o_i = 0$, $D_i^{\mathrm{real}} \geq 1 - \Delta^-/(2\Delta^- + 2) > \frac{1}{2}$; if $o_i \geq 1$, $D_i^{\mathrm{real}} \leq -1$.

Robustness of the sign. Over K^δ, $|D_i(m) - D_i^{\mathrm{real}}| \leq \delta(\deg^+(i) + \eta \deg^-(i)) < \delta(\Delta^+ + \frac{1}{2})$, which is below $\frac{1}{2}$ for $\delta < 1/(2\Delta^+ + 1)$; clipping only shrinks the box. So D_i has the sign of D_i^{real} on the whole box.

All criteria select the realised strict best response. If $D_i > 0$ on the box, *in* is the unique optimum at every belief, hence the only E-admissible plan; *out* is strictly dominated everywhere, so *in* is the only maximal plan; and $\min_K \bar{u}(\text{in}) > 0 = \max_K \bar{u}(\text{out})$, so *out* is interval-dominated and *in* is the unique Γ-maximin plan. The case $D_i < 0$ is symmetric. So σ is admissible under every criterion iff each carrier plays *in* exactly when $o_i = 0$.

Equilibria are kernels. With $S = \{i : \sigma_i = \text{in}\}$, the condition reads $i \in S \Rightarrow N^+(i) \cap S = \emptyset$ (independence) and $i \notin S \Rightarrow N^+(i) \cap S \neq \emptyset$ (absorption). NP-hardness follows from Chvátal [77]; all numbers are polynomially bounded, so hardness is strong; with $\Delta^+ \leq 2$ [78] the threshold is 1/5. □

Proof of Corollary 14. On the instances of Theorem 13, $\delta_0^E = 0$ if D has a kernel; otherwise $\mathrm{ICNE}_E(\delta) = \emptyset$ for all $\delta < 1/(2\Delta^+ + 1)$, so $\delta_0^E \geq 1/(2\Delta^+ + 1)$. A multiplicative approximation outputs 0 in the first case and a positive number in the second; an additive one separates the cases when its error is below $1/(2\Delta^+ + 1)$. □

Proposition 39 (The relaxation step is PPAD-hard under D2)**.** *Every bimatrix game $(R, C) \in \mathbb{Q}^{m\times n} \times \mathbb{Q}^{m\times n}$ is, in polynomial time, the precise game of a D2 instance with $N = 2$ and $d = m + n$. Hence computing a mixed equilibrium under D2 is PPAD-hard.*

Proof. Let $M = I_{m+n}$, no traffic term, $\bar{R} = \max R$ and $\bar{C} = \max C$. Carrier 1 has plans $p \in [m]$ with $a_1(p) = e_p$, $\varphi_1(p) = \sum_q (\bar{R} - R_{pq}) e_{m+q} \geq 0$ and $\pi_1(p) = \bar{R}$; carrier 2 has plans $q \in [n]$ with $a_2(q) = e_{m+q}$, $\varphi_2(q) = \sum_p (\bar{C} - C_{pq}) e_p \geq 0$ and $\pi_2(q) = \bar{C}$. Then $u_1(p; a_2(q)) = R_{pq}$ and $u_2 = C_{pq}$. By Proposition 38, relaxed equilibria are exactly the Nash equilibria of (R, C), and computing these is PPAD-complete [79, 80]. □

Appendix A.5. Section 4.6

Proof of Lemma 15. Write $k^t = (A_{-i}(\sigma), \mu) \in K_i^\delta(\sigma)$ and $\Delta(p', k) = \bar{u}_i(p'; k) - \bar{u}_i(\sigma_i; k)$, affine in k. For any k in the (unclipped) box, $|\Delta(p', k) - \Delta(p', k^t)| \leq \delta \sum_k w_k M_k |\Delta\varphi_{ik}| + \gamma \sum_z \kappa_z |\Delta t_{iz}|$. *E:* take the witness k, at which $\Delta(p', k) \leq 0$. *M:* for each p', maximality and compactness give $k_{p'}$ with $\Delta(p', k_{p'}) \leq 0$. *ID:* $\bar{u}_i(p'; k^t) - \bar{u}_i(\sigma_i; k^t) \leq [\bar{u}_i(p'; k^t) - \min_K \bar{u}_i(p')] + [\max_K \bar{u}_i(\sigma_i) - \bar{u}_i(\sigma_i; k^t)]$, since $\max_K \bar{u}_i(\sigma_i) \geq \min_K \bar{u}_i(p')$, and each bracket is at most $\delta H_i + \gamma \Theta_i$. □

Proof of Proposition 16. The minimum over a growing finite set does not increase and the maximum does not decrease (Lemma 4). Each profile enters $\mathrm{ICNE}_C(\delta)$ at a single threshold and never leaves, so the sets change at most at $\prod_i |\mathcal{P}_i|$ values. For Γ see Example 37. □

Proposition 40 (Price of anarchy under smoothness; conditional)**.** *Assume $W(\sigma) \geq \sum_i U_i(\sigma)$ for all σ, $W^* > 0$, and that the precise game is (λ, μ)-smooth: $\sum_i U_i(\sigma_i^*, \sigma_{-i}) \geq \lambda W(\sigma^*) - \mu W(\sigma)$ for all σ and an optimal σ^*. For $C \in \{E, M, ID\}$ let $\mathcal{E}(\delta) = \max_{\sigma \in \mathrm{ICNE}_C(\delta)} \sum_i \varepsilon_i(\sigma)$ from Lemma 15. If $\lambda W^* > \mathcal{E}(\delta)$, then $\mathrm{PoA}_C(\delta) \leq (1 + \mu)/(\lambda - \mathcal{E}(\delta)/W^*)$.*

Proof. For $\sigma \in \mathrm{ICNE}_C(\delta)$: $W(\sigma) \geq \sum_i U_i(\sigma) \geq \sum_i [U_i(\sigma_i^*, \sigma_{-i}) - \varepsilon_i] \geq \lambda W^* - \mu W(\sigma) - \mathcal{E}(\delta)$. Rearrange. □

Proof of Proposition 17. With π_i as in Section 3 and $x = \tau$, direct expansion gives $W - \Psi_{\hat{\mu}} = \sum_z [\frac{b_z}{2} \sum_i q_{iz}^2 - \frac{\gamma c_z}{2}(T_z^2 - \sum_i \tau_{iz}^2)]$; the α, $\hat{\mu}$ and C_i terms cancel. By Cauchy–Schwarz, $\sum_i \tau_{iz}^2 \geq T_z^2/N$, and $\sum q^2 \geq 0$, so $W(\sigma) \geq \Psi_{\hat{\mu}}(\sigma) - \sum_z \frac{\gamma c_z}{2}(1 - \frac{1}{N}) T_z(\sigma)^2$. Next, $\Psi_{\hat{\mu}}(\sigma) \geq \Psi_{\hat{\mu}}(\sigma^*) - \eta$, and $\Psi_{\hat{\mu}}(\sigma^*) \geq W^* - \sum_z \frac{b_z}{2} \sum_i q_{iz}^{*2}$ because $T^2 \geq \sum \tau^2$ for $\tau \geq 0$. Membership of a potential maximiser in every $\mathrm{ICNE}_E(\delta)$ is Proposition 6, extended to M and ID by Lemma 4. □

Appendix A.6. Section 4.7

Theorem 41 (Exact thresholds and regret brackets)**.** *Let p be a pure profile. (a) The exact E-threshold of carrier i is the optimal value of the linear program $A_i^E(p) = \min\{\delta \geq 0 : m \geq 0,\ \xi \in \Xi,\ |m_k - A_{-i,k}(p)| \leq \delta w_k\ \forall k,\ \bar{u}_i(p_i; m, \xi) \geq \bar{u}_i(p'; m, \xi)\ \forall p' \in \mathcal{P}_i\}$ ($+\infty$ if infeasible), and $p \in \mathrm{ICNE}_E(\delta)$ iff $\delta \geq \max_i A_i^E(p)$. With an exact oracle and implicit $\mathcal{P}_i$, Kelley's cutting-plane method on this program terminates finitely. (b) For anchors $m^i \geq 0$, $\xi^i \in \Xi$ with $p_i \in \arg\max \bar{u}_i(\cdot; m^i, \xi^i)$, $A_i^E(p) \leq \|A_{-i}(p) - m^i\|_{w,\infty}$. (c) For a heuristic plan h_i and a valid bound $\mathcal{U}_i \geq \max_{p'} \bar{u}_i(p'; m, \xi)$, the regret of p_i lies in $[\bar{u}_i(h_i) - \bar{u}_i(p_i),\ \mathcal{U}_i - \bar{u}_i(p_i)]$.*

Proof. (a) By Lemma 3, p_i is E-admissible at width δ iff the system with that δ is feasible; feasibility is upward closed in δ (Lemma 4) and the infimum of an LP is attained when finite. Kelley's method adds the oracle's best plan at the current (m, ξ) as a violated constraint; a plan already present cannot be violated, so cuts are distinct and finitely many. (b) The anchor is feasible with $\delta = \|A_{-i}(p) - m^i\|_{w,\infty}$. (c) Immediate from the definitions. □

Theorem 42 (Rounding guarantee under D2)**.** *Under D2, if the relaxation phase returns a mixed equilibrium x, the sparsify phase (a feasibility LP over $\Lambda(\bar{A}(x))$ with plan sets $\operatorname{supp} x_j$) and the round phase return $p \in \prod_j B_j(x)$ with $p \in \mathrm{ICNE}_E(\delta)$ for all $\delta \geq \hat{\delta}(p)$; $\hat{\delta}(p)$ satisfies the second bound of Theorem 7(b) for any choice in the round phase and the third for a successful sample. Sparsify and round run in time polynomial in $\sum_j |\operatorname{supp} x_j|$ and d.*

Proof. The sparsify LP's vertices have $|J| \leq k^\sharp$ by Theorem 36. The rest is Theorem 7(a), (b) and (d). □

The double-oracle relaxation [81] stops, with exact restricted solves and exact oracles, after at most $\sum_i |\mathcal{P}_i|$ rounds at a relaxed equilibrium: at termination every support plan attains the restricted maximum and no plan exceeds it, and each non-terminal round adds a new plan.

Appendix B. Proofs of the results of Section 5

Appendix B.1. BRL

Proof of Theorem 19. Write $S = \sum_{a \in r} X_a$ (arcs counted with multiplicity). Fix j and $\Theta = \Theta_j$.

Step 1 (pointwise inequality). For reals x_a, d_a with $\sum_a d_a \leq \Theta$, $\left(\sum_a x_a - \Theta\right)^+ \leq \left(\sum_a (x_a - d_a)\right)^+ \leq \sum_a (x_a - d_a)^+$, because $t \mapsto t^+$ is nondecreasing and subadditive.

Step 2 (upper bound). Let $P \in \mathcal{M}$ with marginals F_a. Taking expectations, $\mathbb{E}_P (S - \Theta)^+ \leq \sum_a \mathbb{E}_{F_a} (X_a - d_a)^+$. Since $F_a \geq \underline{F}_a$ pointwise, F_a is first-order dominated by $\underline{F}_a$, and $x \mapsto (x - d)^+$ is nondecreasing, so $\mathbb{E}_{F_a} (X_a - d_a)^+ \leq \overline{\Pi}_a(d_a)$; likewise $\mathbb{E}_{F_a} X_a \leq \bar{x}_a$. Hence $\mathbb{E}_P \phi(S) \leq c_0 + \gamma \sum_a \bar{x}_a + \sum_j \rho_j \sum_a \overline{\Pi}_a(d_a^j)$ for every P and admissible d^j.

Step 3 (attainment). Let U be uniform on $(0, 1)$, $\bar{X}_a = \overline{Q}_a(U)$ and $\bar{S} = \Sigma(U)$ with $\Sigma(u) = \sum_a \overline{Q}_a(u)$; then $\bar{X}_a \sim \underline{F}_a$ and the law of $(\bar{X}_a)$ belongs to $\mathcal{M}$. Let $u^* = \sup\{u \in (0, 1] : \Sigma(u) \leq \Theta\}$ ($\sup \emptyset = 0$). If $u^* = 1$, put $d_a = \overline{Q}_a(1)$; then $\sum_a d_a \leq \Theta$ and both sides of Step 1 vanish almost surely. If $u^* < 1$, put $\ell_a = \overline{Q}_a(u^*)$ (or $\overline{Q}_a(0^+)$ if $u^* = 0$) and $h_a = \overline{Q}_a(u^{*+})$; then $\sum_a \ell_a \leq \Theta$ when $u^* > 0$ and $\sum_a h_a \geq \Theta$. If $u^* = 0$ choose $d_a \leq \ell_a$ with $\sum_a d_a = \Theta$; otherwise choose $d_a \in [\ell_a, h_a]$ with $\sum_a d_a = \Theta$. On $\{U \leq u^*\}$ all $\bar{X}_a \leq d_a$ and both sides of Step 1 are 0; on $\{U > u^*\}$ all $\bar{X}_a \geq d_a$ and $\sum_a \left(\bar{X}_a - d_a\right)^+ = \bar{S} - \Theta = \left(\bar{S} - \Theta\right)^+$. So Step 1 holds with equality almost surely, and the comonotone law attains the bound of Step 2; the minima are attained.

Step 4. $\mathbb{E}\phi(\bar{S}) = \int_0^1 \phi(\Sigma(u)) du$ because $\bar{S} = \Sigma(U)$.

Step 5 (integrality). With integer quantiles and Θ, the numbers ℓ_a, h_a, $\overline{Q}_a(1)$ are integers and integer vectors with the required properties exist in each case of Step 3. □

Proof of Theorem 20. Substitute the budget form of Theorem 19 into the objective and merge the two minimisations. □

Lemma 43 (Concatenation is inf-convolution)**.** *For a label p and a completion q, $\overline{\mathbb{E}}[\phi(T_{p\oplus q})] - c_0 = f_{p\oplus q}(T_{\max}) = \min_D\{f_p(D) + f_q(T_{\max} - D)\}$.*

Proof. The stop-loss terms are separable over arcs. Split the budget vector into the arcs of p and of q, put $D = \sum_{a\in p} d_a$, minimise first over the arcs of p with budget D and then over those of q with budget $T_{\max} - D$, and apply Theorem 19 to each part. The mean terms add. □

Lemma 44 (Completion window)**.** *Let $C_{\min}(v) \le S_{q,1}$ for every completion q of labels ending at v. For every completion q, the minimum in Lemma 43 is attained at some $D^* \le \overline{D}(p)$; if $D^* < \underline{D}(p)$ then $D^* \le S_{p,1}$.*

Proof. Let $\Phi_q(D) = f_p(D) + f_q(T_{\max} - D)$. The map $D \mapsto f_q(T_{\max} - D)$ is convex and nondecreasing with slope in $[0, \rho]$, equal to ρ when $T_{\max} - D < S_{q,1}$; the slope of f_p lies in $[-\rho, 0]$ and is 0 for $D > S_{p,L}$. (i) On $D > T_{\max} - S_{q,1}$, Φ_q has slope ≥ 0, so a minimiser exists with $D \le T_{\max} - S_{q,1} \le T_{\max} - C_{\min}(v)$. (ii) On $D > S_{p,L}$, Φ_q has slope ≥ 0, so decreasing D to $S_{p,L}$ does not increase it. Applying (i) and then (ii) gives $D^* \le \overline{D}(p)$. The last claim is immediate from $\underline{D}(p) \le S_{p,1}$. □

Proof of Theorem 22. Validity. Let q be a completion of p_2; since $\mathcal{V}(p_1) \subseteq \mathcal{V}(p_2)$ and $\ell(p_1) \le \ell(p_2)$, it is a completion of p_1. By Lemma 44 there is a minimiser $D^* \le \overline{D}(p_2)$ for $p_2 \oplus q$. Let g be the difference in Definition 21. If $D^* \in \mathcal{W}(p_2)$, $g(D^*) \le 0$. Otherwise $D^* < \underline{D}(p_2) \le S_{p_2,1}$; on $(-\infty, S_{p_2,1}]$, f_{p_2} has slope $-\rho$ and f_{p_1} slope $\ge -\rho$, so g is nondecreasing there and $g(D^*) \le g(\underline{D}(p_2)) \le 0$. In both cases, by Lemma 43, $f_{p_1\oplus q}(T_{\max}) - y(p_1) \le f_{p_1}(D^*) + f_q(T_{\max} - D^*) - y(p_1) \le f_{p_2\oplus q}(T_{\max}) - y(p_2)$.

Exactness. Call a label processed if it is popped while not deleted. Let OPT be the optimum and Ω the set of pairs (o, q) with o a created label, q a completion of o and reduced cost of $o \oplus q$ equal to OPT. The pair (depot label, optimal route) is in Ω and the depot label is processed. Among pairs in Ω with o processed choose one with the fewest arcs $|q|$. If $|q| = 1$, closing o records a route of value OPT. If $|q| \ge 2$, let (v, j) be the first arc of q, q' the rest and $o_1 = o \oplus (v, j)$, which is created when o is processed; $(o_1, q') \in \Omega$. If o_1 is processed, $|q'| < |q|$ contradicts minimality. Otherwise o_1 was discarded or deleted because a live label o_2 functionally dominates it; by validity $(o_2, q') \in \Omega$, and if o_2 is not processed it was deleted later by some o_3, with $(o_3, q') \in \Omega$ by transitivity. The moments at which these labels cease to be live strictly increase and labels are finitely many, so the chain ends at a processed label with $(o_k, q') \in \Omega$ and $|q'| < |q|$, a contradiction. □

Proposition 45 (Comparison with vector dominance)**.** *Assume $\mathcal{V}(p_1) \subseteq \mathcal{V}(p_2)$, $\ell(p_1) \le \ell(p_2)$ and $y(p_1) \ge y(p_2)$. (a) If $S_{p_1} \le S_{p_2}$ componentwise, p_1 functionally dominates p_2. (b) In the sector model $X_a = \sum_k \tau_{a,k}\xi_k$ with probability boxes on the multipliers, $\tau_{p_1} \le \tau_{p_2}$ componentwise implies $S_{p_1} \le S_{p_2}$. (c) Neither converse holds.*

Proof. (a) f is nondecreasing in each S_l. (b) The upper quantiles of the multipliers are nonnegative. (c) With $L = 2$, $\omega = (\frac{1}{2}, \frac{1}{2})$, $\gamma = 0$, $\rho = 1$ and equal duals: $S_{p_1} = (1, 1)$ and $S_{p_2} = (0, 2.2)$ give $f_{p_1} \le f_{p_2}$ on $\mathbb{R}$ (values $(1, 1.1)$, $(0, 0.6)$, $(0, 0)$ at $D = 0, 1, 2.2$, constant difference -0.1 below 0) although $S_{p_1,1} > S_{p_2,1}$. □

Proposition 46 (A single risk level is not exact)**.** *For every u, $\min_{\text{plan}} \sum_r \overline{\mathbb{E}}[\phi(T_r)] \geq \max_u \min_{\text{plan}} \sum_r [c_0 + \gamma \bar{x}_r + \rho(\mathcal{H}_r(u) - (1-u)T_{\max})]$, where $\mathcal{H}_r(u) = \int_u^1 \sum_{a \in r} \overline{Q}_a(v)\, dv$; the inequality can be strict and the best fixed-u plan need not be optimal.*

Proof. The maximum over u inside each route is at least its value at a common u, and $\min \max \geq \max \min$. Strictness is shown by exact computations by enumeration and set partitioning (Section 7.6); structurally, the optimal level of a route is its worst-case on-time level, which differs between routes of the same plan. □

Appendix B.2. ZMC

Lemma 47 (Price duality)**.** *Let $\bar{\Phi}(x) = \sum_i \sum_p x_i(p) \tilde{c}_i(p) + \frac{1}{2}\|\bar{A}(x)\|_M^2$ on $\prod_i \Delta(\mathcal{P}_i)$, $\bar{\Phi}^* = \min \bar{\Phi}$ and $\mathcal{B} = \prod_k [0, M_k A_k^{\max}]$ with $A_k^{\max} = \sum_i \max_p a_{ik}(p)$. (a) $\mathcal{D}$ is strongly concave with a unique maximiser λ^*, and $\max \mathcal{D} = \bar{\Phi}^* \leq \min_\sigma \Phi(\sigma)$. (b) All minimisers of $\bar{\Phi}$ share the aggregate $\bar{A}^*$, and $\lambda^* = M\bar{A}^* \in \mathcal{B}$. (c) $M^{-1}\lambda^* \in R(\lambda^*)$, where $R(\lambda) = \partial^+ \sum_i V_i(\lambda)$.*

Proof. (a) $V = \sum_i V_i$ is concave and finite and $\frac{1}{2}\|\lambda\|_{M^{-1}}^2$ strongly convex, so $\mathcal{D}$ is strongly concave, continuous and coercive, with a unique maximiser. For $L(x, \lambda) = \sum_i \langle x_i, \tilde{c}_i \rangle + \langle \lambda, \bar{A}(x) \rangle - \frac{1}{2}\|\lambda\|_{M^{-1}}^2$, affine in x on a compact convex set and concave in λ, Sion's theorem gives $\min_x \sup_\lambda L = \sup_\lambda \min_x L$. For fixed x the supremum is at $\lambda = M\bar{A}(x)$ with value $\bar{\Phi}(x)$; for fixed λ, $\min_x L = \mathcal{D}(\lambda)$ because a linear function on a product of simplices is minimised at a vertex carrier by carrier. Pure profiles are vertices, where $\bar{\Phi} = \Phi$. (b) Two minimisers with different aggregates would contradict strict convexity of $\frac{1}{2}\|\cdot\|_M^2$. A minimiser x^* and λ^* form a saddle point, and the unique maximiser of $L(x^*, \cdot)$ is $M\bar{A}(x^*)$. Bounds on $\bar{A}^*$ follow from nonnegativity. (c) This is $0 \in \partial^+ \mathcal{D}(\lambda^*) = R(\lambda^*) - M^{-1}\lambda^*$, by the sum rule for concave functions [104]. □

Proof of Theorem 33. For any pure σ and λ, with $A = A(\sigma)$ and $e = M^{-1}\lambda - A$,

$$\begin{aligned} \sum_i [\tilde{c}_i(\sigma_i) + \langle \lambda, a_i(\sigma_i) \rangle] - \tfrac{1}{2}\|\lambda\|_{M^{-1}}^2 &= \sum_i \tilde{c}_i(\sigma_i) + \langle M(A+e), A \rangle - \tfrac{1}{2}\|A+e\|_M^2 \\ &= \Phi(\sigma) - \tfrac{1}{2}\|e\|_M^2. \end{aligned} \tag{$*$}$$

(a) If $\sigma_i \in O_i(\lambda)$ the left side is $\mathcal{D}(\lambda) \leq \bar{\Phi}^* \leq \min \Phi$. (b) Fix i, $a = a_i(\sigma_i)$, $a' = a_i(p)$, $\Delta = a' - a$ (zero outside $\mathcal{Z}_i$). Since $MA_{-i} = \lambda - M(a+e)$, the change in carrier i's cost $c_i + \langle a_i, MA_{-i} \rangle$ equals $[\tilde{c}_i(p) + \langle \lambda, a' \rangle] - [\tilde{c}_i(\sigma_i) + \langle \lambda, a \rangle] + \frac{1}{2}\|\Delta\|_M^2 - \langle \Delta, Me \rangle \geq -\frac{1}{2}\sum_{k \in \mathcal{Z}_i} M_k e_k^2$, minimising coordinatewise at $\Delta_k = e_k$. (c) For any p, $c_i(p) + \langle a', Mm_i \rangle = [\tilde{c}_i(p) + \langle \lambda, a' \rangle] + \frac{1}{2}\|a' - a\|_M^2 - \frac{1}{2}\|a\|_M^2$, minimised by σ_i. Coordinates outside $\mathcal{Z}_i$ shift all plans' costs equally, so the belief can be replaced by one equal to A_{-i} there; within $\mathcal{Z}_i$ it deviates from A_{-i} by $|e_k| \leq \delta w_k$, and under D1 the Dirac law at it lies in the credal set (Lemma 3). (d) If $e = 0$, $\Phi(\sigma) = \mathcal{D}(\lambda) \leq \bar{\Phi}^* \leq \min \Phi \leq \Phi(\sigma)$; a global minimiser of an exact potential is a pure equilibrium, and uniqueness gives $\lambda = \lambda^*$. (e) If $\Phi(\sigma^\circ) = \bar{\Phi}^*$, Lemma 47(b) gives $\lambda^* = MA(\sigma^\circ)$, so $e = 0$, and $(*)$ shows that $\sum_i \{\tilde{c}_i(\sigma_i^\circ) + \langle \lambda^*, a_i(\sigma_i^\circ) \rangle - V_i(\lambda^*)\} = 0$, a sum of nonnegative terms. □

Proof of Theorem 34. Let $\chi_k = \sup\{\sup_{r \in R(\lambda)} \inf_{r' \in R(\lambda')} |r_k - r'_k| : \lambda, \lambda' \in \mathcal{B},\ \lambda_k = \lambda'_k\}$. *Key inequality.* Let a query $\lambda \in \mathcal{B}$ have $\lambda_k = \mu$ and response $r \in R(\lambda)$. Suppose $s_k > 0$ but $\mu - \lambda_k^* > M_k \chi_k$. Let λ' equal λ^* except $\lambda'_k = \mu$; then $\lambda' \in \mathcal{B}$. With $r^* = M^{-1}\lambda^* \in R(\lambda^*)$,

antitonicity of R gives $r'_k \le \lambda^*_k/M_k$ for every $r' \in R(\lambda')$, and the definition of χ_k gives $r' \in R(\lambda')$ with $r_k \le r'_k + \chi_k + \eta$ for any $\eta > 0$. Hence $s_k = r_k - \mu/M_k \le -(\mu - \lambda^*_k - M_k\chi_k)/M_k + \eta < 0$ for small η, a contradiction. The case $s_k < 0$ is symmetric, and $s_k = 0$ combines both. *Induction.* Initially $\lambda^*_k \in [0, M_k A^{\max}_k]$. If $s_k > 0$ the new bracket is $[\mu, u_k]$; if $\lambda^*_k \ge \mu$ the distance is unchanged, and otherwise it is $\mu - \lambda^*_k \le M_k\chi_k$. The other cases are symmetric. *Rate.* Midpoint rounds halve every bracket and other rounds never enlarge it, so after T rounds with at least $\lceil T/2 \rceil$ midpoints the width is at most $M_k A^{\max}_k 2^{-\lceil T/2 \rceil}$. If plan sets are separable, V_i is separable and the projection of $R(\lambda)$ on coordinate k depends on λ_k only, so $\chi_k = 0$. □

Example 48 (The floor is genuine)**.** Let $d = 2$, $M = I$ and one carrier with plans 0 ($\tilde{c} = 0$, $a = (0, 0)$) and b ($\tilde{c} = -1.5$, $a = (1, 1)$). Then $\lambda^* = (0.75, 0.75)$, $\chi_1 = \chi_2 = 1$, and the queries $(0.7, 0.9)$, $(0.6, 1)$, $(0.5, 1)$ all return plan 0 (tie at the last resolved towards 0), so the bracket update sets $u_1 = 0.5$ and excludes λ^*_1 by 0.25, within the floor but positive.

Proof of Theorem 35. Each round makes one pricing call per carrier. The polish is sequential best-response dynamics in a finite exact-potential game; every accepted change strictly decreases Φ, which takes finitely many values, so it stops at a pure Nash equilibrium. The polish does not increase Φ, and the start has $\Phi \le \Phi(\sigma^{\text{price}}) = \mathcal{D}(\lambda^{t^\star}) + \frac{1}{2}\|e^{t^\star}\|^2_M \le \bar{\Phi}^* + \frac{1}{2}\min_t \|e^t\|^2_M$ by Theorem 33(a) and Lemma 47(a). An exact pure equilibrium makes each plan optimal at the centre belief, which lies in every credal box; the Γ case is Lemma 5. □

Proposition 49 (Lipschitz dependence on charges)**.** *With charges t on the contributions, let $\mu^*(t) = \lambda^*(t) + t$. Then $\|\mu^*(t) - \mu^*(t')\|_{M^{-1}} \le \|t - t'\|_{M^{-1}}$ and $\langle \bar{A}^*(t) - \bar{A}^*(t'), t - t' \rangle \le 0$.*

Proof. With charges, $V^t(\lambda) = V(\lambda + t)$, so the equilibrium condition reads $r = M^{-1}(\mu - t) \in R(\mu)$. For $r' = M^{-1}(\mu' - t') \in R(\mu')$, antitonicity gives $\langle r - r', \mu - \mu' \rangle \le 0$, i.e. $\|\mu - \mu'\|^2_{M^{-1}} \le \langle M^{-1}(t - t'), \mu - \mu' \rangle \le \|t - t'\|_{M^{-1}}\|\mu - \mu'\|_{M^{-1}}$. For the second claim, $\langle r - r', t - t' \rangle = \langle r - r', (\mu - \mu') - M(r - r') \rangle \le -\|r - r'\|^2_M \le 0$. □

Appendix C. Proofs for dual implementation pricing

Notation as in Section 5.3; $m^S_z = b_z + 2\zeta_z$ is the social slope, and $q^2_{iz} = d_{iz}q_{iz}$ is used throughout. The proofs are taken from the theory notes of the repository, file `theory/new_algorithm_C/theory.tex`.

Proof of Proposition 23. (a) $\tau_{iz} = rq_{iz}$. (b) $s + rt_z = u_z$ and $t_z \ge 0$ because $s \le u_z$; zones without lots carry no plan. (c) Substitution of $s_z \equiv s$ and $t_z = \vartheta g_z X^{\text{NE}}_z$. □

Proof of Lemma 24. With $q^2_{iz} = d_{iz}q_{iz}$, $X_z = rQ_z$ and $m^S_z - m_z = \zeta_z$,

$$\Phi_u - \text{SC} = \sum_z \left[\tfrac{m_z}{2} \textstyle\sum_i d_{iz}q_{iz} - \tfrac{\zeta_z}{2} Q^2_z + u_z Q_z \right].$$

Insert $u_z = \zeta_z Q^\circ_z - \frac{m_z}{2}\bar{d}_z + v_z$ and complete the square: $-\frac{\zeta_z}{2}Q^2_z + \zeta_z Q^\circ_z Q_z = -\frac{\zeta_z}{2}(Q_z - Q^\circ_z)^2 + \frac{\zeta_z}{2}(Q^\circ_z)^2$, and $\frac{m_z}{2}\sum_i d_{iz}q_{iz} - \frac{m_z}{2}\bar{d}_z Q_z = \sum_i \frac{m_z}{2}(d_{iz} - \bar{d}_z)q_{iz}$. □

Proposition 50 (Carrier-specific charges; classical)**.** *Let σ^* be a pure profile and $U_{iz} = \zeta_z Q^*_{-i,z} - \frac{b_z}{2} d_{iz}$. Under carrier-specific per-delivery charges U_i, for all i and p, p': $J_i^U(p; \sigma^*_{-i}) - J_i^U(p'; \sigma^*_{-i}) = \mathrm{SC}(p, \sigma^*_{-i}) - \mathrm{SC}(p', \sigma^*_{-i})$. Hence every profile that no single carrier can improve socially is a pure equilibrium.*

Proof. $\mathrm{SC}(p, \sigma^*_{-i}) = R_i(p) + \sum_z q_{iz}(-\alpha_z + m_z^S Q^*_{-i,z} + \frac{m_z^S}{2} d_{iz}) + \text{const}$ and $J_i^U(p) = R_i(p) + \sum_z q_{iz}(-\alpha_z + m_z d_{iz} + m_z Q^*_{-i,z} + U_{iz})$; the brackets coincide because $m_z + \zeta_z = m_z^S$ and $m_z - \frac{b_z}{2} = \frac{m_z^S}{2}$. □

Convexified game. With mixtures $x_i \in \Delta(\mathcal{P}_i)$, $\bar{q}_i = \mathbb{E}_{x_i} q_i$ and $\bar{Q} = \sum_i \bar{q}_i$, the convexified potential and social cost are $\overline{\Phi}_u(x) = \sum_i \mathbb{E}_{x_i}[R_i + \sum_z q_{iz}(-\alpha_z + u_z + \frac{m_z}{2} d_{iz})] + \sum_z \frac{m_z}{2} \bar{Q}_z^2$ and $\overline{\mathrm{SC}}(x) = \sum_i \mathbb{E}_{x_i}[R_i - \sum_z \alpha_z q_{iz}] + \sum_z \frac{m_z^S}{2} \bar{Q}_z^2$. Both are convex, agree with Φ_u and SC at pure profiles, and the identity of Lemma 24 holds for them with Q, q replaced by $\bar{Q}, \bar{q}$.

Proof of Theorem 25. $\overline{\mathrm{SC}}$ is linear in x plus $\sum_z \frac{m_z^S}{2} \bar{Q}_z^2$ with $\bar{Q}$ linear in x, so $\overline{\mathrm{SC}}(x) - \overline{\mathrm{SC}}(x^*) = \langle \nabla \overline{\mathrm{SC}}(x^*), x - x^* \rangle + \sum_z \frac{m_z^S}{2} (\bar{Q}_z - \bar{Q}_z^*)^2$, and the inner product is non-negative by optimality of x^*. Hence $\Delta := \overline{\mathrm{SC}}(x^u) - \overline{\mathrm{SC}}(x^*) \geq \sum_z \frac{m_z^S}{2} (\Delta \bar{Q}_z)^2$. By the identity with $Q^\circ = \bar{Q}^*$, $v = 0$: $\overline{\Phi}_u(x) = \overline{\mathrm{SC}}(x) - \sum_z \frac{\zeta_z}{2} (\bar{Q}_z - \bar{Q}_z^*)^2 + L(x) + \text{const}$. Since x^u minimises $\overline{\Phi}_u$, $\Delta \leq \sum_z \frac{\zeta_z}{2} (\Delta \bar{Q}_z)^2 + L(x^*) - L(x^u)$. Combining, $\sum_z \frac{b_z + \zeta_z}{2} (\Delta \bar{Q}_z)^2 \leq L(x^*) - L(x^u)$; then $\sum_z \frac{\zeta_z}{2} (\Delta \bar{Q}_z)^2 \leq \varrho [L(x^*) - L(x^u)]$ and $\Delta \leq (1 + \varrho)[L(x^*) - L(x^u)]$. Finally $|L(x^*) - L(x^u)| \leq \sum_{i,z} \frac{m_z}{2} |d_{iz} - \bar{d}_z| \, |\bar{q}^*_{iz} - \bar{q}^u_{iz}| \leq E$ because $\bar{q}_{iz} \in [0, d_{iz}]$. □

Proof of Theorem 26. With $\bar{d} = d$, L vanishes on the feasible set, so $E = 0$ in Theorem 25 and every minimiser of $\overline{\Phi}_u$ has aggregate $\bar{Q}^*$ and social cost $\overline{\mathrm{SC}}(x^*)$. Moreover $\overline{\Phi}_u = \overline{\mathrm{SC}} - \sum_z \frac{\zeta_z}{2} (\bar{Q}_z - \bar{Q}_z^*)^2 + \text{const}$; the subtracted term has zero gradient at x^*, so x^* satisfies the first-order optimality condition of the convex function $\overline{\Phi}_u$. □

Proof of Theorem 27. (a) $J_i^u(p) = J_i^0(p) + \langle u, q_i(p) \rangle$, so constraint (i, p) reads $J_i^u(p) \geq J_i^u(\sigma_i^*)$; all of carrier i's constraints hold iff $\min_p J_i^u(p) \geq J_i^u(\sigma_i^*)$, and the violation $J_i^u(\sigma_i^*) - J_i^u(p)$ is maximised by a best response. (b) If a round adds a cut (i, b_i), the current u violates it and satisfies every earlier cut (it is LP-feasible), so the cut is new; there are $\sum_i (|\mathcal{P}_i| - 1)$ non-trivial constraints. Every cut is a valid constraint of $P(\sigma^*)$, so an infeasible LP shows that $P(\sigma^*)$ is empty. □

Proof of Proposition 28. σ is an equilibrium under u iff $u \in P(\sigma)$, so both descriptions of $V^\star$ range over the same pairs (u, σ). The inequality is an inclusion of feasible sets. If the loop ends with $u \in P(\sigma^*)$, the leader attains $\mathrm{SC}(\sigma^*)$. □

Proof of Theorem 29. Best-response dynamics strictly decreases Φ_u on a finite set, so it ends at a pure equilibrium with $\Phi_u(\sigma^u) \leq \Phi_u(\sigma^*)$. Substitute the identity with $Q^\circ = Q^*$ on both sides; the constants cancel and the congestion term vanishes at σ^*. □

Proof of Corollary 30. Count the calls of the three phases; Theorem 27 bounds the rounds, and each improvement of the fallback dynamics costs at most N calls. □

Proof of Theorem 31. By Lemma 5, carrier i's Γ-maximin payoff is its precise payoff minus $\delta\langle a_i(p), Mw\rangle$ (traffic already enters the route loss at its upper expectation). With $a_i = (q_i, rq_i)$, $\delta\langle a_i(p), Mw\rangle = \langle \hat{s}(\delta), q_i(p)\rangle$, so the Γ-maximin cost under u is the precise cost under $u + \hat{s}(\delta)$, and u implements σ^* iff $u + \hat{s}(\delta) \in P(\sigma^*)$. The social cost does not contain the surcharge. The last claim is Lemma 4: an equilibrium of the precise game lies in $\mathrm{ICNE}_C(\delta)$ for $C \in \{E, M, ID\}$ and every δ. □

Proof of Proposition 32. The first claim is Theorem 26 with a uniform tangent charge. For the second, $Q_z^{\mathrm{full}} = n_z d_z$ with equal lots, and without congestion ($m_z \approx b_z$ when $\zeta_z \ll b_z$) $u_z = -\frac{b_z}{2} d_z = -\frac{\alpha_z}{4n_z} = -0.375\,\mathrm{fee}/n_z$. The last claim is Proposition 23. □

Appendix D. Additional tables

Table D.11: Coverage of the exact-oracle game experiments per city and seed: exact grid points of the leader experiment (out of 63), price-of-anarchy experiment computed (yes/–), and exact widths of the imprecision experiment (out of 12); – means not computed with the exact oracle. The base equilibria are exact for all city-seeds.

City	E2 s0	E3 s0	E4 s0	E2 s1	E3 s1	E4 s1	E2 s2	E3 s2	E4 s2
New Delhi	63/63	yes	12/12	63/63	yes	12/12	63/63	yes	12/12
Cairo	13/63	yes	6/12	17/63	yes	6/12	13/63	yes	–
Moscow	12/63	yes	6/12	10/63	yes	6/12	–	–	–
Buenos Aires	63/63	yes	12/12	63/63	yes	12/12	63/63	yes	12/12
New York	7/63	yes	6/12	7/63	yes	6/12	–	–	–
Mumbai	15/63	yes	12/12	14/63	yes	12/12	14/63	yes	12/12
London	63/63	yes	12/12	63/63	yes	12/12	63/63	yes	12/12
Paris	14/63	yes	6/12	13/63	yes	6/12	13/63	yes	–
Berlin	13/63	yes	6/12	13/63	yes	6/12	–	–	–

Table D.12: Calibration per city (base values; ranges and sources in `data/parameters/SOURCES.md`). Parcels per capita are country-level; TomTom congestion is the 2025 city value (Moscow: 2020); drops per vehicle and van capacity are proxies outside the United States; fees and wages are graded C for most cities.

City	parcels/cap./day	cong. %	drops/veh.	shift h	van cap.	fee USD	wage USD/h	fuel USD/L	N	veh./1000
New Delhi	0.00964	60.2	40.0	10	40.0	0.690	1.07	1.15	5	400
Cairo	0.00561	40.5	40.0	8.4	40.0	1.18	0.610	0.440	4	220
Moscow	0.156	54.0	70.0	8.4	126	3.67	4.53	0.900	5	297
Buenos Aires	0.0150	48.1	70.0	8.4	126	0.980	1.28	1.45	4	331
New York	0.184	48.8	120	8.4	226	9.09	22.3	1.22	4	242
Mumbai	0.00964	63.2	40.0	10	40.0	0.690	1.07	1.15	5	330
London	0.166	51.6	72.0	7.3	126	4.21	18.8	2.34	5	288
Paris	0.0650	40.0	70.0	8.4	126	6.04	15.7	2.42	5	575
Berlin	0.143	44.6	126	8.4	206	4.40	18.8	2.39	6	330

Table D.13: Sensitivity of unpriced welfare (% change, low to high value of each weak parameter, seed 0, earlier price-and-branch oracle; not re-run with the exact oracle).

City	fee	wage	drops	N	parcels
New Delhi	-58 to 317	-53 to 43	-90 to 286	-38 to 53	-9 to 0
Cairo	19 to 19	-5 to 2	-31 to 59	-26 to 23	-0 to 0
Moscow	-75 to 48	-26 to 6	-43 to 105	-62 to 20	-0 to 1
Buenos Aires	-61 to 419	-51 to -51	-87 to 443	-18 to 63	-1 to 0
New York	-41 to 55	-8 to 1	-57 to 9	-27 to 23	-0 to 0
Mumbai	-45 to 202	-38 to 24	-72 to 168	-18 to 29	-5 to 1
London	-95 to 363	-3 to 34	-95 to 789	-26 to 107	-2 to 4
Paris	-58 to 83	-4 to 10	-56 to 148	-16 to 27	-0 to 1
Berlin	-59 to 24	-6 to 2	-83 to 4	-3 to -3	-0 to 1

Table D.14: City leader, broad objective with the background-delay proxy (3 seeds). Gains as in Table 3 (mean [min, max]); combined gain per seed; served share at the best point with both instruments. Shut-down optima (served share 0) are an artifact of the proxy.

City	cong. only	deliv. only	both	both by seed	served
New Delhi	77.1 [64, 85]	100	100	100 / 100 / 100	0
Cairo†	0	2.06 [0, 3.3]	4.59 [2.5, 7.5]	2.47 / 7.50 / 3.81	0.67 [0.65, 0.69]
Moscow†	3.47 [3.1, 3.9]	0.0345 [0, 0.069]	4.16 [3.6, 4.7]	3.59 / 4.72	0.92 [0.91, 0.93]
Buenos Aires	2.41 [1.4, 4.0]	0	3.71 [1.1, 5.4]	4.62 / 5.42 / 1.09	0.35 [0.29, 0.46]
New York†	0	0.388 [0.26, 0.52]	0.388 [0.26, 0.52]	0.259 / 0.518	1.0
Mumbai†	66.7 [65, 71]	100	100	100 / 100 / 100	0
London	90.9 [78, 100]	100	100	100 / 100 / 100	0
Paris†	37.7 [32, 44]	74.2 [72, 76]	99.1 [98, 100]	99.3 / 100 / 98.1	0.042 [0, 0.085]
Berlin†	3.14 [2.6, 3.7]	1.98 [1.5, 2.5]	2.40 [2.1, 2.7]	2.13 / 2.66	0.78 [0.68, 0.88]

Table D.15: ZMC on synthetic D1 games with non-separable plan sets (five seeds per row). Entries: mean oracle calls / mean potential gap to the convexified optimum; last column: instances in which ZMC has the smallest gap among FW8, CG8 and multi-start BR.

N	Z	BR	BR multi-start	FW8	CG8	ZMC	CG full	ZMC best
4	30	12 / 5.33	45 / 3.22	38 / 3.64	39 / 3.20	39 / 2.77	58 / 2.69	4/5
5	60	21 / 15.8	56 / 11.8	50 / 7.78	51 / 7.80	49 / 6.90	159 / 6.13	4/5
6	120	28 / 50.2	70 / 38.4	61 / 21.4	63 / 27.5	60 / 17.1	365 / 14.2	4/5
6	250	26 / 91.5	71 / 80.2	64 / 47.9	67 / 55.4	60 / 33.4	797 / 27.8	5/5

Table D.16: Tiny games solved by exhaustive enumeration (earlier price-and-branch oracle): share of instances in which the method's profile is a Nash equilibrium of the enumerated game and reaches the global potential minimum, largest independent-oracle regret and mean excess of the potential over the global minimum.

Method	games	calls	is NE	max regret	global min.	excess
ZMC (6 rounds)	40	17.5	1.00	0	1.00	0
BR	40	4.85	1.00	0	0.875	0.93
FW (6 rounds)	40	16.7	1.00	0	0.950	0.067
CG	40	11.1	1.00	0	0.975	0.020

Table D.17: Checks of ZMC on the city games (seed 0, earlier price-and-branch oracle): largest residual of the clearing identity (Theorem 33(a)); share of rounds in which its bound holds; inverted brackets; active coordinates; largest relative bracket width after eight rounds; in bisection mode, share of coordinates within the bound of Theorem 34 at $T = 8$, largest normalised error and the bound 2^{-4}.

City	identity resid.	bound holds	inverted	active	width_8	within_8	err_8	bound_8
New Delhi	$2.8 \cdot 10^{-14}$	1.00	0	128	0.0435	1.00	0.00385	0.0625
Cairo	$1.6 \cdot 10^{-13}$	0.875	0	64	0.0435	1.00	0.00385	0.0625
Moscow	$8.1 \cdot 10^{-13}$	1.00	0	142	0.0435	1.00	0.00385	0.0625
Buenos Aires	$5.7 \cdot 10^{-14}$	1.00	0	124	0.0435	1.00	0.00385	0.0625
New York	$3.4 \cdot 10^{-12}$	1.00	0	128	0.0435	1.00	0.00385	0.0625
Mumbai	$4.3 \cdot 10^{-14}$	1.00	0	110	0.0435	1.00	0.00385	0.0625
London	$1.1 \cdot 10^{-13}$	1.00	0	178	0.0435	1.00	0.00385	0.0625
Paris	$1.5 \cdot 10^{-12}$	1.00	0	136	0.0415	1.00	0.00385	0.0625
Berlin	$1.8 \cdot 10^{-12}$	1.00	0	164	0.0435	1.00	0.00385	0.0625

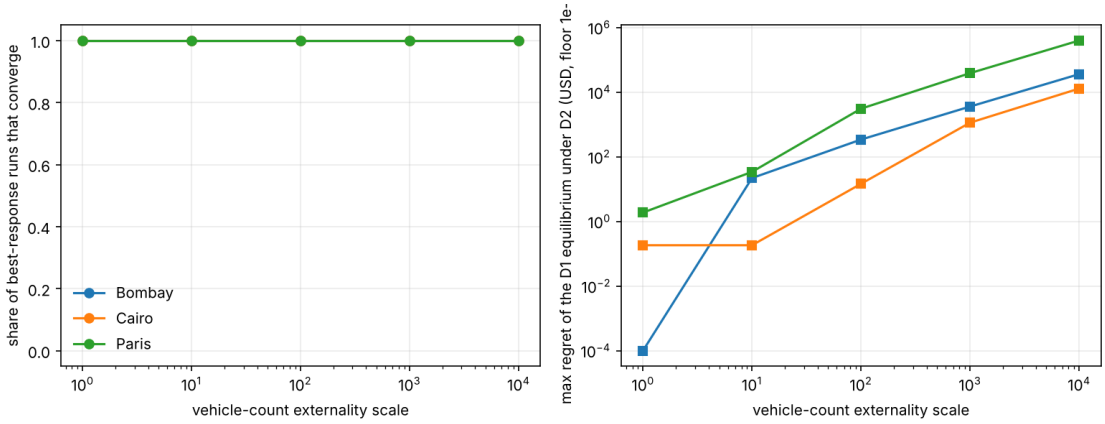


Figure D.8: City games of Cairo, Mumbai and Paris (three seeds) with a vehicle-count externality of increasing scale. Left: share of best-response runs that converge to a pure equilibrium. Right: largest regret of the D1 equilibrium when the game is of type D2 (USD).